

claude-windows / d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\subagents\agent-a705e7a7d538f83b1.jsonl`
- SHA-256: `83e7a0359ba876c450f7fe0fea66f6919b5cc180fa602c2a36d43652ad6bf47a`
- Source modified: `2026-07-04T12:33:13+00:00`
- Imported at: `2026-07-05T16:48:23+00:00`
- project: `subagents`
- session_id: `d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd`
```

Transcript

1. user (2026-07-04T12:17:45.600Z)

You are assembling the FINAL "A13 alpha launch review" document for Khelsutra/badminton-highlight-indexer, working in the git worktree C:\Users\avidu\Projects\khelsutra-guru\wt-a13 (branch docs/a13-launch-review, already checked out at master 122f174). Windows box; use PowerShell for shell commands. Do NOT commit, do NOT push ? leave changes in the working tree; the orchestrator reviews and commits.

Inputs (read ALL three first)

- **Four drafted sections + a completeness critique**** (JSON, from a research fan-out): read the file
`C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\tasks\w3atf3i6j.output`
? it contains `result.drafts` (keys: runbook, observability, privacy, limits ? each a complete markdown section with file:line citations and an evidence_gaps list) and `result.critique` (missing items + contradictions to FIX).
- **Live test-battery evidence**** (measured this session, authoritative):
`C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\scratchpad\A13-live-evidence.md`
- The repo docs it must fit: `docs\ALPHA\_LAUNCH\_READINESS.md` (tracker; §7 row A13, the §"Definition of done" checklist near line 224, and the open owner questions near line 216), `docs\PROJECT\_DOC\_TEMPLATE.md` if present (match the repo's tracked-doc conventions), `docs\DOC\_STATUS.md` (the doc register ? add a row).

Deliverable

Write `docs\ALPHA\_LAUNCH\_REVIEW\_2026-07-04.md` in the worktree: the owner-facing A13 launch review. Structure:

- **TL;DR + GO/NO-GO recommendation**** (recommend, don't decide ? the decision is the owner's). The recommendation shape the evidence supports: *conditional GO for the CURRENT 4-email allowlist* ? conditions: fix the khelsutra copy blockers (khelsutra issue #154) and complete the owner browser-CUJ checklist; *NO-GO for any wider invite wave* until owner closes A10 (ship-gate) + A11 (WASB-C3) and the SA least-privilege + retention items are addressed. Justify from evidence.
- **Scorecard vs the tracker's Definition-of-done checklist**** ? map each DoD checkbox to its current truth with evidence (some are now genuinely done; the browser-CUJ one is NOT ? say so plainly and include a 5-minute owner checklist to complete it: log into api.khelsutra.guru via CF, upload a ?720p clip, process with cloud compute, watch job status, compile, download the reel).
- The four sections from the drafts (runbook, observability & triage, privacy & consent, known limits + rollback), EDITED as follows:
 - **Incorporate the live battery evidence**** (file 2) wherever it strengthens or supersedes a draft claim: the failure-mode table (great for the observability/triage section), the concurrency smoke, the Gemini-leg verification (48?43 segments, real gemini-flash-latest calls ?

goes under known-limits/"validated beyond the alpha config" or a short new subsection; the alpha itself STAYS WASB-only), the secret-accessor grant gap that was found+fixed, the SPA-served-at-root single-origin fact (kills the CORS-preflight worry for the single-origin path), the cert-healthy-today check, and the improved auth-off-window practice (remove allUsers during windows).

- ****Apply every critique fix****: add the cost subsection (state what runs idle: min-instances 1, no-cpu-throttling, 2 vCPU/4Gi CP + L4 per-execution billing with the ~\$1-2/run precedent from docs/archives/decisions/DECISIONS.md:675; mark exact monthly \$ as [unverified ? owner billing-console check]); add the expected-quality statement for testers (WASB-only numbers from the tracker: served toIF1 0.364 / F1@0.5 0.472 vs baseline 0.138/0.237, over-splitting guard still failing ? frame as "roughly-right boundaries, some over-splitting; no quality claims allowed until A10"); add the SA blast-radius paragraph (compute SA holds project roles/editor + self-tokenCreator on an allUsers-invoker service ? recommend least-privilege SA before wider invites); add the tester onboarding one-pager content; assign watch-items an owner + mechanism (recommend a dated GitHub issue or calendar entry for the #472 cert check ~2026-09-30 and a billing check); include the two site-copy privacy items via khelsutra #154; fix the specific contradictions the critique lists (downgrade "used live" to the runlog-cited observation, tag the billing-stops-on-cancel claim [unverified], align the SetupWizard vs policy-page severity, drop 'beta' from the executions-logs command citation, fix the 1-vCPU/2-vCPU quote mismatch).

- ****SCRUB for repo publication****: NO operational secret paths (remove every `G:\My Drive\...` reference ? replace with "owner-held token"), NO tester email addresses, no `memory/session-handoff` citations (replace with "session records" or drop). The doc must be publishable in the repo.

- Keep the drafts' file:line citations; keep [unverified]/[analysis] tags honest.

4. ****Open gates & owner questions**** ? A10/A11/A12 status, the tracker's open owner questions, and the khelsutra #154 copy blockers.

Also edit:

- `docs\ALPHA\_LAUNCH\_READINESS.md`: A13 row ? Status `GATED` (review DELIVERED, GO/NO-GO awaiting owner) with PR column pointing at this doc; update any DoD checklist boxes that are NOW genuinely evidenced (cite), leaving the browser-CUJ box unchecked.

- `docs\DOC\_STATUS.md`: add the new doc's register row following the existing format.

Gates (run from the worktree root, report exact outputs)

1. `python -m tools.check\_md\_links` ? must be clean (fix any broken links you introduce).
2. `python -m pytest tests/ -q -p no:cacheprovider` ? full suite (some tests lint docs).
3. `python -m ruff check .`

Style: match the repo's dense, evidence-cited doc voice (look at the runlogs + tracker). Be honest everywhere ? this doc's whole value is that the owner can trust it. Return: files changed, gate outputs verbatim (final lines), and a 5-bullet summary of the review's verdict/emphases.

2. user (2026-07-04T12:17:51.981Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-44 of 197 total (31482 tokens, cap 25000). Call Read with offset=45 limit=44 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

```
1      {
2      "summary": "Fan out 4 research agents over the repos to draft the A13 alpha launch
review sections",
3      "agentCount": 5,
4      "logs": [],
5      "result": {
6      "drafts": [
7      {
8      "key": "runbook",
9      "markdown": "## Operator runbook ? live A1 alpha (Cloud Run asia-
southeast1)\n\n> Scope note: this is an **operations** runbook for the stack that is already
live. It does not imply launch GO ? A10 (ship-gate target) and A11 (WASB-C3 provenance) are
still **GATED** owner rows (`docs/ALPHA_LAUNCH_READINESS.md:109-110`).\n\n**Live topology** (e2e
runlog `docs/COMPUTE_DECOUPLED_SERVING/runlogs/DEPLOY_REPORT_cloud-
serving_a1-e2e_2026-07-04.md:15-21,95-104`): project `gen-lang-client-0306026826`, region `asia-
```

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southeast1`; control-plane **Service** `rally-control-plane` rev 00007 (edge-auth ON,
`maxScale=1`, image digest `31475f27?` = master `3c97117`); L4 GPU **Job** `rally-worker`
(nvidia-l4, 8 vCPU/32Gi, GCS-FUSE `khelsutra-rally-serving`?`gcs`); storage `gs://khelsutra-
rally-serving`; WASB-only (`indexing.skip_ai_handoff=true` ? no Gemini key anywhere in serving,
`deploy/cloudrun/gen_service_yaml.py:55-56,30-32`). `api.khelsutra.guru` fronts the service via
a Cloud Run domain mapping behind a Cloudflare Access app (4-email allowlist).\n\n### 1. Deploy
/ redeploy the control plane\n\nCanonical path is the **committed manifest generator**
(`deploy/cloudrun/README.md:72-79`; generator defaults bake project/region/service/bucket/aud at
`deploy/cloudrun/gen_service_yaml.py:42-47`):\n\n``powershell\npython
deploy/cloudrun/gen_service_yaml.py --edge-auth on -o service.yaml\ngcloud run services replace
service.yaml --region asia-southeast1\n``\n\n- `--edge-auth` is **required and explicit**;  

`off` is only for smoke/e2e windows ($4 below) (`gen_service_yaml.py:124-131`).\n- Run gcloud
from **PowerShell, not git-bash** ? MSYS mangles `/tmp`- and `/gcs`-style values before gcloud
sees them (`deploy/cloudrun/README.md:129-133`; `gen_service_yaml.py:12-13`).\n- Do NOT add
`CLOUD_RUN_JOB` to the manifest ? it's a reserved runtime env name and the manifest is rejected;
the code default `rally-worker` is already correct (`gen_service_yaml.py:28-29`).\n\n**Gotcha A
? `services replace` with an unchanged manifest is a NO-OP**: it does not re-resolve `:latest`
after a rebuild. Force a new revision with `gcloud run services update rally-control-plane
--image asia-southeast1-docker.pkg.dev/gen-lang-client-0306026826/rally/rally-worker:latest
--region asia-southeast1` (that exact fallback was used live) or stamp a revision suffix; a
generator `--revision-suffix` flag is a filed follow-up candidate (e2e runlog $5,
`a1-e2e_2026-07-04.md:83-85`).\n\n**Gotcha B ? a redeploy WIPES per-instance state** (engine
issue #471, confirmed open): the jobs/videos DB is ephemeral SQLite in the instance's `/tmp`, so
any redeploy invalidates in-flight jobs, job history, and past `/api/highlights` download URLs ?
the handler 404s on the DB record check (`backend/main.py:1052` per the runlog) even though the
reel mirror still exists in gs://; re-processing regenerates state (e2e runlog
`a1-e2e_2026-07-04.md:86-89`; issue #471 body). Corollary: **keep `maxScale=1`** ? >1 instance
splits the queue and `/api/jobs/{id}` polls (issue #471; encoded as the generator default,
`gen_service_yaml.py:145-150`). Operationally: redeploy only when no tester job is in
flight.\n\n### 2. Rebuild the image + re-pin the L4 job\n\nOne image serves both the worker Job
and the control plane (post-#470 it carries the `cloud-run` + `auth` extras;
`gen_service_yaml.py:136-140`).\n\n``powershell\n# 1. Rebuild (~8-10 min heavy CUDA build;
`gcloud builds submit` has NO -f flag, hence the config)\ngcloud builds submit
--config=deploy/cloudrun/cloudbuild.yaml --project gen-lang-client-0306026826
.\n``\n\n(`deploy/cloudrun/cloudbuild.yaml:3-6,26`;  

`deploy/cloudrun/README.md:23-26`)\n\n``powershell\n# 2. Re-pin the Job to the new
digest\ngcloud run jobs update rally-worker --image asia-southeast1-docker.pkg.dev/gen-lang-
client-0306026826/rally/rally-worker:latest --region asia-southeast1\n# 3. Re-point the service
(Gotcha A: replace won't re-resolve `:latest`)\ngcloud run services update rally-control-plane
--image asia-southeast1-docker.pkg.dev/gen-lang-client-0306026826/rally/rally-worker:latest
--region asia-southeast1\n``\n\nThe re-pin step is what the e2e deploy did after each build
(`the Job re-pinned via `gcloud run jobs update`, `a1-e2e_2026-07-04.md:18`); the exact flag
expansion above follows the session record (`gcloud run jobs update rally-worker --image
:latest`, memory `current-state.md:64`) [full flag string unverified against a committed
doc].\n\n- **First execution after a push** sits minutes in `Container image import in
progress` ? visible in `gcloud run jobs executions describe <exec> --region asia-southeast1`
`status.conditions`, NOT in logs; it's progress, not a hang
(`deploy/cloudrun/README.md:134-137`).\n\n### 3. Add / remove an alpha tester (Cloudflare
Access)\n\nTester auth is entirely CF-side ? **no engine redeploy needed** (the app's aud/team-
domain stay constant in the manifest; `gen_service_yaml.py:46-47`).\n\n- Where: Cloudflare Zero
Trust dashboard, team `khelsutra.cloudflareaccess.com` ? Access ? Applications ? **`KhelSutra
Studio (alpha)`** (one self-hosted app covering both `app.khelsutra.guru` and
`api.khelsutra.guru`, aud `c980da67?082b`) ? edit the Allow policy's **Emails** rule (session
notes: memory `session-handoff.md:167-174`; app shape + email-OTP policy per
`khelsutra/deploy/DEPLOY_ALPHA.md:111-115`; the a1-cuj runlog confirms the same app/allowlist
fronts the Cloud Run service, `a1-cuj_2026-07-04.md:80-83`).\n- Current allowlist is 4 emails
(enumerated in memory `session-handoff.md:169`). Free tier covers 250 users; one-time-PIN means
testers need no Cloudflare account (`khelsutra/deploy/DEPLOY_ALPHA.md:113`).\n- API alternative:
the scoped CF token at `G:\My Drive\cloudflare.token.txt` has Access-Apps edit (memory
`session-handoff.md:170-171`).\n- Remove = delete the email from the same policy. [unverified]
whether an already-issued CF session survives until expiry after removal ? verify posture after
removing a tester.\n- Triage quirk to know: testers' uploads below **1280x720 are rejected** by
validation (`resolution_fps_check`) ? a real e2e first-attempt failure, not an auth problem
(`a1-e2e_2026-07-04.md:41-42`).\n\n### 4. Edge-auth OFF/ON windows\n\n**OFF is only for
identity-token smoke/e2e tests** (`deploy/cloudrun/README.md:80-81`;
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`gen_service_yaml.py:18-20,124-131`). Why it's dangerous: the service carries `allUsers`
run.invoker so Cloudflare's proxy can reach it ? with edge-auth ON the app-level Cf-Access JWT
check is what keeps the direct `*.run.app` URL at 401 (`?a1-cuj_2026-07-04.md:94-96`); with
edge-auth OFF that last gate is gone, so keep the window short and restore
immediately.\n\n``powershell\n# open the smoke window\npython
deploy/cloudrun/gen_service_yaml.py --edge-auth off -o service.yaml\ngcloud run services replace
service.yaml --region asia-southeast1\n# drive it with an identity token during the window\ncurl
-H "Authorization: Bearer $(gcloud auth print-identity-token)" https://rally-control-
plane-2bosgeukpa-as.a.run.app/api/health\n# restore\npython deploy/cloudrun/gen_service_yaml.py
--edge-auth on -o service.yaml\ngcloud run services replace service.yaml --region asia-
southeast1\n```\n(commands: `deploy/cloudrun/README.md:77-79`; identity-token pattern: memory
`session-handoff.md:119-120`, e2e runlog $2. Note the replace here is NOT a no-op because the
edge-auth flip changes the baked config. Remember Gotcha B: each flip wipes SQLite
state.)\n\n**Posture verification after restore** (exactly what the e2e verified,
`?a1-e2e_2026-07-04.md:95-104`):\n1. `GET /api/health` on the run.app URL ? **200**
(deliberately exempt; it's the no-auth liveness probe, `backend/main.py:321-325`).\n2. `POST
/api/process` with valid JSON but no CF JWT ? **401** `edge auth: missing Cf-Access-Jwt-
Assertion header`.\n3. `curl -i https://api.khelsutra.guru/api/health` with no CF session ?
**302** to `https://khelsutra.cloudflareaccess.com/cdn-cgi/access/login/api.khelsutra.guru??`
with aud `c980da67?082b` in the login meta.\n4. Triage quirk: a **malformed-JSON** body gets
FastAPI's 422 *before* the edge-auth dependency ? a 422 is not an auth bypass
(`?a1-e2e_2026-07-04.md:91-93`).\n\n### 5. Health / capabilities checks and logs\n\n- `GET
/api/health` ? `200 {"status":"ok",?}`; `GET /api/capabilities` ? `wasb_hybrid` present,
`gemini_key_present=false` (correct for the WASB-only alpha), ffmpeg/ffprobe present
(`?a1-e2e_2026-07-04.md:25-27`; `?a1-cuj_2026-07-04.md:31-33`). Both endpoints are app-level no-
auth probes (`backend/main.py:321-325,359-363`) but sit behind the CF login on
`api.khelsutra.guru`.\n- **Job (L4) logs**: `gcloud run jobs executions list --job=rally-worker
--region=asia-southeast1`, then `gcloud beta run jobs executions logs read <execution>
--region=asia-southeast1` ? the committed pattern for Cloud Run jobs in this repo
(`deploy/gcp/nightly_regression/README.md:111`, adapted from the nightly job to `rally-
worker`).\n- **Cloud Logging filters** [unverified ? no committed filter string exists; standard
resource labels]:\n  - service: `gcloud logging read "resource.type=cloud_run_revision AND
resource.labels.service_name=rally-control-plane" --project gen-lang-client-0306026826 --limit
50`\n  - job: `gcloud logging read "resource.type=cloud_run_job AND
resource.labels.job_name=rally-worker" --project gen-lang-client-0306026826 --limit 50`\n
Polling `gcloud logging read` ~every 60s and grepping
`worker|wasb|rally|ERROR|Traceback|CUDA|OOM|Killed` is the established practice from the live
runs (memory `feedback-tail-long-job-logs.md:52-53`).\n- **Standing owner rule**: any GPU
execution expected to exceed ~20 min must be actively tailed with a ~5-min heartbeat, alerting
on stall ? not just awaited (memory `feedback-tail-long-job-logs.md:18-28,49-53`).\n- Dispatch
completion is a **bucket marker** (`<out_uri>/_dispatch/<token>.done`), not Cloud Run API
status; a failed Job writes a failure marker with the real rc so polls terminate fast
(`deploy/cloudrun/README.md:109-115`).\n\n### 6. Cost hygiene\n\n- **What runs idle**: the
control plane holds `min-instances 1` + `no-cpu-throttling` ? one always-warm ~2-vCPU CPU
instance (`"cheap CPU; NOT the D7 warm **GPU** pool"`, `deploy/cloudrun/README.md:86,90-93`).
The L4 Job is scale-to-zero and **bills only per execution** (`deploy/cloudrun/README.md:5`; one
full CUJ ? 14 min of L4+8vCPU, `?a1-cuj_2026-07-04.md:105`). Owner directive as of 2026-07-04:
keep the stack running (`?a1-cuj_2026-07-04.md:106`).\n- **Teardown** (if the alpha pauses):
`gcloud run services delete rally-control-plane --region asia-southeast1` and `gcloud run jobs
delete rally-worker --region asia-southeast1`; the bucket/secret/Artifact-Registry repo are
retained (`?a1-cuj_2026-07-04.md:107-108`).\n- **GCP VM always-DELETE rule** (applies to any
GPU-VM diagnostics, not the Cloud Run stack): a STOPPED VM still bills its disk;
`deploy/gcp/teardown.sh` defaults to `delete` (owner cost rule 2026-06-20;
`deploy/gcp/README.md:248-257`, `deploy/gcp/teardown.sh:9-12`; memory `MEMORY.md:33` ? also
audit `gcloud compute disks list` for orphans).\n- Budget guardrail: keep the **$100 budget
alert** live (`deploy/gcp/README.md:18`; D17 per `deploy/cloudrun/README.md:8-9`). [unverified]
whether the alert threshold was raised after the owner lifted the spend gate to $300
pre-A1.\n\n### 7. Cert-renewal watch item (issue #472, open)\n\n`api.khelsutra.guru` = Cloud Run
**domain mapping** (CNAME `ghs.googlehosted.com`, **proxied** through Cloudflare) + a Google-
managed cert. Google-managed certs renew ~every 90 days, and renewal validation **can stall
behind the proxied CNAME** (Google's challenge sees Cloudflare's edge, not the expected record)
(issue #472 body).\n\n- **Check before ~2026-09-30**: \n  ``powershell\n  gcloud beta run
domain-mappings describe --domain api.khelsutra.guru --region asia-southeast1\n  ``\n  and
confirm the cert status shows renewal is not stalled (issue #472 body).\n- **If it stalls**:
either temporarily grey-cloud (DNS-only) the CNAME so validation passes, then re-proxy; or

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switch the mapping to a ****Cloudflare Origin CA cert**** so renewal never depends on what Google sees through the proxy (issue #472 body). CF Access (M9) is unaffected either way ? this is only the Google-side TLS cert on the mapping.",

```
10         "evidence_gaps": [  
11             "Exact `gcloud logging read` filter strings are not committed in any repo doc  
? the service/job filters given use standard Cloud Run resource labels and are marked  
[unverified]; only the polling practice (poll ~60s, grep failure signatures) is grounded in  
session memory (feedback-tail-long-job-logs.md:52-53).",  
12             "The full flag expansion of the job re-pin command (`gcloud run jobs update  
rally-worker --image <full path>:latest --region asia-southeast1`) ? the e2e runlog records that  
`gcloud run jobs update` was used (DEPLOY_REPORT_cloud-serving_a1-e2e_2026-07-04.md:18) and  
session memory records the short form (current-state.md:64), but no committed doc carries the  
exact full command string.",  
13             "Whether removing a tester's email from the CF Access policy immediately  
invalidates an already-issued CF session, or the session persists until expiry ? not documented  
anywhere I could open.",  
14             "The precise Cloudflare dashboard click-path for editing the live 'KhelSutra  
Studio (alpha)' app's allowlist policy is grounded in session memory (session-  
handoff.md:167-174) and the droplet-era DEPLOY_ALPHA.md §4 (written for creating a new app), not  
a committed Cloud-Run-era doc.",  
15             "Whether the $100 GCP budget alert (deploy/gcp/README.md:18) is still the live  
threshold after the owner lifted the pre-A1 spend gate to $300 (current-state.md:151) ? could  
not verify without touching GCP, which was out of scope.",  
16             "Cloud Run domain-mapping cert status could not be checked live (READ-ONLY  
constraint, no GCP calls) ? the #472 check command is quoted from the issue body, not re-  
executed."
```

```
17         ]  
18     },  
19     {  
20         "key": "observability",  
21         "markdown": "## Observability & job-failure triage\n\nScope: the live A1 stack ?  
Cloud Run service `rally-control-plane` (rev 00007, edge-auth ON) + L4 Job `rally-worker`,  
bucket `gs://khelesutra-rally-serving` (DEPLOY_REPORT_cloud-serving_a1-e2e_2026-07-04.md:1-22).  
All engine paths below are relative to `badminton-highlight-indexer`. \n\n**Reading a job's  
state.** `GET /api/jobs/{job_id}` returns `{job_id, status, progress, video_id, error, result,  
reports, created_at, started_at, finished_at}` (backend/main.py:1008-1027). Two failure  
channels, deliberately distinct (backend/api/job_worker.py:148-153): \n- `status='done'` +  
`result.status='failed'` ? the **pipeline's verdict** (validation, staging, dispatch, cloud  
rc?0, stitch?) with a human `result.message`; \n- `status='failed'` + `error=\"<ExcType>:  
<msg>\"` ? the **worker itself crashed** (backend/api/job_worker.py:191-193). \n\nThe `progress`  
field is the first triage signal ? the worker stamps every stage: `starting` ? hashing ?  
validating ? staging upload to bucket ? dispatching (cloud_run) ? ingesting (cloud_run) /  
segmenting (<name>) ? finalizing; compile: `resolving` ? stitching ? uploading reel ? finished`  
(backend/infrastructure/database_jobs.py:105-118; backend/orchestration.py:287, 324, 363, 447,  
516, 559, 608, 793, 816, 851, 861). A stuck job's last `progress` value tells you which stage it  
died in. \n\n### Triage table\n\n\"CP logs\" = `rally-control-plane` service logs (Cloud Logging,  
`cloud_run_revision`); \"worker logs\" = `rally-worker` Job execution logs (`cloud_run_job`).  
The whole backend logs via stdout `logging` only (backend/main.py:26-30) ? there is no  
metrics/tracing layer. \n\n| Failure class | CLIENT sees | OPERATOR sees (log pattern ? where) |  
First debugging step | \n|---|---|---| \n| Edge-auth reject (M9) | `401 {\"detail\": \"edge  
auth: missing Cf-Access-Jwt-Assertion header ?\"}` (or verification-failure detail) on  
process/compile/annotations/assets (backend/main.py:910-913, 1107-1110;  
backend/api/auth.py:134-147) | Nothing app-level ? only the request-log 401 on CP. NB: malformed  
JSON gets FastAPI's **422 before** edge-auth runs (deploy report §5, runlogs/?a1-e2e?md:91-93) |  
Confirm the caller came through `api.khelesutra.guru` (CF Access) not the raw run.app URL; check  
the 4-email allowlist in CF | \n| Submit-time input validation | `400` unsupported scheme / null  
byte (backend/main.py:923-933); `404` \"Video file not found at '?'\" (main.py:940-943); `400`  
\"Segmenter '?' not found. Available: [?]\") (main.py:953-958); upload-init `400` bad  
filename/extension (backend/api/uploads.py:91-99) | No app log ? HTTPException raised before any  
logging | Reproduce the request; the `detail` string is self-describing | \n| **465 input-root  
jail** (hosted mode) | `400 {\"detail\": \"hosted mode: 'gs://?' is not under the allowed input  
root 'gs://?'\"}` ? a URI outside `storage.input_root`, or any URI when no uri-root is  
configured (backend/utils/paths.py:97-103, wired at main.py:915-933; PR #465 merged) | No app  
log (pre-work 400) | Compare the submitted URI against `storage.input_root` in the deployed  
config (`GET /api/config` is redacted but shows storage roots) | \n| In-worker video validation
```

failure (e.g. resolution floor ?1280x720) | job `done`, `result.status='failed'`,
`message="\`Video failed validation: resolution_fps_check: ?\`" + per-validator `results`
(backend/orchestration.py:68-76, 520-536) | CP: `Running validations for <path>...`
(orchestration.py:517) then the terminal `[COMPUTE] video\_id=? status=failed` line
(orchestration.py:632-640) | Read `result.results[[]].details` (ffmpeg width/height/fps are
stored); the 640x360 e2e reject is a known floor (deploy report §3.2, lines 41-42) | \n| Upload
sequencing / size / disk | `404` unknown upload_id (uploads.py:134); `409` "Out-of-order part:
expected N, got M" (uploads.py:136-139); `409` "Expected N part(s), received M" on complete
(uploads.py:181-185); `413` per-part/total overflow ? **upload aborted, re-init**
(uploads.py:157-161) or declared-size (uploads.py:100-104); `507` "Not enough free disk space
for upload?" (uploads.py:109; backend/utils/freespace.py:77-83) | Only the 507 logs: `free-
space guard (upload): <path> needs X MB but only Y MB free ? refusing (507).`
(freespace.py:70-76). 409/413 log **nothing** app-side. Upload state is in-process ? an instance
restart silently orphans it ? next part 404s (uploads.py:6-8) | For 409 after a gap: assume
instance recycle ? client re-inits. For 507: check CP instance disk | \n| Staging-to-bucket
failure (#461 path) | job `done`, `result.status='failed'`, `message="\`cloud-serving: failed to
stage the uploaded file to <input_root>: <e>\`"; `compute.path='not_run'`, `dispatched=false`
(orchestration.py:296-307, 198-221) | CP: `[API] cloud-serving: failed to stage <src> ->
<staged_uri>: <e>` with traceback (orchestration.py:297-303) | Check the CP runtime SA's write
access to `storage.input\_root` and that the bucket exists; retry the job | \n| Cloud dispatch
failure (SDK missing / project unset / Run API error) | job `done`, `result.status='failed'`,
`message="\`cloud dispatch error: <e>\`" (orchestration.py:348-350). Distinct pre-flight
RuntimeErrors: `GOOGLE\_CLOUD\_PROJECT is unset?` and `google-cloud-run is required?`
(backend/compute/cloud_run.py:93-106) | CP: `[API] cloud dispatch error: <e>` with traceback
(orchestration.py:349) | If pre-flight: fix CP env (`CLOUD\_RUN\_JOB`/`GOOGLE\_CLOUD\_PROJECT`) or
install the SDK. Else check IAM (run.jobs.run) + `gcloud run jobs executions list --job rally-
worker` | \n| L4 execution failure (worker rc?0 / garbled marker) | job `done`,
`result.status='failed'`, `message="\`cloud job failed (returncode N).\`"
(orchestration.py:351-352). A missing/garbled marker defaults to rc=1, never success
(cloud_run.py:216-226) | CP: `cloud-run: dispatched job=rally-worker exec=<execution>; bucket-
polling <done_uri>` then `cloud-run: job=? done (video_id=? rc=N)` (cloud_run.py:200-205,
229-231). **Worker logs** carry the real failure; the worker writes the failure marker itself so
the poll fails fast (backend/worker/`\_\_main\_\_`.py:236-241) | `gcloud run jobs executions describe
<execution>` then read that execution's logs; inspect `gs://khelsutra-rally-
serving/outputs/<video_id>/` and the `_dispatch/<token>.done` marker JSON | \n| Bucket-poll
timeout (default budget 1800s @10s cadence ? DEFAULT_POLICY, hard-wired:
backend/infrastructure/execution_policy.py:49-54; backend/compute/`\_\_init\_\_`.py:39,76) | job
`done`, `result.status='failed'`, `message="\`cloud dispatch error: cloud-run-infer not ready
within max_wait_sec=1800.0s [cloud-run job=rally-worker execution=<name>; cancel requested ?
verify the execution state in the GCP console]\`" (cloud_run.py:281-285 ?
orchestration.py:348-350) ? the execution name IS in the client-visible message | CP: `cloud-
run-infer: still not ready after Ns (max_wait) ? giving up` (execution_policy.py:184-189); then
either `done-marker found on the final post-timeout check ? rescuing` (cloud_run.py:261-267) or
`requested cancel of execution=?` / `best-effort cancel ? failed` (cloud_run.py:270-280) |
Describe the execution: if it completed just after cancel, the GPU run is lost ? re-submit; if
it hung, read worker logs for the stall stage. Confirm the cancel landed (otherwise L4 bills to
--task-timeout 3600s) | \n| Result-ingest failure (missing/corrupt `intervals.csv`) | job `done`,
`result.status='failed'`, `message="\`cloud-serving: could not read the result from
<outputs_uri>: <e>\`"; partial rows pruned, prior segments preserved (orchestration.py:369-376,
107-117) | CP: `[API] cloud ingest failed: <e>` with traceback (orchestration.py:372) | `gsutil
ls gs://?/outputs/<video_id>/` ? if `intervals.csv` is absent the worker died between marker and
artifact push: read worker logs | \n| Compile / ffmpeg failure | Submit-time: `404` "Indexed
video record not found." / `400` bad segments/filename/min_shot_count floor
(backend/main.py:1111-1120; orchestration.py:699-745). Run-time: job `done`,
`result.status='failed'`, `message="\`Highlight stitching failed: <e>\`"
(orchestration.py:829-837) | CP: `[compile] stitching failed for video\_id=? : <e>` with traceback
(orchestration.py:830-832). Mirror-upload failure is separate and non-fatal: `[compile] could
not mirror reel to <uri>: <e>` (orchestration.py:856) | Check ffmpeg presence via `GET
/api/capabilities` (main.py:384-389); a mirror-only failure means the download will hit the 32
MiB local-fallback trap below | \n| Signed-URL failure (#475?#476) | On success: `307` to a v4
signed URL. On failure: silent fallback to serving the local file ? for a >32 MiB reel that
resurfaces the Cloud Run response-cap edge-500 (deploy report §4, lines 61-79) | CP, post-#476
(merged): `[gcs] keyless signBlob path failed (<e>); falling back to key-based signing` or
`[gcs] keyless signing skipped: runtime SA email unresolved` (backend/storage/gcs.py:104-113),
then `[highlights] signed-url path failed for <video\_id>; serving local: <e>`

(backend/main.py:1062-1063) | Grep CP logs for `[gcs]`; verify the SA still holds self-`roles/iam.serviceAccountTokenCreator` and `iamcredentials.googleapis.com` is enabled (the #475 root-cause pair, deploy report lines 68-73) | \n| Job stuck `queued`/`waiting` | `GET /api/jobs/{id}` stays `queued` (progress null) or `waiting` indefinitely; no timeout ever flips it | Usually **no log at all** ? known O2 risk: a hung job on the drain blocks everything behind it; there is no per-job wall-clock timeout (job_worker.py:40-48). Parked jobs do log `Job <id> parked for Ns (resume when due).` (job_worker.py:189); a broken claim path logs `Failed to claim a due job: <e>` (database_jobs.py:394-396) | Check the head job's `progress`/`started\_at` ? a hung Gemini/GPU/poll call in the head job explains everything queued behind it. A restart converts the backlog to a clean state (next row) | \n| Server restart / instance recycle mid-job | **Self-hosted (persistent DB):** process job ? `status='failed'`, `error="interrupted by server restart"`; compile job ? silently re-queued and finishes (database_jobs.py:420-450). **Cloud Run:** SQLite is per-instance ephemeral ? the new instance has a fresh DB, so `GET /api/jobs/{id}` ? `404 "Unknown job\_id"` and past download URLs 404 on the DB check (main.py:1012-1014, 1052; issue #471 OPEN; deploy report lines 86-89) | Startup log: `[JOBS] Recovered stale jobs from a previous run ? N compile re-queued, M process failed.` (main.py:1475-1480) ? on Cloud Run this line reports 0/0 because the DB is fresh | Correlate the client's job_id timestamp with CP revision/instance-start events in Cloud Logging; the client must re-submit (re-process regenerates state ? outputs and reels in gs:// survive) | \n| Any other worker crash | `status='failed'`, `error="<ExcType>: <msg>"` | CP: `Job <job\_id> crashed: <e>` + full traceback (job_worker.py:191-193) ? the one log line guaranteed to carry the job_id | Grep CP logs for `Job <job\_id>` | \n\n### Correlating a client `job\_id` to a Cloud Run execution\n\nThere is no single trace id ? the chain is: **job_id ? video_id ? execution name**, stitched from the job row plus two log lines:\n\n1. `job\_id ? video\_id`: the worker stamps `video\_id` onto the job row as soon as the result carries one (job_worker.py:177-178); it's in every `GET /api/jobs/{id}` response.\n\n2. **Dispatch token**: each dispatch mints `done\_uri = <output\_root>/\_dispatch/<uuid4-token>.done` (cloud_run.py:179-182) and logs `cloud-run: dispatched job=<job> exec=<execution>; bucket-polling <done\_uri>` (cloud_run.py:200-205) ? this CP line is the token?execution join point. The worker's side logs `[worker] wrote completion marker -> <done\_uri>` in the **execution's** logs (backend/worker/__main__.py:188), closing the loop from the other end.\n\n3. **Provenance on the result**: a successful cloud job's `result.compute` = `{path:"cloud\_run", job, region, execution, outputs\_uri}` and the CP logs `[COMPUTE] cloud\_run dispatched: execution=? job=? region=? out=? video\_id=?` (orchestration.py:384-400); the docstring's own cross-check is `gcloud run jobs executions describe <execution>` (orchestration.py:386). Every pipeline run also ends with `[COMPUTE] video\_id=? requested=? actual=? dispatched=? status=?` (orchestration.py:632-640).\n\n4. On a **poll timeout** the execution name is carried inside the failure message itself (cloud_run.py:281-285), so it reaches the client. But on other dispatch failures the `compute` dict in the failed result has **no execution field** (orchestration.py:206-221) ? the CP `dispatched` log line is then the only record. Note the token/execution are **not persisted on the job row**; if the instance recycles, this mapping survives only in Cloud Logging.\n\n### GAPS ? what an operator cannot currently see\n\n1. **No request tracing/replay** ? issue #326** (OPEN: `"Hosted-app observability: request tracing + replay for debugging production CUJ failures")`. No request id crosses the API?worker?L4 boundary.\n\n2. **No per-video runlog surface** ? issue #328** (OPEN: `"Surface the per-video runlog / execution trace to the user (endpoint + UI), not just server logs")`. `emit\_indexer\_report` writes per-video reports on every run (orchestration.py:645-660) and the paths ride the job result (main.py:1023), but they are **server-local paths on an ephemeral instance** ? unreachable by client or operator after recycle.\n\n3. **`job\_id` is absent from nearly all pipeline logs** ? `[COMPUTE]`, `[API] cloud-serving?`, `cloud-run: dispatched?` key on video_id/execution, not job_id (orchestration.py:325-329, 393-400, 632-640; cloud_run.py:200-205). Only crash/park lines carry it (job_worker.py:189-193). Triage from a user's job_id starts with an extra DB/API hop that dies with the instance.\n\n4. **Job history is ephemeral** (issue #471 OPEN) ? the `error` column and `result` JSON, the primary triage record, vanish on every deploy/recycle; only what was logged survives.\n\n5. **Stuck-queue blindness** ? no per-job wall-clock timeout and no log when a job wedges the single drain lane (job_worker.py:40-48); the failure mode is pure silence.\n\n6. **Client 4xx are unlogged app-side** ? upload 409/413, jail 400s and auth 401s raise HTTPException without a logger call (uploads.py:127-170; main.py:910-913), leaving only status codes in request logs.\n\n7. **No metrics/alerting** ? zero counters; failure detection is entirely reactive log-grepping (nothing beyond stdout logging is wired in the serving backend).\n\n\n**Top 3 cheapest wins** (all log-only, no infra):\n\n1. **One structured terminal line per job** in `\_run\_claimed\_job` ? `[JOB] id=? kind=? status=? video\_id=? compute=? execution=? wall=?s error=?` at done/failed/parked (job_worker.py:177-193). Single choke point; makes every triage class above greppable by the id the client actually has, and closes gap 3 for ~10 lines.\n\n2. **Persist/log the execution name + dispatch token at dispatch time keyed by

job_id** ? e.g. thread job_id into `notify()`/the dispatch logs, or stamp `execution` into `progress` ("dispatching (cloud_run exec=?)"). Cost: a few lines; makes job?execution correlation survive crash-before-result (gap in correlation §4).\n3. **Mirror the per-video report JSON to `gs://?/outputs/<video\_id>/`** next to `intervals.csv` (the worker already pushes artifacts there ? worker/`\_\_main\_\_.py:327`) and/or serve it via a small authenticated endpoint ? a minimal first slice of #328 that also survives instance recycling (gaps 2 and 4).\n\n*Launch-facing note:* these gaps are operability debt, not A13 blockers by themselves ? but with A10 (ship-gate) and A11 (WASB-C3 provenance) still OWNER-GATED (docs/ALPHA_LAUNCH_READINESS.md:109-112), the review should treat gaps 4+5 (silent stuck jobs on an ephemeral store) as the ones most likely to burn an alpha user before the owner gates close.",

```

22         "evidence_gaps": [
23             "Cloud Logging specifics (resource types, retention, whether stdout from
rally-worker executions is actually queryable per-execution) ? READ-ONLY constraint, did not
touch GCP; asserted from Cloud Run defaults and marked by naming the expected log sources
only.",
24             "The deployed rev-00007 poll budget: DEFAULT_POLICY (1800s/10s) is hard-wired
at the call site (backend/compute/`__init__.py:39,76` ? no config override path found), but I did
not inspect the live service env/manifest to confirm nothing patches it at runtime.",
25             "The >32 MiB local-fallback edge-500 behavior is cited from the deploy
report's live observation (runlogs/?a1-e2e?md:61-67), not re-reproduced in this session.",
26             "Claim that recover_stale_jobs reports 0/0 on Cloud Run (fresh ephemeral DB)
is inferred from the per-instance SQLite design (#471) + code reading; not observed in live logs
this session.",
27             "'No metrics/alerting' is based on a grep for
prometheus//metrics/statsd/opentelemetry/structlog across backend/ returning only two eval
scripts (backend/eval/*), whose hits I did not open; possible unrelated matches, but no serving-
path metrics wiring exists in the files traced."
28         ]
29     },
30     {
31         "key": "privacy",
32         "markdown": "## Privacy & consent copy audit (A13)\n\n**Verdict:** the marketing
site has a real, decent privacy policy in all 6 locales ? but the Studio SPA (the surface
testers actually upload through) contains **zero privacy/consent copy**, and its first-run
wizard affirmatively tells users their videos never leave the device, which is false on the
hosted alpha. One code-level guarantee (training capture hard-denied) is *stronger* than the
copy promises; storage location, retention reality, and deletion mechanics are *weaker*.\n\n###
1. What the product says today\n\n**Marketing site** (`khelsutra` repo, `/privacy`, localized in
en/kn/hi/es/da/id ? all 6 catalogs carry the policy keys; English authoritative per
`locales/en/common.json:574`):**\n\n- Full plain-language policy, `Last updated: 22 June 2026`
(`locales/en/common.json:491-577`). Key promises:\n
- *`We do **not** use your footage to
train or improve our models **unless you give us clear, explicit, opt-in permission** ? and even
then, never footage involving minors.`* (`:523`)\n
- *`We may use **trusted third-party AI
and cloud services** (for example, Google's Gemini API) to detect rallies.`* (`:529`)\n
- conditional wording, so not false in the alpha, but see §2.\n
- *`We **never publish** your
footage`* (`:530`); processors named: Cloudflare, Google (Gemini API), email delivery
(`:538-540`). **Google Cloud Storage / Cloud Run is not named; no storage location is given.**\n
- Retention: *`keep your information and footage only as long as we need it to run the alpha?
then delete or anonymise it. You can ask us to delete? at any time`* (`:551`); rights include
*`Delete your data and uploaded footage`* (`:558`); minors never trained/published (`:564`);
*`gated behind an authenticated, invite-only sign-in`* (`:568`).\n
- Access-request form:
email-consent checkbox *`Okay to email me about the KhelSutra alpha.`* + *`We never publish
your footage; footage involving minors is never used to improve the tool.`* with a `/privacy`
link (`locales/en/common.json:180-181`); capture checklist: *`Please record adults with their
okay.`* (`:280`).\n\n**Studio SPA** (`khelsutra/web/src` ? what alpha testers upload
through):**\n\n- **No privacy copy at all**: a repo-wide search of `web/src` for privacy/consent
strings and `/privacy` links returns nothing (grep, 0 matches). No link to the policy, no
upload-moment disclosure.\n
- The only pre-upload acknowledgment is about **cost**, not data:
*`I understand cloud processing may cost a little.`* (`locales/en/common.json:110-112`,
rendered `web/src/components/IngestPanel.tsx:211-224`).\n
- **Contradiction**: the first-run
wizard ? shown once per browser before the ingest panel (`web/src/App.tsx:347-349`) ?
unconditionally renders *`Everything runs on this device ? your videos never leave it.`*
(`web/src/components/SetupWizard.tsx:58`, string at `locales/en/common.json:612`). On the hosted
alpha the cloud path uploads footage to GCS. Wizard step 1 also advertises connecting a Gemini

```


key (`:604-608`), which the alpha build never uses.\n- Upload copy is purely functional: *"\Pick an .mp4 or .mov from this device ? it uploads, then KhelSutra finds the rallies.*"\n(`:101`).\n\n### 2. Engine reality vs. copy\n\n| Alpha reality | Evidence | Disclosed today? |\n|---|---|---|\n| Uploads staged to `gs://khelsutra-rally-serving/videos/?` in **asia-southeast1 (Singapore)**; reels mirrored to `~/highlights/?` in the same bucket | \n`deploy/cloudrun/gen\_service\_yaml.py:42-45,57-62`; e2e runlog \n`docs/COMPUTE\_DECOUPLED\_SERVING/runlogs/DEPLOY\_REPORT\_cloud-serving\_a1-e2e\_2026-07-04.md:37,50` |\n| **No.** Policy says "our own infrastructure"; never names Google Cloud storage or a region. Cross-border (India?Singapore) transfer undisclosed |\n| Reel downloads are **bearer signed URLs, ~1 h expiry** ? anyone holding the link can fetch | runlog `:53` (`X-Goog-Expires=3600`) | No |\n| **Per-instance SQLite** ? a redeploy wipes job/reel records while the bytes **persist in the bucket** (orphaned data; old download URLs die) | runlog `:87-89` (#471) | No |\n| **No retention automation** ? sweep tool exists (7-day default) but nothing schedules it; **no user-facing delete endpoint** (no DELETE routes in `backend/main.py`) | \n`tools/cleanup\_caches.py:10-12`; **301 OPEN** (verified via `gh issue view`, 2026-07-04: "run the upload retention sweep as an automated scheduled job (hosted alpha)") | **Over-promised.** Policy promises deletion on request + limited retention; today fulfillment is manual operator gsutil work |\n| **Training capture HARD-DENIED**: `consent\_attested()` returns `False` for every job, and `flywheel.enabled` defaults off ? two independent guards; claimed consent fields are ignored | `backend/flywheel.py:44-52,144-151`; `tests/test\_flywheel.py:38-55` | Copy's "never train without explicit opt-in" (`:523`) is true and **code-enforced stronger** today. A launch asset ? say it |\n| **Gemini/AI handoff DISABLED** (`indexing.skip\_ai\_handoff=true`; `gemini\_key\_present=false` live) ? no footage reaches any third-party AI | \n`gen\_service\_yaml.py:56`; runlog `:25-26` | Policy's conditional "may use Gemini" over-discloses (fine); SPA wizard still sells a Gemini step (`:604-608`) ? stale for the alpha |\n| Reachability = **Cloudflare Access, same 4-email allowlist policy**, on `api.khelsutra.guru`; edge-auth verified live (401 sans JWT, CF login 302) | `DEPLOY\_REPORT\_cloud-serving\_a1-cuj\_2026-07-04.md:82`; e2e runlog `:99-103`; `gen\_service\_yaml.py:46-47,63-68` | Matches policy's "invite-only sign-in" (`:568`) |\n| **No content-safety/NSFW pre-flight** ? nothing screens what a tester routes into the bucket/GPU | **332 OPEN** (verified via `gh issue view`, 2026-07-04) | No acceptable-use line anywhere in tester-facing copy |\n| Upload validation floor ?1280x720 (rejects smaller) | e2e runlog `:41-42` | No (expectation-setting, minor) |\n| Launch-facing constraint: **A10 (ship-gate/claims) and A11 (WASB-C3 provenance / "no commercial model sharing" copy guardrail) are OWNER-GATED and open** (`docs/ALPHA\_LAUNCH\_READINESS.md:109-110`) ? alpha copy must make no model-quality claims and no model-sharing/commercial claims until resolved.\n\n### 3. What we must tell testers (checklist)\n\n1. **Where footage goes:** uploads and finished reels are stored in a private Google Cloud bucket in **Singapore** (asia-southeast1); Google Cloud must be named as a processor alongside Cloudflare.\n2. **Who can reach it:** invite-only ? a Cloudflare Access email allowlist (currently 4 people) gates the app; operators can access stored footage.\n3. **Download links are shareable:** reel URLs work for ~1 hour for anyone who has them ? don't forward them if the footage is sensitive.\n4. **Retention is manual today:** footage is kept while the alpha runs; deletion is on request via hello@khelsutra.guru (automated cleanup not live ? #301); a redeploy may kill in-flight jobs and old links while the footage bytes remain until swept.\n5. **Training is OFF, in code:** no footage is captured for model training in this build ? hard-disabled, not just policy ? and any future training will be a separate explicit opt-in, never involving minors.\n6. **No third-party AI in this build:** nothing is sent to Gemini or any external AI (skip_ai_handoff).\n7. **Tester responsibility:** adults-only, with their okay, badminton footage only ? there is **no automated content screening** yet (#332), so uploads are trusted.\n8. **Quality/limits framing:** ?720p required; no model-quality or model-sharing claims (A10/A11 gated).\n9. Ship the above in **all 6 locales** (or explicitly English-only with the `:574` authoritative-English caveat).\n\n### 4. Draft alpha disclosure copy (landing page / pre-upload)\n\n> **Before you upload ? what happens to your footage.** This alpha runs in the cloud: when you add a match video, it's uploaded to KhelSutra's private storage on Google Cloud (Singapore region) so a rented GPU can find your rallies, and your finished reel is stored there too. Access is invite-only, and we never publish or share your footage. We keep uploads and reels only while the alpha runs and delete them whenever you ask (email hello@khelsutra.guru ? automatic cleanup is still being built). Your footage is **not** used to train our models ? training capture is switched off in this build, in the code, and would only ever happen later with your separate, explicit opt-in (never footage involving minors).\n\n> - **Stored:** private Google Cloud bucket, Singapore region ? uploads + reels, invite-only access.\n> - **Not sent to AI services:** this build uses no Gemini or other third-party AI on your footage.\n> - **Not training data:** capture is hard-disabled; future training = separate opt-in, adults only.\n> - **Download links:** work for about 1 hour and for anyone who has them ? share with care.\n> - **Deletion:** on request today, automatic later; a redeploy

may cancel in-progress jobs and old links (just re-process).\n> - **Please upload only** badminton footage of adults who are okay being recorded ? there's no automated screening yet, so we're trusting you.\n\n### 5. Gaps, ranked\n\n1. **SPA wizard says \"your videos never leave it\"** (`SetupWizard.tsx:58`) ? factually wrong on the hosted alpha; must be fixed or suppressed for the cloud deployment **before any tester uploads**.\n2. **Storage processor + location undisclosed** ? Google Cloud / Singapore appear nowhere in the policy (`:538-540` names only Cloudflare/Gemini/email); India/Singapore transfer is a counsel-review item.\n3. **Retention/deletion over-promise** ? policy promises timely deletion (`:551,558`) but there is no automation (#301), no delete endpoint, and redeploys orphan bucket data; either land minimum automation or soften/date the promise.\n4. **No privacy surface in the Studio SPA** ? no policy link, no upload-moment disclosure; the only checkbox is about cost (`:110-112`). Add the \$4 copy + link at the upload panel.\n5. **No acceptable-use/content-screening line** (#332 open) ? mitigated by the 4-email allowlist, but the copy should state the trust assumption before the allowlist grows.\n6. **Signed-URL shareability undisclosed** (1 h bearer links).\n7. **Stale Gemini copy** ? SPA wizard Gemini step (`:604-608`) and policy `:529` overstate third-party AI for this build; a one-line \"this build sends nothing to third-party AI\" is both accurate and a selling point.\n8. **A11 copy guardrail** ? until WASB-C3 is resolved, keep all \"own your model\" marketing (`:287`) free of commercial-sharing implications (owner-gated, `ALPHA\_LAUNCH\_READINESS.md:110`),

```

33     "evidence_gaps": [
34         "Live Cloudflare Access policy contents (the '4-email allowlist') were not
inspected directly (no CF/GCP access under the READ-ONLY constraint); cited from
docs/COMPUTE_DECOUPLED_SERVING/runlogs/DEPLOY_REPORT_cloud-serving_a1-cuj_2026-07-04.md:82 ?
[unverified live]",
35         "Absence of a GCS lifecycle/retention rule on gs://khehsutra-rally-serving is
inferred from repo search (no lifecycle config or mention found) ? the live bucket config was
not inspected [unverified live]",
36         "Whether the hosted alpha build actually shows the first-run SetupWizard to
testers was verified only in code (App.tsx:347-349 renders it once per browser when nothing is
loaded); not observed on the deployed site [unverified live]",
37         "kn/hi/es/da/id privacy-policy translations were verified only by key presence
(grep counts of 'privacy'/'consent' keys per catalog), not by reading translated content for
accuracy",
38         "Operator/owner access to bucket contents (stated in the tester checklist)
follows from GCP project ownership; no IAM policy was read [unverified]"
39     ]
40 },
41 {
42     "key": "limits",
43     "markdown": "## Known limits + rollback/disable steps\n\n### K1. Locked alpha
shape (the limits that are *by design*)\n\nThese are deliberate, config-pinned constraints of
the live A1 stack ? launch copy and tester expectations must match them.\n\n| Locked decision |
Evidence |\n|---|---|\n| **WASB-only detection ? no Gemini anywhere in serving.**
`indexing.skip_ai_handoff: true` is baked into the control-plane manifest (\"candidate windows
ARE the rallies; no Gemini key in this service\") | `deploy/cloudrun/gen_service_yaml.py:55-56` ;
live-verified `gemini_key_present=false` (e2e runlog $2,
`docs/COMPUTE_DECOUPLED_SERVING/runlogs/DEPLOY_REPORT_cloud-serving_a1-e2e_2026-07-04.md:25-26`)
|\n| **Single region, single GPU.** Everything is in asia-southeast1 (`gen_service_yaml.py:43`);
the project's GPU quota is `GPUS_ALL_REGIONS=1` ? one L4 execution at a time, globally |
`docs/archives/decisions/DECISIONS.md:675` (\"quota granted (`GPUS_ALL_REGIONS=1`)\" ); worker
Job shape e2e runlog $1 |\n| **One control-plane instance (`maxScale=1`).** Deliberate #471
option-1 mitigation ? per-instance SQLite state means >1 instance splits the job queue |
`gen_service_yaml.py:25-27,97,145-150` ; e2e runlog $1 (rev 00007 deployed with maxScale=1) |\n|
**Upload size cap: 4096 MB whole-file, 95 MB parts.** `SecurityConfig.max_upload_mb: int =
4096`, `max_part_mb: int = 95` (`backend/config/models.py:370-371`), enforced at
`backend/api/uploads.py:57-58` with the 413-style rejection at `uploads.py:153` (\"Upload
exceeds the {N} MB total limit\"). Parts stay ?95 MB because free-tier edge proxies cap bodies
~100 MB (`models.py:368-369`) | files cited |\n| ? [analysis] the 4 GB cap is theoretical on
this deploy: uploads land under `/tmp/apphome/output/uploads` (a1-cuj runlog $5.2) on a **4Gi-
memory** instance (`gen_service_yaml.py:152`) whose `/tmp` is RAM-backed
(`deploy/cloudrun/README.md:47-48`) ? a multi-GB upload would exhaust instance memory well
before the configured cap. Not observed live; treat effective cap as ?4 GB | [analysis] |\n|
**Upload validation floor: ?1280x720 and ?29.9 fps.** A 640x360 clip was rejected live by
`resolution_fps_check` | e2e runlog $3.2 + $5 (lines 41-42, 90); defaults at
`backend/config/models.py:25-27` |\n| **Auth posture:** Cloudflare Access at

```

`api.khelsutra.guru`, same `\"KhelSutra Studio (alpha)\"` Access app, ****4-email allowlist****, plus in-app CF-JWT verification (401 without `CF-Access-Jwt-Assertion`) | a1-cuj runlog \$6
(`DEPLOY\_REPORT\_cloud-serving\_a1-cuj\_2026-07-04.md:78-90`); e2e runlog \$6 (restore verified, rev 00007) | \n\n### K2. Known-limits table (open issues + live rough edges)\n\n| Limit | Impact on the alpha | Source | \n|---|---|---| \n| ****Control-plane state is per-instance ephemeral SQLite**** ? a redeploy/restart wipes jobs/videos/segments; `/api/highlights` then 404s on the DB check (`backend/main.py:1052`) even though the reel still exists in gs://. A deploy invalidates in-flight jobs and past download URLs | Any redeploy mid-tester-session breaks their job polls and download links; re-processing regenerates them | #471 (open, bug); e2e runlog \$5 (lines 86-89) | \n| ****Google-managed cert on `api.khelsutra.guru` can stall at ~90-day renewal**** behind the proxied CF CNAME ? verify before ~2026-09-30 (`gcloud beta run domain-mappings describe --domain api.khelsutra.guru --region asia-southeast1`) | Silent future outage risk; calendar item for the owner | #472 (open) | \n| ****CORS default is `allow\_origins=['\*']`**** ? and the deployed manifest sets no `RALLY\_CORS\_ALLOW\_ORIGINS` (env block `gen\_service\_yaml.py:111-119`), so the wildcard is active on the hosted service too [analysis]. Edge auth + no-credentials CORS blunt it, but it is an unresolved audit residual (R1-12) | Residual cross-origin exposure, mainly on non-PNA browsers | #443 (open) | \n| ****No content-safety/NSFW pre-flight**** before processing. The WASB-only alpha removes the send-to-Gemini ToS risk (no Gemini in serving), but abusive uploads still burn paid L4 time and produce garbage | \"Some users will try\" ? accepted for a 4-email allowlist, not for wider invites | #332 (open) | \n| ****Path-jail vs storage schemes****: gs:// ingest through the jail was unblocked by PR #465 (verified live: gs:// ? 202 with jail enforced, a1-cuj \$6), but the general scheme-exempt design (s3://, gdrive-api://) remains open | gs:// works today; other cloud schemes 400 | #327 (open); a1-cuj runlog \$6 | \n| ****No automated upload retention**** ? the T11 sweep tool exists but nothing schedules it, and it targets the `\*local\*` upload dir. Staged tester uploads persist in `gs://khelsutra-rally-serving/videos/<video\_id>/?` indefinitely; no bucket lifecycle policy found in-repo [unverified] | Tester footage accumulates in the bucket until manually deleted (see R5) | #301 (open); staging path `backend/orchestration.py:284` | \n| ****Observability gaps****: no user-facing per-video trace endpoint (#328); no request capture/replay, and CF-edge failures (302s at Access) never reach engine logs so they're invisible in-app (#326) | Tester-reported failures need manual CF-dashboard + Cloud Logging triage | #328, #326 (open) | \n| ****Performance****: hosted processing is CPU-bound (ffmpeg/cv2) ? measured ~65% of wait is software encode (#331); WASB decode is single-threaded in streaming mode so the paid L4 sits ~80% idle, ~2x realtime detection (#349). Live A1 numbers: ~13m45s process (incl. cold pull) + ~3m26s?4m54s compile for a ~7-min match | Set tester latency expectations in minutes, not seconds; GPU cost per match is inflated | #331, #349 (open); a1-cuj \$3; e2e \$3.4 | \n| ****Golden eval data is machine-split**** (GPU box ? dev sessions, OneDrive sync incomplete) | Slows quality-lane reproduction; no direct tester impact | #175 (open) | \n| ****`gcloud run services replace` with an unchanged manifest is a NO-OP**** ? it does not re-resolve `:latest` after a rebuild; force a new revision with `gcloud run services update --image` | Operator footgun: a `\"redploy\"` can silently ship the old image | e2e runlog \$5 (lines 83-85) | \n| ****422-before-401 quirk****: malformed-JSON bodies get FastAPI's 422 before edge-auth runs; well-formed requests without a CF JWT still 401 | No exposure; don't misread a 422 during auth triage | e2e runlog \$5 (lines 91-93) | \n| ****No video-delete API**** ? no supported way to remove a video/job record; only manual DB/bucket surgery | See R5 for the actual deletion procedure | #447 (open) | \n\n### K3. GATED items that constrain launch copy (A10 / A11 ? both OWNER-GATED, still open)\n\n****A11 ? WASB-C3 provenance (weights NOT commercially cleared).**** The worker image itself carries the warning: `\"the pretrained badminton weights are academic-data-trained ? provenance NOT cleared for commercial sharing? Resolve WASB-C3 before any public, non-owner serving\"` (`deploy/cloudrun/Dockerfile:14-16`). The tracker keeps A11 GATED on owner/legal (`docs/ALPHA\_LAUNCH\_READINESS.md:110`) and names C3 a ****hard model-release gate**** that must not be hidden inside product-launch tasks (\$5 risk table, `ALPHA\_LAUNCH\_READINESS.md:85`). Consequences for the alpha:\n\n- ****Who may use it:**** only the current 4-email invite-only allowlist (a1-cuj \$6). Note honestly for the GO/NO-GO: those testers are already ***non-owner*** users, which sits in tension with the Dockerfile's **\"before any public, non-owner serving\"** ? A13 needs the owner to either resolve C3 or ****explicitly defer it with the documented \"no commercial model sharing\" copy guardrail**** (that's exactly what the A11 row requires). No wider invite wave until then.\n\n- ****What copy must NOT claim:**** an **\"owned\"** or commercially cleared rally-detection model, any model sharing/download, or any commercial license posture. Tracker guardrail: **\"Keep all alpha copy scoped to invite-only processing, not owned-model claims\"** (`ALPHA\_LAUNCH\_READINESS.md:80`; also \$0 TL;DR line 15 and D1, line 31). Mitigating fact for the record: the weights are ****not baked into the image or distributed**** ? they're staged at runtime from the bucket and used server-side only (`Dockerfile:15-16`; `deploy/cloudrun/README.md:15,29-30`).\n\n****A10 ? ship-gate target not owner-ratified (#172).**** The F1@tIoU targets in `gate\_verdict` are explicitly provisional (\"the owner has explicitly stated they did not commit to that figure\" ? #172), the A10 row is GATED on the owner decision

(`ALPHA\_LAUNCH\_READINESS.md:109`), and the latest Gen-0 shadow run is `ship=False` (sign test 9/15, p=0.3036; D6 + \$8 A9 record, `ALPHA\_LAUNCH\_READINESS.md:37,214`). Gate-A served default was ****rejected**** (served mean ?F1 ?0.0328, `ALPHA\_LAUNCH\_READINESS.md:205`). Consequence: ****the alpha can make no quality claims**** ? no accuracy numbers, no `"best-in-class"`, not even the internal R2/tolF1 contrast as a headline (its strict per-video guard still blocks blanket promotion language, `ALPHA\_LAUNCH\_READINESS.md:206`). Allowed framing until A10 is ratified: invite-only alpha that detects candidate rally segments and compiles reels, quality under active measurement.

K4. Rollback / disable inventory

All commands run from PowerShell (not git-bash ? MSYS mangles paths/flags, `deploy/cloudrun/README.md:129-133`), region `asia-southeast1`, project `gen-lang-client-0306026826` (`gen\_service\_yaml.py:42-43`).

****R1. Disable the whole alpha (fast, reversible, keeps GCS data). **** Two independent switches ? throwing either one cuts testers off:

1. ***Edge off:** disable (or delete the policy of) the `"KheISutra Studio (alpha)"` CF Access application in Zero Trust ? Access ? Applications ? the login path dies at the edge (app identified at a1-cuj \$6, lines 81-82; `"disable the Access application"` is the documented teardown lever, `kheIsutra/deploy/DEPLOY\_ALPHA.md:156` ? that runbook is RETIRED but kept as the CF-mechanics reference, line 3). Dashboard click-op; exact click path [unverified].

2. ***Origin off:** remove the public invoker that CF-proxied traffic rides in on (the inverse of the documented enable step, a1-cuj \$6 remaining-(a), lines 93-95):

```

`gcloud run services remove-iam-policy-binding rally-control-plane --region asia-southeast1 --member=allUsers --role=roles/run.invoker`
`Requests then 403 at Google Front End;`
`min-instances 1` CPU cost continues until you also scale down or delete.
3. *Full teardown* (documented in the runlog; remember state is ephemeral anyway ? but staged videos/reels in gs:// survive):
`gcloud run services delete rally-control-plane --region asia-southeast1`
`gcloud run jobs delete rally-worker --region asia-southeast1`
(a1-cuj $7, lines 107-108: bucket/secret/repo retained.)

```

****R2. Roll back to a previous image digest. **** Do NOT use `services replace` with an unchanged manifest ? it's a no-op (e2e \$5). Pin the digest explicitly:

```

`ngcloud run services update rally-control-plane --region asia-southeast1 --image asia-southeast1-docker.pkg.dev/gen-lang-client-0306026826/rally/rally-worker@sha256:<digest>`
`ngcloud run jobs update rally-worker --region asia-southeast1 --image asia-southeast1-docker.pkg.dev/gen-lang-client-0306026826/rally/rally-worker@sha256:<digest>`

```

The `jobs update` re-pin is the exact mechanism used in the e2e run (e2e \$1); the service image path/name from `gen_service_yaml.py:157` (one image serves both roles, #470). ****Last-known-good digest:** `sha256:31475f27?` (rev 00007, edge-auth ON ? e2e \$1/\$6). Do not roll back to `sha256:ed2fe448?`: it predates #476 and reintroduces the broken >32 MiB signed-URL download (#475, e2e \$4). Caveats: any new revision ****wipes the per-instance SQLite state**** ? in-flight jobs and existing download URLs die (#471; e2e \$5); and if config (not code) is what regressed, regenerate the manifest instead: `python deploy/cloudrun/gen_service_yaml.py --edge-auth on --image <pinned-digest-ref> -o service.yaml` then `gcloud run services replace service.yaml --region asia-southeast1` (`gen\_service\_yaml.py:6-8,136-140`).

****R3. Revoke a tester. **** Zero Trust ? Access ? Applications ? `"KheISutra Studio (alpha)"` ? edit the Allow policy's `*Emails*` rule and remove the address (policy mechanics per `kheIsutra/deploy/DEPLOY\_ALPHA.md:113`; same-app/allowlist confirmed live at a1-cuj \$6). Also revoke the user's active Access session in Zero Trust so an existing cookie doesn't outlive the policy edit ? [unverified: exact dashboard path; not in repo docs]. Defense-in-depth already holds: every `/api/process` re-verifies the CF JWT in-app (`backend/config/models.py:374-379`; e2e \$6 shows 401 without it), so a revoked user can't fall back to hitting the run.app URL directly. There is no app-side account/data scoping to clean up (per-user subdirs are a future PR-3 item, `models.py:364-366`).

****R4. Emergency-stop a running L4 execution. ****

```

`ngcloud run jobs executions list --job rally-worker --region asia-southeast1`
`ngcloud run jobs executions cancel <execution-name> --region asia-southeast1`

```

Execution names look like `rally-worker-k49kg` (a1-cuj \$3.2). The `cancel` verb is standard gcloud but not exercised in any repo runlog [unverified in-repo]. Known consequence chain, from repo sources: a killed container writes ****no completion/failure marker****, so the control plane's bucket-marker poll runs to its bounded timeout ?

```

`ExecutionPolicy.max_wait_sec` default **1800 s**
(`backend/infrastructure/execution_policy.py:54`; `backend/compute/cloud_run.py:245`) ? then surfaces a clean `cloud dispatch error` job failure, not a hang
(`deploy/cloudrun/README.md:112-115`). GPU billing stops with the execution; the job row on the CP resolves ~30 min later or dies with the instance on the next redeploy (#471).

```

****R5. Delete a user's data for a given `video\_id`. **** Bucket is the durable store; the DB is ephemeral.

Bucket objects (all three prefixes, plus the user's own source object if they supplied a raw gs:// path outside `videos/<video_id>/`):

```

`gcloud storage rm -r gs://kheIsutra-rally-serving/videos/<video_id>/` # staged upload ? orchestration.py:284 (input_root
gen_service_yaml.py:61)
`gcloud storage rm -r gs://kheIsutra-rally-serving/outputs/<video_id>/` # worker outputs ? a1-cuj $3.2 (
`outputs` at gs://?/outputs/915473?/\`)
`gcloud storage rm -r`

```

```

gs://khelsutra-rally-serving/highlights/<video_id>/ # mirrored reels ?
orchestration.py:748-755\n ```\n Also sweep the dispatch done-markers
(`<out_uri>/_dispatch/<token>.done`, `deploy/cloudrun/README.md:109-110`) ? exact prefix in the
serving bucket [unverified].\n- *DB reality:* rows live in per-instance ephemeral SQLite (#471)
with **no delete API** (#447 open ? the only known method is manual row surgery, impractical
inside a Cloud Run instance). Practically: after the bucket `rm`, any surviving DB record is
inert (`/api/highlights` 404s once the mirror check fails / on the `main.py:1052` record check
after the next restart), and **any redeploy wipes all DB rows anyway** (e2e §5) ? a blunt but
real `"forget everything"` lever. This is adequate for a 4-email alpha but is not a real GDPR-
style deletion story; flag alongside #447/#301 in the launch review.\n\n**Cross-cutting caution
for every lever above:** rollback ? data rollback. `gs://khelsutra-rally-serving` (weights,
staged videos, outputs, reels) is untouched by service/job changes; state loss is confined to
the CP's ephemeral SQLite (#471).",
44         "evidence_gaps": [

```

3. user (2026-07-04T12:17:52.863Z)

```

1      # A13 live test-battery evidence (2026-07-04, this session ? all measured, none
inferred)
2
3      Run against the deployed stack during a controlled test window: `allUsers` invoker
REMOVED first
4      (public probes 403 ? zero public exposure; NB this also takes api.khelsutra.guru down
for testers
5      during the window since CF rides that binding), then edge-auth-off revisions; restored
to rev 00010
6      (digest `31475f27?`, auth ON, allUsers re-added) and posture re-verified at the end:
7      health(token)=200, health(public)=200 (exempt), process-sans-JWT=401,
api.khelsutra.guru=302,
8      gemini_key_present=false.
9
10     ## Failure-mode battery (all via identity token, rev 00008)
11
12     | Probe | Result |
13     |---|---|
14     | process gs://evil-bucket/x.mp4 (foreign bucket) | **400** `hosted mode: '?' is not
under the allowed input root 'gs://khelsutra-rally-serving/videos'` (#465 live at the API
surface) |
15     | process gs://khelsutra-rally-serving/videos-evil/x.mp4 (separator-boundary attack) |
**400** same jail message (the boundary check holds) |
16     | process nonexistent object under input_root | **202** job queued ? job `failed`, error
`NotFound: 404 ? No such object: khelsutra-rally-serving/videos/definitely-not-here.mp4` (clean
async failure, client-visible via poll) |
17     | process bad segmenter | **400** `Segmenter 'not_a_segmenter' not found. Available
segmenters: [?]|
18     | GET /api/jobs/deadbeef | **404** |
19     | compile bad video_id | **404** `Indexed video record not found.` |
20     | GET /api/highlights/x/..%2F..%2Fconfig.json (traversal) | **404** (rejected; no file
disclosure) |
21     | GET /api/highlights/<missing>/reel.mp4 | **404** |
22     | upload init filename=malware.exe | **400** `Extension '.exe' not allowed. Allowed:
['mov', 'mp4']` |
23     | upload init total_size 9999999999 | **413** `File exceeds the 4096 MB upload limit.`
|
24     | upload PUT part/5 first (out-of-order) | **409** `Out-of-order part: expected 0, got
5` ; DELETE cleanup ? 200 |
25
26     ## Concurrency smoke (#468 live, rev 00008)
27
28     3 distinct 720p clips (4s/5s/6s) uploaded + submitted `compute=cloud` back-to-back ? ALL
THREE jobs
29     claimed and completed `success` (0 segments each ? synthetic content). No false-empty,
no stranded
30     job, GPU-quota=1 serialized the L4 executions but the drain claimed everything.
31

```

```

32     ## Gemini handoff leg (rev 00009 ? one-off test config, NOT the alpha posture; restored
after)
33
34     - Prep finding: the `gemini-api-key` secret existed but had **NO accessor grant** ? the
documented
35     secret-mount path had never been exercised. Fixed live:
36     `gcloud secrets add-iam-policy-binding gemini-api-key
--member=serviceAccount:492532266377-compute@developer.gserviceaccount.com
--role=roles/secretmanager.secretAccessor`
37     (grant left in place; the secret is only mounted when a config asks for it).
38     - Job: `gcloud run jobs update rally-worker --set-secrets GEMINI_API_KEY=gemini-api-
key:latest` ? worked.
39     - CP: manifest with `skip_ai_handoff=false` + secret env ? booted clean
(`AI_HANDOFF_KEY_MISSING`
40     startup check passed; `gemini_key_present=true` in /api/capabilities).
41     - Run: MahadevpuraSingles, wasb_hybrid, compute=cloud ? **success, 43 play segments**
(vs 48 in
42     WASB-only mode ? the AI pass genuinely refined/filtered).
43     - Worker logs prove real Gemini traffic: `backend.providers.gemini [Gemini Handoff
1..48] Submitting
44     multimodal request to gemini-flash-latest` ? `POST
https://generativelanguage.googleapis.com/v1beta/
45     models/gemini-flash-latest:generateContent "HTTP/1.1 200 OK"` for all 48 windows;
`Phase 3
46     completed. AI confirmed 43 semantic play segments.`; execution `rally-worker-khx58`,
wall ~similar
47     to WASB-only + ~4-11s per handoff window.
48     - Restored after: secret unmounted from the job (`--remove-secrets`), alpha config back
49     (skip_ai_handoff=true), `gemini_key_present=false` re-verified.
50
51     ## Other live facts gathered today
52
53     - The CP serves the bundled SPA at its root: `GET /` ? 200 `text/html` (single-origin ?
the tester
54     path has NO cross-origin CORS dependency; the CF-Access "bypass OPTIONS" concern
applies only to a
55     split-origin deploy).
56     - Domain-mapping cert (issue #472): `Ready;CertificateProvisioned;DomainRoutable =
True;True;True`
57     today (2026-07-04) ? healthy; the ~90-day renewal watch date stands (~2026-09-30).
58     - Better auth-off window practice (used today, recommend for the runbook): REMOVE the
allUsers
59     invoker for the duration of any edge-auth-off window (public 403; identity tokens
still work),
60     re-add on restore. Cost: api.khelsutra.guru is down for testers during the window.
61     - khelsutra copy launch-blockers filed as khelsutra issue #154 (SetupWizard "never
leaves your
62     device" false on hosted; policy "our own infrastructure" on GCP; SPA has zero consent
copy).
63

```

4. user (2026-07-04T12:17:59.392Z)

```

44     "evidence_gaps": [
45     "Exact Cloudflare Zero Trust click-path for disabling the Access application
and for revoking an individual tester's active Access session ? dashboard operations, not
captured in any repo doc (the policy/app mechanics are cited from the retired DEPLOY_ALPHA.md
and the a1-cuj runlog).",
46     "The `gcloud run jobs executions cancel` command is standard gcloud but is not
exercised or documented anywhere in the repo; its consequence chain (no marker -> bounded poll
timeout -> clean dispatch-error failure) IS repo-cited.",
47     "Whether gs://khelsutra-rally-serving has any bucket lifecycle/retention
policy configured ? nothing found in-repo; #301 covers only the local upload-dir sweep tool.",
48     "Exact object prefix of the `_dispatch/<token>.done` markers inside the
serving bucket (the pattern is documented in deploy/cloudrun/README.md but the concrete out_uri

```

layout was not confirmed).",

49 "The four allowlisted email addresses themselves (CF dashboard state; only the count '4-email allowlist' is repo-cited from the a1-cuj runlog).",

50 "The claim that a multi-GB upload would OOM the 4Gi control-plane instance is [analysis] from config values (max_upload_mb=4096, memory 4Gi, RAM-backed /tmp), not an observed failure.",

51 "Wildcard CORS being active on the hosted control plane is [analysis] (no RALLY_CORS_ALLOW_ORIGINS in the generated manifest env block) ? not verified against the live service's response headers."

52]

53 }

54],

55 "critique": {

56 "missing": [

57 "BIGGEST GAP - browser/SPA CUJ never verified, and it is the tracker's own DoD #1: docs/ALPHA_LAUNCH_READINESS.md:226 '[] Invited tester can complete the hosted record/upload -> reel download CUJ without SSH' is UNCHECKED, and D2 (:32) mandates a page-driven alpha. Both runlogs (a1-cuj, a1-e2e) drove the CUJ via curl with identity tokens/JWTs only. No section audits the SPA's live deployment state: is Cloudflare Pages still serving app.khelsutra.guru with VITE_API_BASE=https://api.khelsutra.guru/api (khelsutra/deploy/DEPLOY_ALPHA.md:102-103 - a RETIRED doc), is the CF Access app's 'Bypass OPTIONS/CORS preflight' setting on (DEPLOY_ALPHA.md:115 says without it EVERY cross-origin API call fails), does the SPA's cloud compute toggle send compute=cloud against the new Cloud Run backend, does the wizard's /api/capabilities fetch work behind CF? The review must either include a browser-CUJ verification or state explicitly that the alpha is currently curl-only.",

58 "No GO/NO-GO scorecard or recommendation: the tracker has an explicit Definition-of-done checklist (ALPHA_LAUNCH_READINESS.md:224-235, 6 of 10 boxes unchecked) and open owner questions (:216-222, incl. 'acceptable alpha wording while C3 unresolved'). No section maps the review to that checklist or states the actual decision shape the owner must sign (e.g. GO for the current 4-email allowlist conditional on fixing SetupWizard copy; NO-GO for any wider invite wave until A10/A11 close). A12 (GATED, :111 - flywheel/feedback loop deferred) is never mentioned in any section.",

59 "No cost/budget projection for the standing stack: idle cost of the always-on control plane (minScale=1 + cpu-throttling off, 2 vCPU/4Gi - gen_service_yaml.py:96-98,151-152), L4 cost precedent ~\$1-2/run (docs/archives/decisions/DECISIONS.md:675), expected monthly figure for the 4-tester wave, and the [unverified] budget-alert threshold (\$100 per deploy/gcp/README.md:18 vs the claimed \$300 spend gate) - that unverified is a single console check and should be resolved before the owner decision.",

60 "No expected-quality statement for testers: the sections say what may NOT be claimed (A10) but never what testers WILL experience from the WASB-only path (skip_ai_handoff=true means candidate windows ARE the rallies). The measured numbers exist: served milestone tolF1 0.364 / F1@0.5 0.472 vs baseline-off 0.138/0.237 with the strict split-count guard still failing (ALPHA_LAUNCH_READINESS.md:142-153,206), 48 segments on the ~7-min e2e match. Owner needs 'testers will see roughly-right rally boundaries with over-splitting' framing to judge alpha-burn risk.",

61 "Runtime service-account blast radius unaudited: the compute SA (492532266377-compute@) holds project-wide roles/editor plus self-serviceAccountTokenCreator (memory session-handoff.md:159-160) on a service with allUsers run.invoker. No section inventories IAM or recommends a least-privilege SA before the allowlist grows - launch-review-grade security gap beyond the CORS residual already covered.",

62 "No tester onboarding/support content: what an invited tester is actually sent (app URL, CF email-OTP flow, the >=720p/29.9fps floor, 4GB/mp4|mov caps, expect ~15-20 min per match) collected in one place, and where testers report failures (hello@khelsutra.guru?) given A12 (feedback loop) is GATED and observability gap 2 means they cannot self-serve diagnostics.",

63 "Watch items have no owner/calendar mechanism: the #472 cert check before ~2026-09-30 and the budget watch are documented as commands but nobody is assigned; an A13 review should name the owner and the reminder mechanism (the repo owner runs multiple parallel sessions - a calendar entry or issue-due-date is the difference between documented and done).",

64 "Privacy audit misses two site-copy items: (a) policy line locales/en/common.json:528 'We process your video on our own infrastructure' is arguably FALSE on rented Google Cloud infrastructure - same class as the SetupWizard issue but on the public policy page, stronger than the draft's 'not disclosed' framing; (b) the site footer trust line :488 'Run from Bengaluru, India' sits in tension with all footage stored/processed in Singapore.",

65 "Doc-hygiene decision: the runbook reproduces the operational secret location

G:\\My Drive\\cloudflare.token.txt (and memory session-handoff.md:169's tester emails are one hop away). Decide whether the assembled A13 review is an owner-only doc or must be scrubbed before any wider sharing."

```

66         ],
67         "contradictions": [
68             "Runbook Gotcha A overstates its provenance: 'that exact fallback was used live'
- the cited e2e runlog section (DEPLOY_REPORT_cloud-serving_a1-e2e_2026-07-04.md:83-85) records
the no-op and the remedy as an observation, not as an executed live command, and the full flag
string is sourced only from memory current-state.md:64 which the runbook itself marks
[unverified]. Downgrade 'used live' to [analysis] or cite the runlog line that actually shows
the command ran.",
69             "Limits R4 states 'GPU billing stops with the execution' unconditionally, but
cancel is best-effort (backend/compute/cloud_run.py:268-277) and R4 itself marks 'executions
cancel' as [unverified in-repo]; the observability timeout row correctly warns the opposite
('Confirm the cancel landed - otherwise L4 bills to --task-timeout 3600s'). The billing claim
should carry the same [unverified] tag.",
70             "Privacy section applies inconsistent severity: SetupWizard.tsx:58 copy is
called 'factually wrong... must be fixed before any tester uploads' (correct - verified at
locales/en/common.json:612), but policy :528 'our own infrastructure' - false by the same
standard on the hosted alpha - is graded only as a nondisclosure in the section 2 table
('Disclosed today? No').",
71             "Runbook section 5 attributes 'gcloud beta run jobs executions logs read' to
'the committed pattern' at deploy/gcp/nightly_regression/README.md:111 - the committed line has
no 'beta' release track and targets nightly_regression_launcher in asia-south1; the adaptation
is fine but the citation implies more than the source carries.",
72             "Trivial: runbook section 6 quotes deploy/cloudrun/README.md:86,90-93 while
saying '~2-vCPU' - the quoted README line 92 actually says '~1-vCPU CPU instance'; the deployed
manifest is cpu=2 (gen_service_yaml.py:151), so the runbook's number is right but does not match
the source it quotes.",
73             "(Verification note, not a defect) All other spot-checked claims held: 18
issue/PR states re-confirmed via gh on 2026-07-04
(#471/#472/#301/#332/#326/#328/#447/#443/#327/#175/#331/#349/#172 OPEN; #465/#476/#470 merged;
#461/#463 closed); ~40 file:line cites verified across gen_service_yaml.py, backend/main.py,
uploads.py, job_worker.py, cloud_run.py, execution_policy.py, orchestration.py, flywheel.py
(consent hard-deny confirmed at 44-52), config/models.py, Dockerfile:14-16, both runlogs, the
tracker, khelsutra locales/SetupWizard/App.tsx, and the web/src privacy-string zero-match claim
(grep reproduces: only two code comments, no user-facing privacy copy).",
74         ],
75         "verdict": "needs_additions"
76     }
77 },
78     "workflowProgress": [
79         {
80             "type": "workflow_phase",
81             "index": 1,
82             "title": "Research"
83         },
84         {
85             "type": "workflow_phase",
86             "index": 2,
87             "title": "Critique"
88         },
89         {
90             "type": "workflow_agent",
91             "index": 1,
92             "label": "draft:runbook",
93             "phaseIndex": 1,
94             "phaseTitle": "Research",
95             "agentId": "a5fffddeaf184a077",
96             "model": "claude-fable-5",
97             "state": "done",
98             "startedAt": 1783165705934,
99             "queuedAt": 1783165705919,
100             "attempt": 1,
101             "lastToolName": "StructuredOutput",

```



```

102     "lastToolSummary": "## Operator runbook ? live A1 alpha (Cloud Run asia-southea?",
103     "promptPreview": "You are drafting ONE section of the A13 \"alpha launch review\"
for the KhelSutra badminton-highlight alpha\n(repo Khelsutra/badminton-highlight-indexer, engine
checkout at C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer; SPA/site repo at
C:/Users/avidu/Projects/khelsutra-guru/khelsutra).\nContext: the A1 serving stack is LIVE on
Cloud Run asia-southeast1 ? control plane service\n'ra?",
104     "lastProgressAt": 1783165983146,
105     "tokens": 108196,
106     "toolCalls": 26,
107     "durationMs": 277212,
108     "resultPreview": "{\n  \"markdown\": \"## Operator runbook ? live A1 alpha (Cloud Run
asia-southeast1)\n\n> Scope note: this is an **operations** runbook for the stack that is
already live. It does not imply launch GO ? A10 (ship-gate target) and A11 (WASB-C3 provenance)
are still **GATED** owner rows (`docs/ALPHA_LAUNCH_READINESS.md:109-110`).\n\n**Live
topology** (e2e runlog `docs/COMPUTE_DECOUPLED_SERVING/runlogs/DEPLOY?\"
109   },
110   {
111     \"type\": \"workflow_agent\",
112     \"index\": 2,
113     \"label\": \"draft:observability\",
114     \"phaseIndex\": 1,
115     \"phaseTitle\": \"Research\",
116     \"agentId\": \"a4b697c820517aa72\",
117     \"model\": \"claude-fable-5\",
118     \"state\": \"done\",
119     \"startedAt\": 1783165705936,
120     \"queuedAt\": 1783165705919,
121     \"attempt\": 1,
122     \"lastToolName\": \"StructuredOutput\",
123     \"lastToolSummary\": \"## Observability & job-failure triage Scope: the live A1 st?\",

```

5. assistant (2026-07-04T12:18:05.881Z)

Now let me inspect the worktree and the repo docs.

6. user (2026-07-04T12:18:08.645Z)

On branch docs/a13-launch-review

Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean

122f174 docs(runlog): A1 post-merge e2e ? CUJ green on the final image; flip A1 DONE (#477)

3c97117 fix(storage): GCS keyless signing mints a cloud-platform-scoped token for signBlob

(closes #475) (#476)

195932f deploy(cloudrun): commit the control-plane service-manifest generator (closes #473)

(#474)

docs/a13-launch-review

7. user (2026-07-04T12:18:08.716Z)

No files found

8. user (2026-07-04T12:18:19.612Z)

Name

.github

.vscode

backend

deploy

docs

eval_baselines

scripts

tests

tools
training
.coveragerc
.gitattributes
.gitignore
CLAUDE.md
config.json
mypy.ini
pyproject.toml
pytest.ini
README.md
requirements-trackers.txt
requirements.txt
ruff.toml
run_all_tests.py
run_eval.py
run_regression.py
SESSION_HANDOFF.md
archives
COMPUTE_DECOUPLED_SERVING
data_pipeline
mockups
ABLATION_LAYER.md
ALPHA_LAUNCH_READINESS.md
ANALYZERS_AND_RECONCILERS.md
BACKLOG.md
CLOUD_GPU_DRYRUN_GUIDE.md
CODE_MAP.md
COMMERCIALIZATION.md
CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md
CREATING_REELS.md
DATA_IN_GCS.md
DECISION_SNAPSHOT_REGRESSION.md
DESKTOP_APP_PLAN.md
DOC_STATUS.md
EVALUATION.md
EXECUTION_POLICY.md
EXPERIMENT_HARNESS.md
GOLDEN_REGRESSION_FIXTURES.md
HOSTING_PLAN.md
HOW_RALLY_DETECTION_WORKS.md
I18N_PLAN.md
KHELSUTRA_PRODUCT_OVERVIEW.md
MULTI_SIGNAL_FUSION_PLAN.md
NEXT_STEPS.md
NIGHTLY_ENGINE_REGRESSION.md
OWNED_MODEL_IMPLEMENTATION_PLAN.md
PACKAGING_PLAN.md

9. user (2026-07-04T12:18:23.662Z)

```
1      # Alpha Launch Readiness Plan
2
3      > **Status:** `DRAFT` - cross-repo execution tracker; **Owner:** @avidullu; **Created:**
2026-07-03; **Last updated:** 2026-07-04
4      > **Lifecycle:** `DRAFT` -> IN PROGRESS -> DONE -> archived`
5      > **Tracking anchors:** Section 7 is the progress tracker and source of truth; indexed
in `docs/README.md` and `docs/DOC_STATUS.md`.
6      > **Relation to existing docs:** consolidates the launch-relevant pending items from
`NEXT_STEPS.md`, `ROADMAP.md`, `docs/data_pipeline/GOLDEN_VIDEOS.md`,
`R2_EVAL_METRIC_DESIGN.md`, `COMPUTE_DECOUPLED_SERVING/`, and the sibling `khelsutra` Studio
docs.
7      > **Honesty note:** `[verified]` means checked against the local synced repos on
2026-07-03. `[analysis]` means derived from a local command run on that snapshot. This is not
legal, financial, or launch approval.
```

```

8
9      ---
10
11      ## 0. TL;DR
12
13      The corpus is now large enough to move from "grow labels first" into alpha-readiness
work, but the 15-video headline is not itself a launch gate pass. `[verified]` The golden
manifest has 15 badminton videos; `[analysis]` A2-A9 made the local manifest/path, full-
fusion/audio, decision, and shadow-training surfaces explicit; `[analysis]` A4/A5 refreshed the
n=15 nightly and heuristic floors; `[analysis]` A8/A9 still reject served Gate A default
promotion and owned-model serving claims on the current evidence.
14
15      The first controlled alpha should be a hosted, authenticated product loop: upload a
video, process it, select rallies, compile, and download a reel. The owned-model/commercial
rally-detection claim remains a separate NO-GO until the ship-gate target is ratified and the
WASB checkpoint provenance is resolved.
16
17      ## 1. Problem & goal
18
19      The previous launch blockers were spread across roadmap docs and mixed together three
different questions:
20
21      - **Can an invited tester use the product without SSH?**
22      - **Can the 15-video golden corpus now support honest quality gates?**
23      - **Can we claim or ship an owned/commercial rally detector?**
24
25      This doc separates those into executable workstreams. The goal is to reach an invite-
only alpha where the product is usable end-to-end, the quality claims are honest and
reproducible, and the remaining model/legal gates are explicit rather than accidentally implied
by launch activity.
26
27      ## 2. Decisions locked
28
29      | # | Decision | Source / date | Implication |
30      |---|---|---|---|
31      | D1 | **Product alpha and owned-model launch are separate decisions.** |
`NEXT_STEPS.md` NO-GO banner, 2026-06-30; verified 2026-07-03 | We can launch a controlled
hosted product loop without claiming the owned model is commercially cleared. |
32      | D2 | **First alpha must be page-driven, not SSH/CLI-driven.** | `NEXT_STEPS.md` P0 UI-
driven/shareable item; verified 2026-07-03 | Hosted service, edge auth, upload, job polling,
compile, and download are launch-critical. |
33      | D3 | **The 15-video corpus is now a measurement asset, not a free launch pass.** |
`[verified]` `docs/data_pipeline/GOLDEN_VIDEO.md`; `[analysis]` local eval runs, 2026-07-03 |
Rebase/portability, feature completeness, baseline refresh, and gate ratification come before
quality claims. |
34      | D4 | **Do not promote served Gate A by default from the current n=15 evidence.** |
`[analysis]` `backend.eval.serve_contrast` run, 2026-07-04 | Proposal-side lift does not survive
served-windowing safety; keep it as an opt-in/analysis lever until owner ratification. |
35      | D5 | **P10/P11 corpus promotion and train/re-eval are alpha-plus unless owner reopens
D13.** | sibling `khehsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`; verified 2026-07-03 | First alpha
can use manual/offline corpus promotion; automated flywheel follows after serving and gates are
stable. |
36      | D6 | **Gen-0 n=15 shadow training is non-serving.** | `[analysis]`
`training.gen0.harness` run, 2026-07-04 | Learned mean beats the heuristic floor, but the paired
sign test is not significant; #172 and WASB C3 remain hard blockers. |
37
38      ## 3. Verified snapshot, 2026-07-03
39
40      All five Git checkouts were fast-forwarded before analysis:
41
42      | Repo | Branch | Synced HEAD |
43      |---|---|---|
44      | `badminton-highlight-indexer` | `master` | `e4914d1` |
45      | `khehsutra` | `master` | `b0f70a5` |
46      | `rally-annotator` | `main` | `ef9e088` |

```

```

47 | `sports-data-collector` | `master` | `bfc014f` |
48 | `sports-obsreport` | `main` | `a0c7252` |
49
50 Corpus/eval facts:
51
52 - `[verified]` `docs/data_pipeline/GOLDEN_VIDEOS.md` says
`output/human_lovo_manifest.json` is the ingested golden corpus at `n = 15`.
53 - `[analysis]` The local manifest had stale absolute paths. A temporary rebased manifest
found all 15 trajectories and labels in this workspace.
54 - `[analysis]` A7 generated full shuttle/person fusion schema for all 15 videos and A8
audio augmentation populated audio features for all 15.
55 - `[analysis]` `python -m backend.eval.fusion_compare --features-dir output` now runs
over all 15 and measures person-fusion `Delta F1 = +0.008`; the paired CI crosses zero, so
person fusion is not promotable by itself.
56 - `[analysis]` `python -m backend.eval.fusion_audio --features-dir output --compare`
measures audio `Delta F1 = +0.018`; the paired CI crosses zero, so audio is a shadow/suggestive
cue, not a default-flip.
57 - `[analysis]` `python -m backend.eval.serve_contrast --manifest
output\human_lovo_manifest.json` measured proposal Gate A mean `Delta F1 +0.0287`, but served
mean `Delta F1 -0.0328`; not served-safe.
58 - `[analysis]` `python -m training.gen0.harness --manifest
output\human_lovo_manifest.json --floor eval_baselines\heuristic_lovo_n15.json --version
0.1.1-n15-shadow.20260704 --out artifacts\rally_tal_gen0` produced learned LOVO mean F1 `0.551`,
floor passed versus heuristic `0.446`, but sign test `9/15`, `p = 0.3036`, `ship=False`.
59 - `[analysis]` `python scripts/nightly_regression.py --manifest <rebased-n15>
--features-dir output --no-report` reported default/milestone tolF1 `0.3636` vs baseline-off
tolF1 `0.1382`, but marked the frozen baseline stale due to config/corpus changes.
60 - `[analysis]` Before A6, `python -m backend.eval.ablation --features-dir output
--granularity group` failed at import due to a circular import between
`backend/eval/ablation.py` and `backend/eval/ablation_pdf.py`; A6 fixes the module entrypoint
and keeps full-fusion ablation scoped to the 6 videos with full schema until A7.
61
62 ## 4. Launch strategy
63
64 ### 4.1 Alpha lane: make the product usable
65
66 The alpha lane is successful when an invited tester can open the app, authenticate,
upload a video, watch the job progress, pick rallies, compile a reel, and download it. This lane
should avoid model-quality promises beyond the measured state and should default to the most
reliable serving path.
67
68 ### 4.2 Quality lane: make the 15-video corpus gate-ready
69
70 The quality lane is successful when the 15-video corpus is portable, reproducible,
feature-complete, and backed by refreshed baselines. Only then should R2 tolF1,
Gen-0/TemporalMaxer, or other model/promotion decisions be ratified.
71
72 ### 4.3 Flywheel lane: make tester feedback useful
73
74 The flywheel lane is successful when user corrections can flow into the collector/vault
path with consent/provenance and a visible re-eval delta. This is valuable for alpha learning,
but it should not block the first controlled alpha if manual promotion can bridge the gap.
75
76 ## 5. Risks and guardrails
77
78 | Risk | Why it matters | Guardrail |
79 |---|---|---|
80 | Launch activity implies model launch approval | The engine docs explicitly say rally-
detection product launch is NO-GO until ship target and WASB C3 clear | Keep all alpha copy
scoped to invite-only processing, not owned-model claims |
81 | Stale local manifests create false confidence | Manifest-driven tools skipped or
failed until paths were rebased | Make the corpus manifest portable and add a path/fixture smoke
check |
82 | 15-video corpus is only partially feature-complete | Fusion/person/audio eval is still
n=6 today | Regenerate or upconvert full-fusion features for all 15 before fusion decisions |

```

83 | Baseline refresh accidentally rubber-stamps a regression | Nightly runner correctly reported stale baselines | Refresh baselines only after reviewer sign-off and record the command/output |

84 | P10/P11 expands scope before hosted alpha is usable | Corpus flywheel is valuable but can consume the alpha launch window | Keep P10/P11 behind explicit owner reopen unless alpha lane is already stable |

85 | WASB checkpoint provenance remains unresolved | Blocks commercial sharing of a trained model regardless of F1 | Treat C3 as a hard model-release gate; do not hide it inside product launch tasks |

86

87 ## 6. Honest limits

88

89 Completing this doc's alpha lane does ****not**** prove the owned model is ready, that WASB weights are commercially clean, that the product can serve unbounded public traffic, that multi-sport is ready, or that every tester correction automatically improves the model.

90

91 Completing the quality lane does ****not**** by itself deploy a service. It only makes the measurement surface honest enough for a launch decision.

92

93 ## 7. Deliverables & progress tracker

94

95 Legend: TODO = not started, IN PROGRESS = actively being worked, DONE = shipped/verified, GATED = blocked on owner/legal/platform decision. One PR per row unless the row explicitly names a cross-repo batch.

96

97 | ID | Deliverable | Repo / owner | Depends on | Gated? | Status | PR |

98 |----|-----|-----|-----|-----|-----|----|

99 | A0 | Create this alpha readiness tracker and index it in docs | `badminton-highlight-indexer` | - | No | DONE | #451 |

100 | A1 | Stand up alpha serving endpoint: Cloud Run/GPU or approved host, health check, edge auth ON, upload -> process -> jobs -> compile -> download verified | `khealsutra` + `badminton-highlight-indexer` | A0 | Owner spend / CF config | DONE | #462/#464/#465/#467/#468/#469/#470/#474/#476; e2e runlog `docs/COMPUTE\_DECOUPLED\_SERVING/runlogs/DEPLOY\_REPORT\_cloud-serving\_a1-e2e\_2026-07-04.md` |

101 | A2 | Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke check that fails loudly when labels or trajectories are missing | `badminton-highlight-indexer` | A0 | No | DONE | #452 |

102 | A3 | Promote the 15-video source-video/MD5 records into the collector/vault path; reconcile collector's six-video source manifest with the 15-video eval corpus | `sports-data-collector` + vault | A2 | Owner upload/storage scope | DONE | collector #62 |

103 | A4 | Refresh the n=15 nightly regression baseline intentionally and record the exact command/output; keep stale-baseline warnings until reviewed | `badminton-highlight-indexer` | A2 | Reviewer sign-off | DONE | #453 |

104 | A5 | Recompute the heuristic n=15 floor and replace the stale `heuristic\_lovo\_n6` floor for future Gen-0 comparisons | `badminton-highlight-indexer` | A2 | No | DONE | #454 |

105 | A6 | Fix `backend.eval.ablation` CLI circular import, then run the R2/tolF1 ablation on the n=15 corpus | `badminton-highlight-indexer` | A2 | No | DONE | #455 |

106 | A7 | Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so fusion/person/audio analyses run over all 15 | `badminton-highlight-indexer` | A2 | GPU/data availability | DONE | #458 |

107 | A8 | Re-run `fusion\_compare`, group ablation, and served contrast after A7; explicitly promote/reject person fusion, Gate A served-default, and any other R-series default flips | `badminton-highlight-indexer` | A6, A7 | Reviewer sign-off | DONE | #458 |

108 | A9 | Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor; produce a non-serving artifact bundle and gate verdict | `badminton-highlight-indexer` | A5, A8 | Owner approval for M0.4 scope | DONE | #458 |

109 | A10 | Decide and record the alpha ship-gate target: metric, corpus slice, minimum threshold, significance rule, and what claims are allowed | Owner + `badminton-highlight-indexer` | A4, A5, A8 | Owner decision | GATED | - |

110 | A11 | Resolve or explicitly defer WASB C3 provenance for alpha messaging; if deferred, document "no commercial model sharing" copy guardrail | Owner/legal | A10 | Legal/provenance | GATED | - |

111 | A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector promotion, `train`/`reeval` job, delta-F1 surface, and alignment with `COMPUTE\_DECOUPLED\_SERVING` M11 | `khealsutra` + `badminton-highlight-indexer` + collector | A1,

```

A3 | Owner D13 reopen (`khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`) | GATED | - |
112 | A13 | Alpha launch review: runbook, privacy/consent copy, observability/job-failure
trriage, known limits, rollback/disable steps, and owner GO/NO-GO | Cross-repo | A1, A4, A10, A11
| Owner launch decision | TODO | - |
113
114 ## 8. Concrete next PR sequence
115
116 1. **Docs PR:** land this tracker.
117 2. **Corpus portability PR:** rebase/portable manifest tooling plus missing-file smoke
test.
118 3. **Hosted alpha PR/batch:** health/auth/upload/process/job/compile/download
verification and runbook updates.
119 4. **Baseline PR:** intentional n=15 nightly baseline refresh and heuristic floor
refresh.
120 5. **Ablation/fusion PRs:** fix ablation import, complete 15-video features, rerun
promotion/rejection decisions.
121 6. **Launch decision PR/docs update:** record ship-gate target, C3 status, alpha copy
limits, and launch checklist.
122
123 A2 smoke command:
124
125 ```powershell
126 python -m backend.eval.golden_manifest --manifest output/human_lovo_manifest.json --min-
count 15
127 python -m backend.eval.golden_manifest --manifest output/human_lovo_manifest.json --min-
count 15 --write output/human_lovo_manifest.local.json
128 ```
129
130 The checker resolves stale absolute paths against the repo's sibling `Annotation Setup`
folders and `output/`, then exits non-zero if any required `trajectory` or `labels` file is
absent.
131
132 A4 nightly baseline refresh record:
133
134 ```powershell
135 python scripts\nightly_regression.py --manifest output\human_lovo_manifest.json
--features-dir output --update-baseline
136 ```
137
138 Result:
139
140 ```text
141 INFO baseline snapshot written to C:\Users\avidu\Projects\khelsutra-guru\badminton-
highlight-indexer\eval_baselines\nightly_baseline.json
142 INFO default: n=15 tolF1=0.364 f1@0.5=0.472 merge_rate=0.135
143 INFO milestone: n=15 tolF1=0.364 f1@0.5=0.472 merge_rate=0.135
144 INFO baseline_off: n=15 tolF1=0.138 f1@0.5=0.237 merge_rate=0.638
145 ```
146
147 Pass-mode proof:
148
149 ```powershell
150 python scripts\nightly_regression.py --manifest output\human_lovo_manifest.json
--features-dir output --no-report
151 ```
152
153 Result: `PASS`, exit code `0`, baseline snapshot `2026-07-03`, default/milestone `tol_f1
= 0.3636`, `f1@0.5 = 0.4723`, `merge_rate = 0.1350`; baseline-off `tol_f1 = 0.1382`, `f1@0.5 =
0.2366`, `merge_rate = 0.6380`.
154
155 A5 heuristic floor refresh record:
156
157 ```powershell
158 python -m backend.eval.calibrate_local --manifest output\human_lovo_manifest.json
--metric f1 --floor-out eval_baselines\heuristic_lovo_n15.json

```

```

159     ``
160
161     Result:
162
163     ```text
164     baseline (untuned) mean held-out f1: 0.237
165     calibrated mean held-out f1:          0.446 +/- 0.239 (gap +0.000)
166     ``
167
168     The committed floor is `eval_baselines/heuristic_lovo_n15.json`: `n = 15`, `mean_f1 =
0.4456`, `std = 0.2390`, `metric = f1`, `iou = 0.5`. Every leave-one-video-out fold selected
`(0.05, 1.0, 2.0, 0)`.
169
170     A6 ablation/R2 record:
171
172     ```powershell
173     python -m backend.eval.ablation --features-dir output --granularity group
174     python -m backend.eval.rally_seg_eval --manifest output\human_lovo_manifest.json
--features-dir output --preset served --max-window-duration 0 --inactivity-end 0 --vs served
--vs-max-window-duration 6 --vs-inactivity-end 1.5 --json-out output\a6_r2_ablation.json
175     ``
176
177     Result: the ablation CLI now runs, but full-fusion feature ablation still reports `n=6`
because only the original six feature JSONs carry the full fusion schema; A7 must
regenerate/upconvert the newer nine before fusion/person/audio promotion decisions. The n=15
R2/tolF1 contrast is positive for the served milestone versus baseline-off: baseline/off `tolF1
= 0.138`, served `tolF1 = 0.364`, `Delta tolF1 = +0.225` with CI `[+0.165, +0.283]`, sign
`14/0`, `p = 0.000`; `Delta F1@0.5 = +0.236` with CI `[+0.161, +0.313]`; merge rate improves
from `0.638` to `0.135`. Net over-segmentation guard passes (`1.276 -> 0.378`), while the strict
segment-count rule remains blocked by increased splits, so R2/tolF1 is the supported gate for
the baseline-off to served milestone contrast rather than a blanket fusion/default-flip
approval.
178
179     A7 full-fusion/audio feature refresh record:
180
181     ```powershell
182     python -m backend.eval.fusion_golden --manifest output\human_lovo_manifest.json --out-
dir output --source-manifest ..\sports-data-collector\data\golden_source_videos.json --missing-
fusion-only --smallest-first --max-gpu-temp-c 87
183     python -m backend.eval.fusion_audio --manifest output\human_lovo_manifest.json
--features-dir output --source-manifest ..\sports-data-collector\data\golden_source_videos.json
--augment
184     ``
185
186     Result: the GPU pass completed without a thermal abort after fans were set to max
performance. Full-fusion schema is now present for `15/15` feature JSONs, and audio features are
present for `15/15`. The large regenerated clips completed as follows: `mahadevpura_1` `1`
window / `0` positives / `10` GT, `GX020094` `99` / `30` / `40`, `GX030094` `79` / `21` / `41`,
`GX010141` `26` / `11` / `29`, `GX010137` `56` / `31` / `34`, `Boxhill_Doubles` `59` / `12` /
`43`, `mahadevpura_2` `14` / `2` / `40`, `GX010128` `51` / `15` / `52`, `Badminton_BXH_2` `29` /
`1` / `44`.
187
188     A3 source-manifest follow-up: `sports-data-collector` PR #62 is merged. The
authoritative collector manifest now records `13/15` present local source/proxy paths and two
Boxhill run-log-only eval inputs: `GX010141_1080p.mp4` (`daaf8af198640516d14b217f50445d99`) and
`GX010137_1080p.mp4` (`74eedbc35d739c231344b8929df4b4a7`). The F-drive 4K originals are
preserved there only as `original_` provenance, not as eval join targets, because the golden
trajectories/labels are 1920-wide. Freshly re-extracting those two Boxhill full-fusion files
requires recovering the exact 1080p eval inputs or adding a coordinate-normalization fix; do not
substitute the 4K originals for person-cue extraction.
189
190     A8 fusion/default-decision record:
191
192     ```powershell
193     python -m backend.eval.fusion_compare --features-dir output

```

```

194     python -m backend.eval.fusion_audio --features-dir output --compare
195     python -m backend.eval.ablation --features-dir output --granularity group
196     python -m backend.eval.serve_contrast --manifest output\human_lovo_manifest.json
197     python -m backend.eval.rally_seg_eval --manifest output\human_lovo_manifest.json
--features-dir output --preset served --max-window-duration 0 --inactivity-end 0 --vs served
--vs-max-window-duration 6 --vs-inactivity-end 1.5 --json-out output\A6_r2_ablation.json
198     ```
199
200     Result:
201
202     - Person fusion is not promoted: shuttle-only F1 `0.302`, shuttle+person F1 `0.311`,
`Delta F1 +0.008`, paired CI `[-0.040, +0.055]`, sign `8W/6L`, `p = 0.791`.
203     - Audio is shadow/suggestive, not promoted: shuttle+person F1 `0.311`,
shuttle+person+audio F1 `0.329`, `Delta F1 +0.018`, paired CI `[-0.017, +0.053]`, sign `10W/4L`,
`p = 0.180`.
204     - Group ablation remains non-significant for cue promotion: audio `+0.018` CI `[-0.017,
+0.053]`; shuttle `+0.013` CI `[-0.002, +0.029]`; gate_a `-0.001` CI `[-0.012, +0.008]`; person
`-0.009` CI `[-0.045, +0.021]`.
205     - Gate A served default is rejected: proposal mean `Delta F1 +0.0287`, CI `[-0.0222,
+0.0680]`; served mean `Delta F1 -0.0328`, CI `[-0.0799, +0.0052]`, `regresses=True`.
206     - R2/cap/inactivity remains the strong quality result: served candidate F1 `0.472` vs
baseline-off `0.237`, `Delta F1 +0.236`, CI `[+0.161, +0.313]`; tolF1 `0.364` vs `0.138`, `Delta
tolF1 +0.225`, CI `[+0.165, +0.283]`; merge rate `0.135` vs `0.638`. Net over-seg guard passes,
strict per-video split-count guard still blocks blanket promotion language.
207
208     A9 Gen-0 shadow-training record:
209
210     ```powershell
211     python -m training.gen0.harness --manifest output\human_lovo_manifest.json --floor
eval_baselines\heuristic_lovo_n15.json --out artifacts\rally_tal_gen0 --version
0.1.1-n15-shadow.20260704
212     ```
213
214     Result: current Gen-0 LogReg/TAL harness produced local bundle
`artifacts\rally_tal_gen0\0.1.1-n15-shadow.20260704` with learned LOVO mean F1 `0.551` versus
heuristic floor `0.4456`, so `floor_passed=True`. The gate remains non-serving: paired sign
test `9/15`, `p = 0.3036`, `ship=False`; blockers are uncalibrated ship-gate target #172 and
WASB checkpoint provenance C3. TemporalMaxer itself is not implemented in this branch; the
evidence supports either closing A9 as a Gen-0 shadow baseline or opening a follow-up M0.4
scorer-swap PR.
215
216     ## 9. Open questions
217
218     1. Owner: Is the first alpha allowed to use manual/offline corpus promotion, or
should P10/P11 be reopened before any tester wave?
219     2. Owner/platform: Which hosted compute target is the next stand-up: Cloud Run GPU,
GCE L4 VM, or another approved GPU host?
220     3. Owner/legal: What is the acceptable alpha wording while WASB C3 remains
unresolved?
221     4. Engineering: Should the portable corpus manifest live as a generated artifact, a
checked-in redacted manifest, or both?
222     5. Engineering: Should A9 close on the current Gen-0 non-serving bundle, or should
the actual TemporalMaxer scorer swap be implemented before marking A9 done?
223
224     ## 10. Definition of done
225
226     - [ ] Invited tester can complete the hosted record/upload -> reel download CUJ without
SSH.
227     - [ ] Alpha service is gated behind the chosen auth posture and has a documented
disable/rollback path.
228     - [x] n=15 corpus is portable enough that manifest-driven quality commands do not rely
on stale local absolute paths.
229     - [x] n=15 nightly baseline and heuristic floor are refreshed with recorded commands.
230     - [x] R2/tolF1 gate is promoted or rejected with a recorded ablation result.
231     - [x] Fusion/person/Gate-A served-default decisions are explicitly promoted or rejected

```



```

on the 15-video corpus.
232     - [ ] Ship-gate target and allowed alpha claims are recorded.
233     - [ ] WASB C3 status is recorded and reflected in alpha messaging.
234     - [ ] P10/P11 are either reopened with a committed sequence or explicitly deferred for
alpha-plus.
235     - [ ] Owner records GO/NO-GO for the first alpha wave.
236
237     ## 11. References
238
239     Internal docs:
240
241     - `docs/NEXT_STEPS.md`
242     - `docs/ROADMAP.md`
243     - `docs/data_pipeline/GOLDEN_VIDEOS.md`
244     - `docs/R2_EVAL_METRIC_DESIGN.md`
245     - `docs/RALLY_DETECTION_FIXES_PLAN.md`
246     - `docs/COMPUTE_DECOUPLED_SERVING/README.md`
247     - sibling `khelsutra/NEXT_STEPS.md`
248     - sibling `khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`
249     - sibling `khelsutra/docs/PER_RALLY_SELECTION.md`
250     - sibling `sports-data-collector/data/golden_source_videos.json`
251
252     Local verification commands used for the 2026-07-03 snapshot:
253
254     ```powershell
255     python -m backend.eval.fusion_compare --features-dir output
256     python -m backend.eval.serve_contrast --manifest <rebased-n15-manifest>
257     python -m training.gen0.harness --manifest <rebased-n15-manifest> --floor
eval_baselines\heuristic_lovo_n15.json --version analysis-n15 --out <temp-out>
258     python scripts\nightly_regression.py --manifest <rebased-n15-manifest> --features-dir
output --no-report
259     python -m backend.eval.ablation --features-dir output --granularity group
260     python -m backend.eval.rally_seg_eval --manifest output\human_lovo_manifest.json
--features-dir output --preset served --max-window-duration 0 --inactivity-end 0 --vs served
--vs-max-window-duration 6 --vs-inactivity-end 1.5 --json-out output\a6_r2_ablation.json
261     ```
262
263     ---
264
265     ### Changelog
266
267     - 2026-07-03 - Created from a synced five-repo review of `NEXT_STEPS`, `ROADMAP`,
golden-corpus docs, local eval artifacts, and the current 15-video corpus state.
268     - 2026-07-04 - Recorded A7 full-fusion/audio regeneration, A8 n15 fusion/default
decisions, and A9 Gen-0 n15 shadow-training gate verdict.
269     - 2026-07-04 - Marked A3/A7/A8/A9 complete after collector #62 and indexer #458
review/CI; clarified that the two Boxhill eval inputs are run-log-only 1080p hashes and that
F-drive 4K originals are provenance-only.
270

```

10. user (2026-07-04T12:18:31.314Z)

```

1     # Documentation Status & Archival Register
2
3     > **Status:** ? **LIVING** ? the single board for doc **lifecycle + archival due-
diligence**. **Last full review:** 2026-07-02.
4     > **Update cadence:** whenever a doc changes lifecycle state (created / shipped a
milestone / delivered / archived), and re-review at each clean-slate.
5     > **Why this exists:** so we can (a) **review completeness** at a glance, (b) **update
it as work finishes** to prove we're done, and (c) **never archive a foundational doc without
honestly updating it first** (owner directive, 2026-06-21).
6
7     ---
8
9     ## 0. How to use this

```

10 - **Reviewing now?** Read §4 (the open findings/actions) ? that's what needs attention.
 §3 is the full register.

11 - **Marking something complete?** Update the doc's own **§7** progress tracker (the source of truth for its completeness), then flip its row here, then ? if terminal ? run the §2 archival checklist.

12 - **This register does NOT duplicate completeness detail** ? it indexes lifecycle + archive-readiness and points to each tracked doc's §7.

13

14 ## 1. Lifecycle legend

15 `DRAFT ? IN PROGRESS ? DONE/DELIVERED ? ARCHIVED`. Plus: **LIVING** (permanent reference, never "done"), **PROPOSAL/DESIGN** (owner-gated, not started), **REPORT** (a terminal record of a run/launch), **CONVENTION** (a standing rule/guide).

16

17 ## 2. Archival due-diligence checklist ? the repeatable process (do NOT just `git mv`)*

18 Before a foundational/tracked doc moves to `docs/archives/`:

19 1. **VERIFY** claims against the code ? no stale assertions; tag load-bearing facts `[verified]`. (A doc that says "nothing built" when code shipped is **not** archive-ready ? fix it first.)

20 2. **Flip Status** ? DELIVERED / DONE / REJECTED + a completion note ? what shipped, where (PRs, paths, artifacts), and what was **deferred** / carried forward.

21 3. **Update ALL** inbound references ? `docs/README.md` index, `CLAUDE.md`, `docs/NEXT\_STEPS.md`, and any cross-doc links ? the new path + status.

22 4. **Keep the lineage** ? ADR forward/back-links + "supersedes / superseded-by" so the evolution stays traceable.

23 5. **git mv** the whole dir/file ? `docs/archives/past\_projects/` (terminal-state ONLY; live docs stay at `docs/` top level). Relative links inside become historical (note this in the doc's header).

24 6. **Record the move** ? flip the row here to §3e + a line in memory `current-state`.

25

26 > **Worked example** (the template): **DECOUPLED_COMPUTE** ? #270 ? verified M0?M2+M3.5 delivered on GCP, flipped ??? DELIVERED with a completion note (incl. carried-forward WASB-C3/DPA/`md5`), reindexed `docs/README` + `past\_projects/README`, kept the ADR-010?014 lineage, then `git mv` to `docs/archives/past\_projects/DECOUPLED\_COMPUTE/`.

27

28 ## 3. The register

29

30 ### 3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's §7 is the completeness source-of-truth)

Doc	Status	Archive-ready?	Next action
`ALPHA_LAUNCH_READINESS.md`	DRAFT ? cross-repo alpha readiness tracker created 2026-07-03 after the golden corpus reached n=15; separates hosted product-alpha work from owned-model/commercial rally-detection gates	No	Execute §7 in order: corpus portability + n=15 baselines, hosted alpha serving, ablation/fusion reruns, then launch GO/NO-GO
`COMPUTE_DECOUPLED_SERVING/`	IN PROGRESS ? M0 approved + M1?M9 LARGELY SHIPPED* (#279?#290; M5 #286, M6 #287, M8 #288, M9-auth #290 all merged, default-OFF); Gen-0 weights pushed (`weights/0.1.0-l4/`) No M11 flywheel + M12 gdrive-api (owner residual: counsel/prices/OAuth/real-run within the D17 cap)		
`RALLY_QUALITY_RESEARCH.md`	Foundation, tracked & ACTIVE (§7) No Drives the R-series; complete when §7.1 single quality number is attributed		
`R2_EVAL_METRIC_DESIGN.md`	IMPLEMENTED (augmenting) No (gates R4 + dual-eval; ablation pending) Promote-or-reject toLF1 as the gate (ablation) ? then archive with the R-cluster		
`R1_MILESTONE_LAUNCH_REPORT.md`	LAUNCHED (#204) ? terminal record Soon (with the R-cluster) Keep with R-series for narrative; archive when the track wraps		
`RALLY_DETECTION_GAP_CLOSURE.md`	DRAFT (owner-gated) No G1?G3 levers; part of the R/quality cluster		
`RALLY_DETECTION_FIXES_PLAN.md`	DONE ? findings A/B/C resolved; P2b promoted `inactivity_temporal_density` default-ON (#396) With the R-cluster (deferred per its header; gate = R2 toLF1 ablation + §7.1 number) Archive with the R-series when the track wraps		
`RALLY_DETECTION_CODE_REVIEW.md`	Terminal snapshot ? findings A (#394), B (#394 ? promoted #396), C (`sdc#50`) all resolved With the R-cluster Archive with the R-series		
`PER_VIDEO_WORKSPACE.md`	DRAFT, IN PROGRESS (P0 = #264) No ? supersede-reconcile (finding #1) DONE; P3 de-hardcode base ? via #282; remaining P1?P6 (P1 v6 migration coordinates		

with M8's v6 cost-ledger ? version bump authored once) |

42 | `MULTI\_SIGNAL\_FUSION\_PLAN.md` | P1 DONE; P2+ owner-gated | No | P3/P4 (learned fusion, audio) ? needs GPU golden features |

43 | `EXPERIMENT\_HARNESS.md` | P1 IMPLEMENTED | No | A/B + promotion gate; supports the quality track |

44 | `PROMINENT\_COURT\_DETECTION.md` | DRAFT (owner-gated) | No | C-series; multi-track selection design |

45 | `EXECUTION\_POLICY.md` | P1 SHIPPED; P3b next | No | Async job policy; reused by Cloud Run dispatch |

46 | `GOLDEN\_REGRESSION\_FIXTURES.md` | IN PROGRESS (owner-gated) | No | Golden eval fixtures |

47 | ****Data layer**** ? `docs/data\_pipeline/` (`GOLDEN\_SET\_IMPLEMENTATION` . `GOLDEN\_SET\_PHASE1\_VIDEOS` . `GOLDEN\_VIDEOS` . `VIDEO\_RECORDING\_GUIDELINES` . `GOLDEN\_DATA\_SHARING`) | LIVING (data, ****no-ML****) | n/a | Relocated 2026-06-21 to the isolated data pillar; tracked via `data\_pipeline/README.md`. Roora **vendor** docs archived ? `archives/roora-vendor/`. |

48 | `archives/past\_projects/AUDIT\_REPORT\_khelsutra-guru\_2026-07-02-COMPLETED-2026-07-02.md` ****umbrella**** | ****COMPLETE / ARCHIVED (2026-07-02)**** ? the five-repo code audit + folded-in **Remediation tracker** (P0?P28 all closed) + absorbed prior audits. All non-owner-gated residuals landed; owner-gated/future items filed as issues (#443/#444/#445, + #301/#327/#332/#384). | Archived | Owner confirmed archival 2026-07-02; moved to `archives/past\_projects/` per §2. Residuals now live as GitHub issues, not in this doc. |

49 | `DECISION\_SNAPSHOT\_REGRESSION.md` | ****DRAFT ? decisions LOCKED**** (D1?D11 locked 2026-06-25, #383). Zero code shipped for P1?P7. | No | Start P1+P2 (schema + stitch-plan factor); de-risk the P10 main.py split |

50 | `DESKTOP\_APP\_PLAN.md` | DRAFT (owner-gated, nothing built) | No | P0 compute-feasibility spike; relates to truly-local profile |

51 | `OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` | PLAN/roadmap (owner-gated) | No (authoritative roadmap) | Gen-0?Gen-2; the owned-model track |

52 | `PERSONALIZATION\_PLAN.md` | DESIGN ? owner-adopted; Phase-0 | No | Per-user models (DB v5 store landed) |

53 | `UI\_ALPHA\_EXECUTION\_PLAN.md` | ACTIVE | No | khelsutra UI alpha track |

54 | `I18N\_PLAN.md` | Design (P0); ****in implementation**** (adapted) | No | Executed in `khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md` (P1?site/, P3?web/); P2 backend stays here, owner-gated |

55 | `POST\_166...RISK\_ASSESSMENT.md` | Plan for owner review | No | Milestones/risk doc |

56

57 **### 3b. Foundational / strategy references (LIVING ? archive only if ****superseded****, not when "done")**

58 | Doc | Status | Note |

59 | ---|---|---|

60 | `PLATFORM\_ARCHITECTURE.md` | Strategy/research | The `ComputeBackend`/`StorageBackend` registries ? realized by COMPUTE_DECOUPLED_SERVING |

61 | `COMMERCIALIZATION.md` | ? LIVING | Strategy; carries WASB-C3 |

62 | `VIDEO\_LOCALITY\_MODEL.md` | THINKING DOC | L0 fully-local = the truly-local profile |

63 | `HOSTING\_PLAN.md` | PROPOSAL | ****? partly overtaken by ADR-013 + the shipped site ? verify (finding #5)**** |

64 | `UI\_PROPOSAL.md` | PROPOSAL (partly shipped) | UX half ****in implementation**** ? `khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md`; currency banner added (finding #5 ? for this doc) |

65 | `STORAGE\_SHARING\_MODEL.md` | DRAFT (co-design pending) | ? |

66 | `ANALYZERS\_AND\_RECONCILERS.md` | DESIGN (owner-gated) | ? |

67 | `ROADMAP.md` / `CODE\_MAP.md` / `HOW\_RALLY\_DETECTION\_WORKS.md` / `KHELSUTRA\_PRODUCT\_OVERVIEW.md` | LIVING references | Keep current; not archive targets |

68

69 **### 3c. Reports & terminal records (archive candidates once their track wraps)**

70 | Doc | Status | Note |

71 | ---|---|---|

72 | `REAL\_FOOTAGE\_VALIDATION\_REPORT.md` | validation complete | Motivates R2; archive with the R-cluster |

73 | `RALLY\_DETECTION\_QUALITY\_REPORT.md` | report | Quality snapshot; part of R-cluster |

74 | `QUALITY\_ITERATIONS.md` | LIVING empirical log | ****Keep**** ? the paper-backbone ablation log ([[paper-coauthor-aim]]) |

75 | `CODE\_AUDIT\_AND\_TEST\_HARDENING` . `HARDENING\_LOOP\_HANDOFF` . `DEFERRED\_HARDENING\_PLAN` | COMPLETED ? ****finish-archived 2026-07-02**** (top-level stubs removed; absorbed into the

umbrella's *Prior audits*) | ? Full copies under `docs/archives/\*-COMPLETED-2026-06-14.md`;
inbound links repointed (finding #4 resolved)|

76

77 ### 3d. Conventions / guidelines / templates / runbooks (permanent ? do not archive)

78 `PROJECT\_DOC\_TEMPLATE.md` · `NEXT\_STEPS.md` (? refreshed 2026-06-21) · `BACKLOG.md` ·
`EVALUATION.md` · `ABLATION\_LAYER.md` · `SETUP\_NEW\_MACHINE.md` · `\*\*CREATING\_REELS.md\*\*` (? end-
user usage guide ? install ? process ? pick ? reel; LIGHT tier; added 2026-06-23) ·
`\*\*CONVERT\_GCS\_VIDEOS\_TO\_HIGHLIGHTS.md\*\*` (? operator runbook ? `gs://` videos ? reels on L4
Cloud Run; verified e2e 2026-06-23) · `TRACKNET\_WSL\_SETUP.md` · `README.md`. *(2026-06-21
relocation: the data-layer guides moved ? `data\_pipeline/`; the field-ops + Roora docs ?
`archives/{field-pmf,roora-vendor}/`.)*

79

80 ### 3e. Already archived (`docs/archives/`)

81 `past\_projects/DECOUPLED\_COMPUTE/` (? 2026-06-21, #270) · `past_projects/low-regret-
moves/` · `**field-pmf/**` (field-ops + customer-feedback, 2026-06-21) · `**roora-vendor/**`
(annotation-vendor engagement, 2026-06-21) · `PER\_VIDEO\_OUTPUT\_LAYOUT-SUPERSEDED-2026-06-21.md`
· `CODE\_AUDIT\_AND\_TEST\_HARDENING-COMPLETED-2026-06-14.md` · `DEFERRED_HARDENING_PLAN-
COMPLETED-2026-06-14.md` · `HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` ·
`COMMERCIAL\_READINESS\_REVIEW.md` · `MONETIZATION\_AUDIT.md` · `TEST\_RUN\_SUMMARY.md` · `research/`
· `checkpoints/` · `quality-iterations/` ·
`past\_projects/CODE\_CLEANUP\_AUDIT\_2026-06-24-SUPERSEDED-2026-07-02.md` (absorbed ? umbrella) ·
`past\_projects/KHELSUTRA\_GURU\_HEALTH\_REMEDIATION\_PLAN-FOLDED-2026-07-02.md` (folded ? umbrella).

82

83 ### 3f. ADR log ? `docs/archives/decisions/DECISIONS.md`

84 `\*\*ADR-001 ? ADR-014\*\*` (14, all ratified). The compute/serving lineage: `**ADR-005 ? 008 ?
009 ? 010 ? 011 ? 012 ? 013 ? 014**`. ADR-014 was `**RATIFIED 2026-06-21**` (owner M0 sign-off for
COMPUTE_DECOUPLED_SERVING).

85

86 ## 4. Open due-diligence findings / actions (2026-06-21 review)

87

88 > `\*\*? Doc-restructuring done (2026-06-21, owner-directed):\*\*` field-ops/customer-feedback
? `archives/field-pmf/`, Roora annotation-vendor ? `archives/roora-vendor/`, and a no-ML `**data
layer**` carved into `docs/data_pipeline/`. Inbound + outbound links rewired; three dir READMEs
added (the `data\_pipeline` one sets the "no segmentation/stitching" boundary).

89 1. `\*\*? RESOLVED (2026-06-21) ?` `PER\_VIDEO\_WORKSPACE.md` supersedes
`PER\_VIDEO\_OUTPUT\_LAYOUT.md`.**` Folded forward the `video_slug` naming option (workspace \$8.5) +
the `output/GX010125\_run/` manual-prototype precedent; predecessor archived ?
`docs/archives/PER\_VIDEO\_OUTPUT\_LAYOUT-SUPERSEDED-2026-06-21.md`; `CODE\_MAP.md` repointed.

90 2. `\*\*? RESOLVED (2026-06-21) ?` `NEXT\_STEPS.md` refreshed**` with a current 2026-06-21
state + a `\*\*cross-track prioritized pending list\*\*`; the 2026-06-13 owned-model/quality body is
kept (clearly marked) as that track's detail.

91 3. `\*\*R-series / quality cluster is ACTIVE\*\*` (`R1`, `R2`, `RALLY\_DETECTION\_GAP\_CLOSURE`, `RA
LLY_QUALITY_RESEARCH`, `QUALITY_ITERATIONS`, `REAL_FOOTAGE_VALIDATION_REPORT`, `RALLY_DETECTION_QUA
LITY_REPORT`). `**Do NOT archive yet**` ? archive as a *cluster* once R2 is promoted-or-rejected
as the gate + \$7.1's single number is attributed. (Owner already confirmed "leave as is",
2026-06-21.)

92 4. `\*\*? RESOLVED (2026-07-02) ?` finish-archived the hardening stubs.**` The three top-
level pointer-stubs (`CODE\_AUDIT\_AND\_TEST\_HARDENING`, `HARDENING\_LOOP\_HANDOFF`,
`DEFERRED\_HARDENING\_PLAN`) were removed; full archived copies remain under `docs/archives/`,
inbound links (README, PLATFORM_ARCHITECTURE, DESKTOP_APP_PLAN) were repointed, and the sweep is
summarized in the umbrella's *Prior audits (absorbed)*.

93 5. `\*\*HOSTING\_PLAN.md` / `UI\_PROPOSAL.md` are PROPOSAL but partly overtaken**` by ADR-013
(DO dropped) + the shipped khelsutra site. `\*\*?` `UI\_PROPOSAL.md` banner added (2026-06-21)**` ?
its UX half is now tracked in `khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md`; the `I18N\_PLAN.md`
localization track is likewise in implementation there. `\*\*HOSTING\_PLAN.md` banner still
pending**` (ADR-013 already noted in it).

94 6. `\*\*? RESOLVED (2026-06-21) ?` `COMPUTE\_DECOUPLED\_SERVING` M0 approved + M1?M4 SHIPPED**`
(#279/#280/#282/#283); all \$8 gating Qs locked (D7?D17); M5+ default-OFF code batch authorized
(owner residual: counsel ToS/DPA, real GCP prices, OAuth app, real cloud run within the D17
\$100/?10k cap).

95

96 ## 5. Maintenance

97 Update a row when its doc changes lifecycle; run \$2 before any archive; re-review \$4 at
each clean-slate. This register + the \$2 checklist are the operational form of the `CLAUDE.md`
archive policy and [[working-agreement-tracked-project-docs]].

11. user (2026-07-04T12:18:32.118Z)

No files found

12. user (2026-07-04T12:18:41.060Z)

No files found

13. user (2026-07-04T12:18:42.054Z)

No files found

14. user (2026-07-04T12:18:42.489Z)

No files found

15. assistant (2026-07-04T12:18:50.946Z)

Glob seems not to work against this path; using PowerShell instead.

16. user (2026-07-04T12:18:53.489Z)

FullName

C:\Users\avidu\Projects\khelsutra-guru\wt-a13\docs\PROJECT_DOC_TEMPLATE.md

C:\Users\avidu\Projects\khelsutra-

guru\wt-a13\docs\COMPUTE_DECOUPLED_SERVING\runlogs\RUNLOG_truly-local_TEMPLATE.md

True

17. user (2026-07-04T12:19:02.550Z)

DEPLOY_REPORT_cloud-serving_2026-07-04.md

DEPLOY_REPORT_cloud-serving_a1-cuj_2026-07-04.md

DEPLOY_REPORT_cloud-serving_a1-e2e_2026-07-04.md

README.md

RUNLOG_truly-local_TEMPLATE.md

18. user (2026-07-04T12:19:02.557Z)

```

1      <!--
2      PROJECT DOC TEMPLATE ? copy this file to docs/<YOUR_PLAN>.md and fill it in.
3
4      A "tracked project doc" is how this repo carries substantial design/project work: it can
be
5      ITERATED on, its progress TRACKED, and it is MARKED DONE against explicit deliverables,
then
6      archived. Standardizes the emergent pattern in docs/RALLY_QUALITY_RESEARCH.md §7 and
7      docs/GOLDEN_REGRESSION_FIXTURES.md.
8
9      Rules of thumb:
10     - §7 (the progress tracker) is the SOURCE OF TRUTH ? one row per independently-shippable
PR.
11     - Keep it HONEST: tag each claim [verified] (checked against code), [researched] (needs
sign-off),
12     or [design] (proposed). The doc reflects real shipped state, never aspirational.
13     - It POINTS to detail (code anchors, research artifacts), it does not duplicate it.
```

```

14 - Sections marked OPTIONAL are included only when relevant (e.g. Foundation/Threat-model
for
15 research- or security/privacy-heavy work). Delete sections that do not apply.
16 - Move to docs/archives/ once Status = DONE (per the archive policy: only terminal-state
work).
17 -->
18
19 # <Project / Plan Title>
20
21 > **Status:** `DRAFT` ? `<owner-gated? | in progress>` . **Owner:** `<name>` .
**Created:** `<YYYY-MM-DD>` . **Last updated:** `<YYYY-MM-DD>`
22 > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
23 > **Tracking anchors:** $7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in memory `current-state` (and `docs/NEXT_STEPS.md` once work starts).
24 > **Relation to existing docs:** `<extends | supersedes | peer-of ?>`
25 > **Honesty note:** mark each claim `[verified]` / `[researched]` / `[design]`. Not
legal/financial advice where applicable.
26
27 ---
28
29 ## 0. TL;DR
30 <2?5 sentences: the goal, the approach, and the single most important caveat. Call out
any finding that is already actionable on its own.>
31
32 ## 1. Problem & goal
33 <Why now; the concrete user-facing outcome; what "good" looks like. Link the code/docs a
reader must read to get current.>
34
35 ## 2. Decisions locked
36 <Table ? capture answered questions so they are not relitigated.>
37
38 | # | Decision | Source / date | Implication |
39 |---|-----|-----|-----|
40 | D1 | ? | ? | ? |
41
42 ## 3. Foundation ? research / prior art *(OPTIONAL)*
43 <Include when the plan rests on external facts: cited research, legal/compliance,
benchmarks, prior-art survey. Mark researched-vs-verified; list any sign-off gates this
creates.>
44
45 ## 4. Design / architecture
46 <The actual plan. Data contracts, diagrams, and a REUSE MAP ? name the existing
modules/code this builds on so it is clear what is new vs reused.>
47
48 ## 5. Threat model / risk table *(OPTIONAL ? security / privacy / compliance work)*
49 <Who/what is protected vs NOT; residual risks stated plainly.>
50
51 ## 6. Honest limits ? what this does NOT do
52 <The false-confidence guardrail: what a green suite / a DONE status does NOT prove.>
53
54 ## 7. Deliverables & progress tracker ? **source of truth**
55 Legend: ? Todo . ? In progress . ? Done . ? Blocked/gated. **One small PR per row**,
branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.
56
57 | ID | Deliverable | Depends on | Gated? | Status | PR |
58 |---|-----|-----|-----|-----|----|
59 | P0 | ? | ? | No | ? | ? |
60
61 ## 8. Open questions ? owner / external *(split blocking-gates from nice-to-knows; name
who decides)*
62 1. ?
63
64 ## 9. Definition of done
65 <Explicit acceptance checklist. When all ticked ? flip `Status ? DONE` and archive.>

```

```

66     - [ ] ?
67
68     ## 10. References
69     **Internal code anchors `[verified]`:** `` . **Internal docs:** `` .
70     **External:** `` .
71
72     ---
73
74     ### Changelog
75     - `` ? `` .

```

19. user (2026-07-04T12:19:12.959Z)

```

1     # A1 alpha serving ? post-merge e2e deploy test (cloud-serving)
2
3     **Date:** 2026-07-04 . **Region:** asia-southeast1 . **Project:** gen-lang-
client-0306026826
4     **Scope:** A1 (ALPHA_LAUNCH_READINESS §7) closure ? prove the merged serving fixes
5     (#465 input jail, #467 upload staging, #468 drain concurrency, #469 signed-URL download,
6     #470 auth-in-base-image) plus the committed manifest generator (#474) on a REAL deployed
build,
7     then restore the live edge-auth posture.
8
9     > Companion to `DEPLOY_REPORT_cloud-serving_a1-cuj_2026-07-04.md` (PR #464): that report
stood the
10    > stack up and FILED #461/#463/#466; THIS report proves their fixes end-to-end on merged
master and
11    > records the one new bug the run surfaced (#475, fixed by #476 mid-run).
12
13    ## 1. What was deployed
14
15    | Resource | Value |
16    |---|---|
17    | Source | merged `master` ? `195932f` (first build) ? **`3c97117`** (final build, incl.
#476) |
18    | Worker image | `?/rally/rally-worker:latest` ? build 1 `sha256:ed2fe448?` (195932f),
build 2 **`sha256:31475f27?`** (3c97117); the Job re-pinned via `gcloud run jobs update` after
each build |
19    | L4 worker **Job** | `rally-worker` ? unchanged shape (nvidia-l4, 8 vCPU/32Gi, FUSE
`khelsutra-rally-serving`?`/gcs`) |
20    | Control-plane **Service** | `rally-control-plane` ? rev **00005** (auth-off, build 1)
? rev **00006** (auth-off, final digest) ? rev **00007** (auth-ON restore, final digest).
Deployed FROM THE BASE IMAGE directly ? no derived PyJWT overlay (**#470 proven**); `maxScale=1`
(#471 option 1) |
21    | Manifest | generated by the **committed** `deploy/cloudrun/gen_service_yaml.py` (#474)
? `--edge-auth off` for the test window, `--edge-auth on` to restore. The scratchpad-artifact
gap (#473) is closed: this deploy used only repo-committed tooling |
22
23    ## 2. Boot / health (rev 00005+, identity-token auth)
24
25    - `GET /api/health` ? `200 {"status":"ok",?}`; `GET /api/capabilities` ? `wasb_hybrid`
present,
26    `gemini_key_present=false` (skip_ai_handoff honored), ffmpeg/ffprobe present.
27    - Before-state captured on the outgoing rev 00004: `/api/process` without a CF JWT ?
**401**
28    `edge auth: missing Cf-Access-Jwt-Assertion header` ? the M9 posture was live as
documented.
29
30    ## 3. CUJ on the FINAL image (rev 00006, digest `31475f27?`) ? all green
31
32    1. **gs:// process** ? `POST /api/process {gs://?/MahadevpuraSingles.MP4, wasb_hybrid,
33    compute=cloud}` ? 202 ? `validating ? dispatching (cloud_run) ? done` ? **48 play
segments**
34    (same count as the #464 run; job `18dc6502?`).

```

```

35      2. **#461 upload staging** ? chunked upload (`init ? part/0 ? complete`) of a
105,345-byte 720p
36      clip ? `POST /api/process` with the returned **local** path + `compute=cloud` ? the
job staged
37      it to `gs://khelsutra-rally-serving/videos/d8335f51?/upload-e2e-720p.mp4` (byte-exact
size
38      verified in the bucket) and dispatched to the L4 ? **success** (0 segments ? correct
for a
39      synthetic test pattern). NOT the pre-fix "cloud-serving needs a cloud input" hard-
fail.
40      Verified on BOTH builds (`ed2fe448?` and the final `31475f27?`).
41      - ? Upload validation floor: a 640x360 test clip is rejected (`resolution_fps_check`,
42      min 1280x720) ? the first attempt failed on this; alpha docs should mention the
floor.
43      3. **#468 drain concurrency, observed live** ? the upload job staged + dispatched
**while** the
44      48-segment reel compile was stitching on the CP: remote dispatch overlapped the local
compile
45      lane exactly as designed (no serialization, no false-empty stall). On rev 00006 the
two process
46      jobs (gs:// + upload) also ran the drain concurrently.
47      4. **compile** ? `POST /api/compile {video_id, segment_ids:[1..48], "e2e.mp4"}` ? 202 ?
done;
48      measured stitch wall on rev 00005: created 10:39:32 ? finished 10:44:26 (**~4m54s**,
incl. the
49      gs:// source re-fetch); rev 00006 re-run equally green. Reel **mirrored to
50      `gs://khelsutra-rally-serving/highlights/915473?/e2e.mp4` ? 174,046,228 bytes (~166
MiB)**.
51      5. **#463 signed-URL download ? VERIFIED**:
52      - `GET /api/highlights/915473?/e2e.mp4` ? **~307** ?
53      `https://storage.googleapis.com/khelsutra-rally-
serving/highlights/?/e2e.mp4?X-Goog-Algorithm=G00G4-RSA-SHA256&X-Goog-
Credential=492532266377-compute@?&X-Goog-Expires=3600&X-Goog-Signature=?`
54      (v4, signed AS the runtime SA via keyless signBlob ? no key file anywhere).
55      - `curl -L` through the redirect: **full 174,046,228 bytes in 8.1s** (single GET ? no
32 MiB
56      edge-500, no Range workaround), and the file **ffprobes valid**: H.264 1920x1080 @
29.97 fps
57      + AAC, duration 244.29s (the 48-rally reel).
58
59      ## 4. Bug surfaced mid-run: #475 ? fixed by #476 (merged, redeployed)
60
61      On the FIRST attempt (rev 00005, build `ed2fe448?` = pre-#476), the #463 check
**failed**:
62      `GET /api/highlights/...` ? **500**; log:
63      `[highlights] signed-url path failed?: you need a private key to sign credentials`.
64
65      - The reel WAS mirrored correctly; the **keyless signBlob path** failed and the handler
fell back
66      to serving the 166 MiB file locally ? Cloud Run's ~32 MiB cap ? edge-500 (the original
#463
67      symptom, resurrected by a different root cause).
68      - **Root cause (verified live):** `GCSStorage.signed_url()` reused the storage client's
own token
69      for `iamcredentials.signBlob`; `google.cloud.storage.Client.SCOPE` is **devstorage-
only**, which
70      signBlob rejects (`ACCESS_TOKEN_SCOPE_INSUFFICIENT`) ? and the branch's bare `except:
pass`
71      swallowed the real error, so the fallback raised the misleading private-key
message.
72      Environment was clean: the SA holds self-`serviceAccountTokenCreator` and
73      `iamcredentials.googleapis.com` is enabled ? code bug, not config.
74      - **Why tests never saw it:** the fakes carry no `_credentials`; this was the FIRST run
of #469's
75      path on real compute credentials. Filed **#475** ? fixed by **#476**

```



```

(`_signblob_identity()`)
76         mints a fresh `cloud-platform`-scoped ADC credential; failures now LOGGED before
falling back;
77         regression tests reproduce the credential shape). Suite 1659? / cov 85.46% / ruff+mypy
clean;
78         merged on green CI per the standing agreement; rebuilt (`31475f27?`) and redeployed
(rev 00006);
79         $3.5 is the pass on the fixed build.
80
81     ## 5. Notes / observations (rough edges captured)
82
83     - **`services replace` with an unchanged manifest is a NO-OP** ? it does not re-resolve
`:latest`
84         after a rebuild. Force a new revision (`gcloud run services update --image ? :latest`)
or stamp a
85         revision suffix. Follow-up candidate: `--revision-suffix` in `gen_service_yaml.py`.
86     - **A redeploy wipes the per-instance SQLite state** ? `/api/highlights` then 404s on
the DB
87         record check (`backend/main.py:1052`) even though the reel mirror exists in gs://.
Known, filed
88         as **471**; the CUJ was re-run on rev 00006 to regenerate state. Alpha-operator
implication: a
89         deploy invalidates in-flight jobs and past download URLs (re-process regenerates
them).
90     - Upload validation floor (?1280x720) ? $3.2.
91     - 422-before-401 quirk: a **malformed-JSON** body gets FastAPI's 422 before the edge-
auth
92         dependency runs; any well-formed request without a CF JWT still 401s. No exposure;
noted so a
93         422 in edge-auth triage isn't misread.
94
95     ## 6. Edge-auth restore + live posture (rev 00007, digest `31475f27?`)
96
97     - Redeployed with the generator's `--edge-auth on` manifest ? rev **00007**.
98     - `GET /api/health` ? **200** (deliberately exempt).
99     - `POST /api/process` (valid JSON, no CF JWT) ? **401**
100         `edge auth: missing Cf-Access-Jwt-Assertion header`.
101     - `https://api.khelsutra.guru/api/health` (no CF session) ? **302** ?
102         `https://khelsutra.cloudflareaccess.com/cdn-cgi/access/login/api.khelsutra.guru??`
with the
103         Access-app aud `c980da67?082b` in the login meta ? the exact pre-test posture,
restored.
104
105     ## 7. A1 DoD checklist
106
107     | DoD item | Status |
108     |---|---|
109     | Serving endpoint deployed + healthy | ? rev 00007, final digest `31475f27?` |
110     | Upload ? process ? jobs ? compile ? download CUJ | ? end-to-end on the final image |
111     | Edge-auth (M9) restored + verified | ? 401 sans JWT; api.khelsutra.guru ? CF login 302
|
112     | >32 MiB download via signed URL (#463) | ? 307 + full 174 MB single-GET + ffprobe-
valid |
113     | Browser-upload staging to gs:// (#461) | ? staged byte-exact + dispatched + success |
114     | Parallel drain (#466/#468) observed live | ? dispatch overlapped compile; 2 process
jobs drained concurrently |
115
116     **A1 ? DONE** (flipped in `docs/ALPHA_LAUNCH_READINESS.md` §7 in this PR).
117

```

20. user (2026-07-04T12:19:23.282Z)

```

668     2. **Standing default for ALL future compute work = GCP** (a GPU VM today via
`deploy/gcp`; Cloud Run+GPU for serving per ADR-012). Do **not** re-evaluate D0/Paperspace for
compute without a new ADR.

```

```

669     3. The `$200` DO credits, if used at all, go only to non-compute that DO bills
cleanly (CPU droplets / a site host / Spaces storage / bandwidth) ? never GPU.
670
671     ### Rationale (researched 2026-06-20, against DO's own docs)
672
673     - DO GPU is neither enabled on the account nor a clean free-credits path. The DO API
`/v2/sizes` returns 0 GPU sizes (GPU Droplets not enabled); GPU Droplets are North-
America-only (NYC2/TOR1/ATL1/RIC1/AMS3 ? no India/Singapore); and DO's own docs state that for
newer customers DO charges the team's primary payment method when GPU Droplets are created
(anti-abuse) ? so GPU spend hits the card, not the promo credit. The ADR-011 premise ("200
credits fund DO training") does not hold.
674     - Paperspace (now DO Gradient) free GPU is subscription-gated (a paid "G"/Growth
plan; high-end A100/H100 needs the $39/mo Growth plan), and Paperspace bills separately from
DO Droplet credits ? the $200 credit doesn't fund it either.
675     - GCP works today: quota granted (`GPUS_ALL_REGIONS=1`), bucket + SA + `$100` budget
provisioned, and M1/M2/M3.5 ran on an L4 for ~$12/run with no subscription. Real money, but
trivial per run ? and one portable PyTorch+CUDA stack (local ? GPU VM ? Cloud Run), no
second provider to maintain.
676
677     ### Consequences
678
679     - Compute provisioning = `deploy/gcp/` only. `SETUP_NEW_MACHINE.md` repointed to
GCP; `deploy/digitalocean/README.md` marked DEPRECATED; `HOSTING_PLAN.md`'s "All-
DigitalOcean" compute option retired.
680     - `deploy/digitalocean/` scripts are KEPT ? `bootstrap.sh` / `mount_scratch.sh` /
`gpu_setup.sh` are provider-agnostic Linux-GPU setup the GCP runbook reuses; only the DO-
specific provisioning (doctl droplet create, Spaces) is dropped. (Optional later cleanup: move
them to `deploy/common/`.)
681     - The shared DO CPU droplet `claude-do-rally-test` (another session's testbed) is
owner-coordinated cleanup, separate from this decision; the freshly-rotated DO API token
can be retired once that droplet is gone.
682     - Storage stays bucket-abstracted (`gcs://` primary). The `s3` backend still speaks
any S3-compatible store, so DO Spaces remains a possible storage target even though DO
compute is dropped.
683     - Capacity reality: GCP GPU stockouts are common (asia-south1/Mumbai was fully
stocked-out 2026-06-20 ? the M2 run landed an L4 in asia-southeast1/Singapore; weights still
land in the asia-south1 bucket). Sweep zones/regions; scale-to-zero (ADR-012) is part of why
Cloud Run+GPU fits.
684     - DECOUPLED_COMPUTE archives on M2 completion with a note that the DO leg was
dropped (this ADR), not "delivered on DO".
685

```

21. user (2026-07-04T12:19:23.515Z)

```

1     """Markdown internal-link checker ? fail CI on a broken internal link in a LIVE doc.
2
3     Scans tracked ``*.md`` files (via ``git ls-files``) and, for each ``[text](dest)`` /
4     ``![alt](dest)``
5     link whose destination is INTERNAL (a repo path and/or a ``#anchor``), verifies that
6     - the target file exists (resolved relative to the linking file, or the repo root for
7     ``/abs``), and
8     - any ``#anchor`` matches a GitHub-style heading slug in the target markdown file.
9
10    SKIPPED: external links (``http(s)``/``mailto``/``tel``/``ftp``/protocol-relative
11    ``/``), links inside
12    fenced code blocks, and any file under ``docs/archives/`` ? archived docs are frozen
13    snapshots whose
14    relative links are historical-by-design (see ``docs/DOC_STATUS.md`` §2). Mirrors
15    ``tools/leak_scan.py``:
16    run ``python -m tools.check_md_links`` (exit 1 on any broken internal link in a live
17    doc).
18
19    """
20
21    from __future__ import annotations

```

```

16
17     import argparse
18     import re
19     import subprocess
20     import sys
21     from pathlib import Path
22     from typing import Dict, List, Optional, Sequence, Set, Tuple
23
24     #: Source docs under these prefixes are NOT scanned ? archived = frozen snapshots whose
relative
25     #: links are historical-by-design (DOC_STATUS.md §2 step 5).
26     EXCLUDE_PREFIXES: Tuple[str, ...] = ("docs/archives/",)
27     #: This tool + its test legitimately embed sample/broken links ? never scan them.
28     SELF_EXCLUDE: Tuple[str, ...] = ("tools/check_md_links.py",
"tests/test_check_md_links.py")
29
30     _LINK_RE = re.compile(r"!?\[[^\]]*\]\(\s*([^\)]+?)\s*\)")
31     _HEADING_RE = re.compile(r"^(#{1,6})\s+(\.??)\s*#\s*$")
32     _FENCE_RE = re.compile(r"^\s*(?:``|~~~)")
33     #: a scheme (`http:`, `mailto:`, ?) or protocol-relative `//` ? external, not our
concern.
34     _EXTERNAL_RE = re.compile(r"^(?:[a-zA-Z][a-zA-Z0-9+.\-]*:|//)")
35
36
37     def slugify(heading: str) -> str:
38         """GitHub-style heading ? anchor slug (matches `github-sluggger`): lowercase, drop
punctuation
39         (keep word chars / spaces / hyphens), then replace EACH space with a hyphen ?
crucially NOT
40         collapsing, so a heading with removed punctuation (e.g. 'a / b' ? 'a b') yields
'--' like GitHub."""
41         s = re.sub(r"^\w\s-", "", heading.strip().lower())
42         return s.replace(" ", "-")
43
44
45     def heading_slugs(text: str) -> Set[str]:
46         """The set of GitHub anchor slugs for a markdown file's headings (with -1/-2 dedup
suffixes)."""
47         counts: Dict[str, int] = {}
48         out: Set[str] = set()
49         in_fence = False
50         for line in text.splitlines():
51             if _FENCE_RE.match(line):
52                 in_fence = not in_fence
53                 continue
54             if in_fence:
55                 continue
56             m = _HEADING_RE.match(line)
57             if not m:
58                 continue
59             base = slugify(m.group(2))
60             n = counts.get(base, 0)
61             out.add(base if n == 0 else f"{base}-{n}")
62             counts[base] = n + 1
63         return out
64
65
66     def _split_dest(dest: str) -> Tuple[str, str]:
67         """`path#anchor` (with an optional `title` suffix) ? `(path, anchor)`."""
68         dest = dest.strip().split()[0] # drop a trailing "title"
69         path, _, anchor = dest.partition("#")
70         return path, anchor
71
72
73     def check_file(repo: Path, rel: str, slug_cache: Dict[str, Set[str]]) -> List[Tuple[int,

```

```

str]]:
74     md = repo / rel
75     base = md.parent
76     findings: List[Tuple[int, str]] = []
77     in_fence = False
78     for i, line in enumerate(md.read_text(encoding="utf-8",
errors="replace").splitlines(), 1):
79         if _FENCE_RE.match(line):
80             in_fence = not in_fence
81             continue
82         if in_fence:
83             continue
84         for m in _LINK_RE.finditer(line):
85             dest = m.group(1).strip()
86             if not dest or _EXTERNAL_RE.match(dest):
87                 continue
88             path, anchor = _split_dest(dest)
89             if path == "":
90                 target = md # pure same-file anchor
91             elif path.startswith("/"):
92                 target = repo / path.lstrip("/")
93             else:
94                 target = (base / path).resolve()
95             if path != "" and not target.exists():
96                 findings.append((i, f"broken link -> {dest} (no file {path})"))
97                 continue
98             if anchor and target.suffix.lower() == ".md" and target.is_file():
99                 key = str(target)
100                 if key not in slug_cache:
101                     slug_cache[key] = heading_slugs(
102                         target.read_text(encoding="utf-8", errors="replace")
103                     )
104                 if slugify(anchor) not in slug_cache[key]:
105                     findings.append((i, f"broken anchor -> {dest} (no heading
#{anchor})))")
106     return findings
107
108
109 def tracked_md(repo: Path) -> List[str]:
110     out = subprocess.run(
111         ["git", "ls-files", "*.md"], cwd=str(repo), capture_output=True, text=True
112     )
113     files = sorted({f.strip() for f in out.stdout.splitlines() if f.strip()})
114     return [
115         f for f in files if not f.startswith(EXCLUDE_PREFIXES) and f not in SELF_EXCLUDE
116     ]
117
118
119 def scan_repo(repo: Path) -> List[Tuple[str, int, str]]:
120     slug_cache: Dict[str, Set[str]] = {}
121     results: List[Tuple[str, int, str]] = []
122     for rel in tracked_md(repo):
123         try:
124             for lineno, reason in check_file(repo, rel, slug_cache):
125                 results.append((rel, lineno, reason))
126         except OSError:
127             continue
128     return results
129
130
131 def main(argv: Optional[Sequence[str]] = None) -> int:
132     ap = argparse.ArgumentParser(
133         description="Check internal markdown links in LIVE docs (excludes
docs/archives/)."
134     )

```

```

135     ap.add_argument(
136         "--root",
137         type=Path,
138         default=Path(__file__).resolve().parents[1],
139         help="repo root (default: the parent of tools/)",
140     )
141     args = ap.parse_args(list(argv) if argv is not None else None)
142     results = scan_repo(args.root)
143     if not results:
144         print("[check-md-links] clean ? no broken internal links in live docs.")
145         return 0
146     print(f"[check-md-links] {len(results)} broken internal link(s):", file=sys.stderr)
147     for rel, lineno, reason in results:
148         print(f"  {rel}:{lineno}: {reason}", file=sys.stderr)
149     return 1
150
151
152 if __name__ == "__main__":
153     raise SystemExit(main())
154

```

22. user (2026-07-04T12:19:38.433Z)

```

1     """Leak-scan guard ? refuse to commit obvious PII / attribution leaks into the code
surface.
2
3     A *structural* CI guard (deliberately NOT a name detector): it scans the TRACKED text
files under
4     the code/data roots for
5
6     (a) a 32-hex MD5 token ? the ``video_id`` shape
(``backend/utils/hashing.py::compute_video_id``);
7     16-hex ``fx`` fixture slugs and 12-hex ``gt_hash`` values do NOT match, by
design.
8     (b) a personal absolute user path ? ``C:\\Users\\<name>``, ``/home/<name>/``,
``/Users/<name>/`` ?
9     where ``<name>`` is a real username (documented placeholders like
``<name>/``/``<user>`` are OK).
10
11     Optionally, given a PRIVATE (gitignored) name-list via ``--names`` (default
``<root>/leakscan_private``
12     if present), it ALSO flags those tokens ? so the owner can scan for real player/venue
names locally
13     while CI runs only the name-free structural guard (you can't commit a name-list to scan
for).
14
15     A line carrying ``# leakscan-allow`` is skipped (escape hatch for a legitimate token,
e.g. an MD5
16     test-vector). Run ``python -m tools.leak_scan`` (exit 1 on any finding). SCOPE = the
shipped-code
17     surface (``backend/ training/ scripts/ tools/ eval_baselines/``); ``tests/`` (legit
example
18     fixtures), ``docs/``, and gitignored ``scratch/``/``output/`` are NOT scanned (scan
those locally
19     with ``--names`` against a private list if desired).
20     """
21
22     from __future__ import annotations
23
24     import argparse
25     import re
26     import subprocess
27     import sys
28     from pathlib import Path
29     from typing import List, Optional, Sequence, Tuple

```

```

30
31     #: Shipped-code + committed-data surfaces scanned. NOT ``tests/`` (fixtures legitimately
embed
32     #: example paths/usernames/MD5s ? noise for a structural guard; scan those locally with
--names),
33     #: and NOT ``docs/``/``scratch/``/``output/`` (intentional operator notes).
34     SCAN_ROOTS: Tuple[str, ...] = (
35         "backend",
36         "training",
37         "scripts",
38         "tools",
39         "eval_baselines",
40     )
41
42     #: This tool + its test legitimately contain sample patterns ? never scan them.
43     SELF_EXCLUDE: Tuple[str, ...] = ("tools/leak_scan.py", "tests/test_leak_scan.py")
44
45     #: A run of EXACTLY 32 hex chars (the MD5 video_id shape). The negative look-around
means a 16-hex
46     #: ``fx_`` slug, a 12-hex ``gt_hash``, or a 40/64-hex SHA do NOT match.
47     _MD5_RE = re.compile(r"(?![0-9a-fA-F])[0-9a-fA-F]{32}(?![0-9a-fA-F])")
48     _WIN_USER_RE = re.compile(r"[A-Za-z]:[\\\/]+Users[\\\/]+([A-Za-z0-9._-]+)")
49     _NIX_USER_RE = re.compile(r"/(?:home|Users)/([A-Za-z0-9._-]+)/")
50
51     #: Tokens that are a DOCUMENTED placeholder / generic example, not a real username ? so
a doc'd
52     #: path like ``C:\\Users\\x`` or ``/home/u`` does not trip the guard.
53     _PLACEHOLDERS = frozenset(
54         {
55             "<name>",
56             "<user>",
57             "<username>",
58             "<you>",
59             "you",
60             "youruser",
61             "yourname",
62             "user",
63             "username",
64             "x",
65             "u",
66             "a",
67             "b",
68             "c",
69             "n",
70             "foo",
71             "bar",
72             "baz",
73             "test",
74             "testuser",
75             "tmp",
76             "home",
77             "example",
78         }
79     )
80
81     _ALLOW = "# leakscan-allow"
82     _TEXT_EXT = {
83         ".py",
84         ".json",
85         ".txt",
86         ".csv",
87         ".yaml",
88         ".yml",
89         ".cfg",
90         ".ini",

```

```

91         ".toml",
92         ".md",
93     }
94
95
96     def _tracked_files(root: Path) -> List[str]:
97         """The tracked files under SCAN_ROOTS (via ``git ls-files`` so only committed paths
are scanned)."""
98         out = subprocess.run(
99             ["git", "ls-files", *SCAN_ROOTS], cwd=str(root), capture_output=True, text=True
100         )
101         return [f.strip() for f in out.stdout.splitlines() if f.strip()]
102
103
104     def _load_names(path: Optional[Path]) -> List[str]:
105         if path is None or not path.is_file():
106             return []
107         return [
108             ln.strip()
109             for ln in path.read_text(encoding="utf-8", errors="replace").splitlines()
110             if ln.strip() and not ln.lstrip().startswith("#")
111         ]
112
113
114     def scan_text(text: str, *, names: Sequence[str] = ()) -> List[Tuple[int, str]]:
115         """Return ``(lineno, reason)`` findings for one file's text (1-based line
numbers)."""
116         findings: List[Tuple[int, str]] = []
117         lname = [n.lower() for n in names if n]
118         for i, line in enumerate(text.splitlines(), start=1):
119             if _ALLOW in line:
120                 continue
121             if _MD5_RE.search(line):
122                 findings.append((i, "32-hex token (video_id / MD5 shape)"))
123             for m in _WIN_USER_RE.finditer(line):
124                 if m.group(1).lower() not in _PLACEHOLDERS:
125                     findings.append(
126                         (i, f"personal Windows user path (?Users\\{m.group(1)})")
127                     )
128                     break
129             for m in _NIX_USER_RE.finditer(line):
130                 if m.group(1).lower() not in _PLACEHOLDERS:
131                     findings.append((i, f"personal POSIX user path ({m.group(1)})"))
132                     break
133             low = line.lower()
134             for n in lname:
135                 if n in low:
136                     findings.append((i, f"private name-list token {n!r}"))
137                     break
138         return findings
139
140
141     def scan_repo(
142         root: Path, *, names_file: Optional[Path] = None
143     ) -> List[Tuple[str, int, str]]:
144         """Scan the tracked code-surface files under ``root``; return ``(relpath, lineno,
reason)``."""
145         names = _load_names(names_file)
146         results: List[Tuple[str, int, str]] = []
147         for rel in _tracked_files(root):
148             if rel in SELF_EXCLUDE or Path(rel).suffix.lower() not in _TEXT_EXT:
149                 continue
150             try:
151                 text = (root / rel).read_text(encoding="utf-8", errors="replace")
152                 except OSError:

```

```

153         continue
154     for lineno, reason in scan_text(text, names=names):
155         results.append((rel, lineno, reason))
156     return results
157
158
159     def main(argv: Optional[Sequence[str]] = None) -> int:
160         ap = argparse.ArgumentParser(
161             description="Leak-scan guard: 32-hex MD5 + personal paths (+ optional private
162             name-list).")
163         ap.add_argument(
164             "--root",
165             type=Path,
166             default=Path(__file__).resolve().parents[1],
167             help="repo root (default: the parent of tools/)",
168         )
169         ap.add_argument(
170             "--names",
171             type=Path,
172             default=None,
173             help="optional gitignored private name-list (default: <root>/leakscan_private
174             if present)",
175         )
176         args = ap.parse_args(list(argv) if argv is not None else None)
177         root: Path = args.root
178         names_file = args.names if args.names is not None else (root / ".leakscan_private")
179         findings = scan_repo(root, names_file=names_file)
180         if not findings:
181             print(
182                 "[leak-scan] clean ? no MD5 / personal-path / private-name leaks in the code
183                 surface."
184             )
185             return 0
186         print(
187             f"[leak-scan] {len(findings)} potential leak(s) "
188             f"(add '# leakscan-allow' on a line to suppress a legitimate one):",
189             file=sys.stderr,
190         )
191         for rel, lineno, reason in findings:
192             print(f" {rel}:{lineno}: {reason}", file=sys.stderr)
193         return 1
194
195     if __name__ == "__main__":
196         raise SystemExit(main())

```

23. user (2026-07-04T12:19:39.444Z)

Found 2 files
C:\Users\avidu\Projects\khelsutra-guru\wt-a13\tests\test_leak_scan.py
C:\Users\avidu\Projects\khelsutra-guru\wt-a13\tests\test_check_md_links.py

24. user (2026-07-04T12:19:51.471Z)

No matches found

25. user (2026-07-04T12:19:57.095Z)

```

21: def test_slugify_matches_github_double_hyphen_on_removed_punct():
27: def test_heading_slugs_dedup_and_skips_fences():
33: def test_valid_link_and_anchor_pass(tmp_path):
41: def test_missing_file_flagged(tmp_path):

```



```

46: def test_dead_anchor_flagged(tmp_path):
51: def test_external_and_fenced_are_skipped(tmp_path):
56: def test_link_to_existing_archived_file_is_ok(tmp_path):
61: def test_title_suffix_is_stripped(tmp_path):

```

26. assistant (2026-07-04T12:20:11.533Z)

Unit tests only ? good. Now let me read the a1-cuj runlog since the drafts cite it heavily.

27. user (2026-07-04T12:20:12.616Z)

```

1      # A1 alpha serving deploy ? hosted CUJ run report (cloud-serving)
2
3      **Date:** 2026-07-04 . **Region:** asia-southeast1 . **Project:** gen-lang-
client-0306026826
4      **Scope:** A1 (ALPHA_LAUNCH_READINESS $7) ? stand up the alpha serving endpoint and
verify the
5      hosted `process ? jobs ? compile ? download` CUJ against the WASB-only owned detector.
6
7      > Companion to `DEPLOY_REPORT_cloud-serving_2026-07-04.md` (PR #457, the GPU-batching
runlog). That
8      > report proved the L4 worker + measured the batching win via the M6 dispatch **shim**;  
THIS report
9      > proves the deployed **control-plane service** end-to-end. They do not overlap.
10
11     ## 1. What was deployed
12
13     | Resource | Value |
14     |---|---|
15     | L4 worker **Job** | `rally-worker` ? nvidia-l4, 8 vCPU / 32Gi, `--no-gpu-zonal-
redundancy`, `--max-retries 0 --task-timeout 3600`, GCS-FUSE mount `khelsutra-rally-
serving`?`/gcs`, `WASB_WEIGHTS=/gcs/models/wasb_badminton_best.pth.tar` |
16     | Control-plane **Service** | `rally-control-plane` ? rev 00002, `--no-cpu-throttling`,
`min/maxScale 1/3`, cpu 2 / mem 4Gi, port 8080, **private (IAM-gated, no allUsers)** |
17     | Service image | `asia-southeast1-docker.pkg.dev/?/rally/rally-control-plane:latest` (=
`rally-worker:latest` + `google-cloud-run`; see $5 bug 1) |
18     | Service URL | `https://rally-control-plane-2bosgeukpa-as.a.run.app` |
19     | Config (via base64 startup wrapper ? `RALLY_APP_HOME=/tmp/apphome/config.json`) |
`deployment.profile=cloud-serving`, `compute_target=cloud_run`, `indexing.skip_ai_handoff=true`,
`storage.backend=gcs`, `output_root=gs://khelsutra-rally-serving`, `input_backend=gcs`,
`input_root=?/videos` |
20     | Env | `DEPLOYMENT_PROFILE=cloud-serving`, `CLOUD_RUN_REGION=asia-southeast1`,
`GOOGLE_CLOUD_PROJECT=?`, `RALLY_ALLOWED_HOSTS=*` (CLOUD_RUN_JOB is reserved ? code default
`rally-worker` is correct) |
21
22     **Boot-crash trap avoided:** the `cloud-serving` preset forces
`indexing.skip_ai_handoff=False`
23     (Gemini ON), which would trip the `AI_HANDOFF_KEY_MISSING` startup check (no Gemini key)
and
24     crash-loop. Verified `loader.py:194` deep-merges `defaults < preset < config.json`, so
the
25     explicit `skip_ai_handoff=true` wins. Also pinned `storage.backend=gcs` explicitly to
satisfy the
26     `STORAGE_INCOHERENT` check. Boot log confirmed: `deployment profile ACTIVE: cloud-
serving
27     (compute_target=cloud_run)` + `Uvicorn running on http://0.0.0.0:8080`.
28
29     ## 2. Health / capabilities (authenticated via `gcloud auth print-identity-token`)
30
31     - `GET /api/health` ? `200`
{"status":"ok","sport":"badminton","default_segmenter":"yolo_hybrid"}`
32     - `GET /api/capabilities` ? `segmenters` incl. `wasb_hybrid`;
`deployment.cloud_available=true`;
33     `gemini_key_present=false` (no crash ? skip_ai_handoff honored); `gcs` backend usable.
34

```

```

35     ## 3. gs:// bucket-path CUJ ? PROVEN end-to-end
36
37     1. `POST /api/process {"video_path":"gs://khelsutra-rally-
serving/videos/MahadevpuraSingles.MP4","segmenter":"wasb_hybrid","compute":"cloud"}` ? `202
{job_id: d16654d1?}`
38     2. Poll `GET /api/jobs/{id}`: `validating ? dispatching (cloud_run) ? done`.
39         - CP pulled + validated the video (format/resolution/pts/scene/blur/relevance all
pass).
40         - Dispatched L4 Job execution `rally-worker-k49kg`; worker log: `WASB in FULLY-LOCAL
mode
41             (skip_ai_handoff=True) ? no gemini handoff, no API key needed`, `torch 1.11.0+cu113
cuda True`.
42         - `**Result: `success` ? "Successfully indexed 48 play segments (cloud_run /
'wasb_hybrid')***;
43             outputs at `gs://khelsutra-rally-serving/outputs/915473?/`.
44         - Wall: created 02:12:42 ? finished 02:26:27 (**~13m45s**, incl. L4 cold image pull +
inference).
45     3. `POST /api/compile {video_id, segment_ids:[1..48],
output_filename:"mahadevpura_wasb_highlights.mp4"}` ? `202 {job_id: d947a9e6?}` ? poll ?
`**success`**. In-process ffmpeg stitch (trim 48 clips w/ padding ? concat ? watermark) on the
CP; `**Dynamic highlight reel saved ? (166.0 MB)`**; wall ~3m26s (02:27:47?02:31:13, incl. re-
fetch of the gs:// source).
46     4. `GET /api/highlights/{video_id}/mahadevpura_wasb_highlights.mp4` ? the reel is served
(correct `206`/`video/mp4`/etag/size headers), and the `**full 174,038,655-byte reel was
retrieved via <32 MiB Range requests and fprobes as a VALID video: H.264 1920x1080 @ 29.97fps +
AAC, duration 244.3s**` (48 rallies). ? `**A single full `GET` returns edge-500**` ? Cloud Run's
~32 MiB response cap vs. the `FileResponse` fixed `Content-Length` (see $5 bug 3).
47
48     ## 4. Notes / observations
49
50     - `**In-process compile serializes behind process jobs**` (drain
`ThreadPoolExecutor(max_workers=1)`):
51         a queued compile waits while a process job holds the worker in its bucket-poll. Fine
for the alpha
52         (low concurrency); worth noting for load.
53     - `**WASB batch tuning is NOT forwarded on the dispatch path.**` `_maybe_dispatch_remote`
forwards only
54         `skip_ai_handoff` (orchestration.py:274-283); the L4 job runs its baseline batch (not
the preset's
55         batch64/prefetch4), so the measured #457 ?30.5% win is not applied via the deployed
CP. Minor
56         (correctness unaffected); follow-up = forward `indexing.wasb.*` overrides like
`skip_ai_handoff`.
57     - `player_pool` / `max_frames` are dropped on the cloud path (already noted at
orchestration.py:273).
58
59     ## 5. Bugs surfaced + handled
60
61     1. `**google-cloud-run` missing from the image**` (BLOCKER). First real CP deploy failed
every dispatch
62         with `**google-cloud-run is required for CloudRunCompute's real client**` ? the image
installed only
63         `[storage-gcs]`; the reused-as-control-plane role needs `google.cloud.run_v2`.
`**Unblocked**` with a
64         derived image (`rally-worker` + `pip install google-cloud-run`). `**Durable fix: PR
#462**` (adds a
65         `cloud-run` extra + `[storage-gcs,cloud-run]` in the Dockerfile + test guard; 19
tests pass).
66     2. `**Browser-upload ? gs:// staging gap.**` Uploads land on the CP's local FS
(`/api/upload/complete` ?
67         `/tmp/apphome/output/uploads/?`), and `_maybe_dispatch_remote` hard-rejects local
input under
68         cloud-serving (orchestration.py:250-255). Upload endpoints verified working; the gap
is staging.
69         `**Filed: issue #461**` (fix = stage local?gs://input_root before dispatch).

```

```

70      3. **`/api/highlights` download 500s for reels >32 MiB.** `FileResponse` sets a fixed
`Content-Length`;
71      Cloud Run caps a single buffered HTTP/1 response at ~32 MiB (empirically: 29.6 MB
Range ? 206,
72      32.4 MB ? 500). App logs 200; Google Frontend returns 500. Every real reel exceeds 32
MiB.
73      **Filed: issue #463** (fix = serve reels via a gs:// signed-URL redirect; also fixes
the ephemeral
74      `/tmp` reel location). Workaround verified: ranged download reassembles a byte-exact,
valid reel.
75
76      ## 6. M9 edge auth ? ENABLED + gate verified (DNS routing remains, owner-handoff)
77
78      Cloudflare Access edge-auth is **ON** (rev `rally-control-plane-00004-tvr`) and
enforcing:
79      - Image rebuilt to add `PyJWT[crypto]` (the `auth` extra) + the `google-cloud-run`
client.
80      - Config: `security.edge_auth_enabled=true`,
`cf_team_domain=https://kheIsutra.cloudflareaccess.com`,
81      `cf_access_aud=c980da67?082b` (the existing "KheIsutra Studio (alpha)" CF Access app
already lists
82      `api.kheIsutra.guru` in `self_hosted_domains` ? no new app needed; same 4-email
allowlist policy),
83      `allowed_video_root=/tmp/apphome/output`.
84      - **Bug found + fixed to make M9 compatible with cloud-serving:** enabling edge-auth
requires
85      `allowed_video_root`, and its jail (`validate_input_path`?`resolve_under_root`)
`realpath()`-mangled
86      and **rejected gs:// inputs** ? so edge-auth would have 400'd the proven gs:// CUJ.
**PR #465** makes
87      the local-path jail skip storage URIs. Overlaid into the deployed image (build-time
asserted);
88      runtime-verified (Phase 1): gs:// ? 202 accepted; a local out-of-root path ? 400 (jail
still enforced).
89      - **Verified (Phase 2):** `/api/health` (exempt) ? 200; `/api/process` without a CF JWT
? **401**
90      ("missing Cf-Access-Jwt-Assertion header"). Service boots (no crash-loop from the
exposure checks).
91
92      **Remaining (owner, CF/GCP dashboard ? outward-facing production DNS):**
`api.kheIsutra.guru` is still a
93      stale CNAME to a dead CF tunnel. To make the alpha reachable-with-auth by allowlisted
testers: (a) add
94      `allUsers`?`roles/run.invoker` on the Cloud Run service (so CF's proxied requests reach
the app ? the
95      app still requires the CF JWT, so direct run.app access stays 401); (b) route
`api.kheIsutra.guru` to the
96      service (Cloud Run domain-mapping in asia-southeast1, or a CF origin-rule Host
override); (c) re-point the
97      CF DNS record (proxied) off the dead tunnel. Then the authed CUJ works end-to-end
through the CF login.
98      Full end-to-end authed CUJ verification needs (a)+(b) (a CF-issued JWT only arrives via
the CF edge).
99
100     Rollback (make it owner-usable again without CF): redeploy with `EDGE_AUTH` unset (edge-
auth off) via
101     `scratchpad/gen_service_yaml.py` ? `gcloud run services replace`.
102
103     ## 7. Cost ledger + teardown
104
105     - One L4 execution (~14 min, 8 vCPU + 1xL4) for the CUJ. CP service holds `min-instances
1` (small
106     always-on CPU cost). Owner directive: **keep running** (closed alpha).
107     - Teardown (if needed): `gcloud run services delete rally-control-plane`, `gcloud run
jobs delete

```

```
108 rally-worker`, both `--region asia-southeast1`; bucket/secret/repo retained.
109
```

28. user (2026-07-04T12:20:27.633Z)

```
105 updated baseline. The runner reports added/removed clips as **STALE** (exit 4) so a new
clip can't silently
106 pass un-baselined.
107
108 ## Observe ? where results go
109 - **Email** to `NIGHTLY_EMAIL_TO` every run (subject `[KhelSutra nightly]
PASS|REGRESSION|STALE ?`).
110 - **Reports in GCS**: `gs://khelsutra-rally-nightly/reports/<date>/<date>.{md,json}`
(kept 30 days).
111 - **Launcher logs**: `gcloud run jobs executions list --job=nightly-regression-launcher
--region=asia-south1` then `? executions logs read <exec> --region=asia-south1`.
112 - **The GPU run**: `gcloud compute instances get-serial-port-output <vm> --zone=<zone>`
(while it lives) or its Cloud Logging.
113 - **Exit codes**: `0` pass · `1` regression (reel-count changed / a clip errored /
quality dropped) · `2` operational · `3` no baseline · `4` stale (corpus/segmenter changed ? re-
freeze).
114
115 ## Pause / disable / resume / remove
116 ```bash
```

29. user (2026-07-04T12:20:28.400Z)

```
80 `--edge-auth off` is only for the pre-CF identity-token smoke/e2e window; redeploy with
`on`
81 before testers touch the service again. The flag-based equivalent (for reference ? a
manifest
82 round-trips the bash wrapper's metacharacters, flags don't on Windows):
83 ```bash
84 gcloud run deploy rally-control-plane --image <image> --region $REGION \
85 --no-cpu-throttling \          # CPU stays allocated between requests ? the drain thread
finishes
86 --min-instances 1 \          # one always-warm instance drains reliably (cheap CPU;
NOT the D7 GPU pool)
87 --cpu 2 --memory 4Gi --port 8080 --allow-unauthenticated \
88 --set-env-vars DEPLOYMENT_PROFILE=cloud-
serving,CLOUD_RUN_REGION=$REGION,GOOGLE_CLOUD_PROJECT=$PROJECT
89 ```
90 Without `--no-cpu-throttling`, a `/api/compile` returns 202 and then **stalls** ? the
reel never
91 finishes and startup crash-recovery would mark it terminal. `min-instances 1` does NOT
contradict
92 D7 (that was the ~$500/mo warm **GPU** pool); this is a ~1-vCPU CPU instance. Defense-
in-depth is in
93 the code: `Database.recover_stale_jobs()` **re-queues** a compile interrupted by a
restart (idempotent
94 stitch) rather than failing it, so even an unlucky scale-down mid-stitch self-heals on
the next drain.
95 Reserve `--allow-unauthenticated` for the pre-CF-auth smoke test only; lock ingress to
the CF proxy
96 (M9) before inviting testers. The image ships the `auth` extra (PyJWT[crypto]), so
flipping
97 `security.edge_auth_enabled` on needs no derived image ? the M9 Cf-Access JWT verify
works out of
98 the box.
99
100 ## 4. M7 acceptance (the owner runlog)
101 Capture a `DEPLOY_REPORT_cloud-serving_<date>.md` under
102 `docs/COMPUTE_DECOUPLED_SERVING/runlogs/` (mirrors the truly-local template): the exact
commands,
103 rally counts, `gpu_active_seconds`/`wall_seconds`/`bytes_out`, a reel smoke-test
```

```

(`ffmpeg -f null -`
104     exit 0 + a 5-frame contact sheet), gotchas, and the **$0 teardown audit**.
105
106     ## Gotchas
107     - The dispatched container must run inline ? `CloudRunCompute` forces
`compute_target=local_gpu`
108     + `indexing.detector_impl=native` in the per-execution args so the Job never re-
dispatches itself.
109     - Completion is a bucket marker (`<out_uri>/_dispatch/<token>.done`), not the Cloud
Run API status
110     ? so it reuses the storage layer + `execution_policy.poll_until` and survives a
control-plane restart.
111     - A failed Job (validation rc 2 / segmenter rc 3) now writes a failure marker
carrying the
112     real non-zero rc, so the dispatcher's poll terminates FAST + accurately (no 30-min
hang on a bad
113     video). Only a hung/crashed container (no marker at all) makes the bounded poll
time out
114     (`ExecutionPolicy.max_wait_sec`); the worker CLI maps that to a clean rc 4, and
the API path
115     surfaces it as a `cloud dispatch error` job failure (not a traceback).
116     - libx264 encode (parity, D16) + NVDEC decode-only; the image keeps ffmpeg's default
libx264.
117     - GOOGLE_CLOUD_PROJECT is required for the real client ?
`GoogleCloudRunJobClient.run_job`
118     fails fast with a clear `RuntimeError` if it is unset/empty (it must resolve a real
`job_path`;
119     an unset project would otherwise build a bogus `projects/None/...` name). Export it
(step 3) before
120     dispatching. The guard runs before the `google-cloud-run` import, so it is exercised
in CI even
121     though the SDK isn't a CI/test dependency.
122     - Hand-rolled `jobs execute --args` with `gs://` URIs: gcloud's default comma-split
plus its
123     alternate-delimiter syntax both bite ? repeated `--set` tokens trip the comma parser,
and a
124     `^: ^` delimiter splits the `:` inside `gs://`. Use a delimiter that appears nowhere in
the args,
125     e.g. `--args='^@@^-v@@infer@@--video-uri=gs://?'`. (The dispatcher path doesn't have
this
126     problem ? prefer it; this is for one-off diagnostics.) Container-arg overrides must
include the
127     dispatcher's two forced `--set`s: `deployment.compute_target=local_gpu`,
128     `indexing.detector_impl=native`.
129     - Git Bash (MSYS) on Windows mangles Unix-looking values before gcloud sees them:
130     `--add-volume-mount ?mount-path=/gcs` became `C:/Program Files/Git/gcs`, and
131     `--command=/bin/bash` became `C:/Program Files/Git/usr/bin/bash` (the container then
exec-fails).
132     Run gcloud commands with such flags from PowerShell (`MSYS_NO_PATHCONV=1` breaks
gcloud's own
133     wrapper). PowerShell has its own trap: quote any flag value containing commas.
134     - First execution after an image push spends minutes in "Container image import in
progress"
135     (visible in `executions describe status.conditions`, NOT in logs) ? that's progress,
not a hang;
136     later executions reuse the imported image.
137

```

30. user (2026-07-04T12:20:47.856Z)

```

1     # Cloud Run GPU serving ? build / deploy / run (COMPUTE_DECOUPLED_SERVING M6/M7)
2
3     The cloud-serving compute path: a control plane (`CloudRunCompute`,
`backend/compute/cloud_run.py`)
4     dispatches a Cloud Run Job execution running this image's worker on an L4, then

```

```

**bucket-polls**
5      a done-marker for completion (D16). Scale-to-zero (D7) ? no warm pool.
6
7      > **Status:** the dispatch shim + image are **landed + mock-tested** (M6). The **real
build + push +
8      > run on an L4** is the **M7** owner step, within the **$100/?10k** budget (D17) ?
confirm cmd + hourly
9      > cost first, **delete every resource after** ([[gcp-vm-always-delete]]). CI does
**not** build this image.
10
11     ## 0. Prerequisites (owner)
12     - A GCP project with billing on + the Cloud Run, Artifact Registry, and Cloud Build APIs
enabled.
13     - GPU quota for L4 (`GPU_ALL_REGIONS` ? 1) in the chosen region (asia-south1, else
Singapore ? see
14     `deploy/gcp/README.md`; L4 is often stocked-out, fall back per that runbook).
15     - The WASB weights staged in the bucket (WASB-C3 unresolved ? owner-gated):
`gs://<bucket>/models/wasb_badminton_best.pth.tar`.
16
17     ## 1. Build + push the image (Artifact Registry)
18     ```bash
19     PROJECT=<gcp-project>; REPO=rally
20     # L4 GPU is region-limited ? use asia-southeast1 (Singapore), NOT asia-south1 (Mumbai).
21     REGION=asia-southeast1
22     gcloud artifacts repositories create $REPO --repository-format=docker --location=$REGION
2>/dev/null || true
23     # `gcloud builds submit` has NO -f flag, so a non-root Dockerfile needs a build config:
24     gcloud builds submit --config=deploy/cloudrun/cloudbuild.yaml --project $PROJECT .
25     ```
26     *(A CUDA + micromamba image is large; the first build is ~8?10 min. Budget-watch per
D17.)*
27
28     ## 2. Create the Cloud Run **Job** (GPU, scale-to-zero)
29     The proven shape (first real M7 deploy, 2026-07-03): weights staged in a **same-region
bucket**
30     mounted read-only via GCS FUSE (no weights in the image ? WASB-C3; no init-fetch step
needed).
31     ```bash
32     gcloud run jobs create rally-worker \
33     --image $REGION-docker.pkg.dev/$PROJECT/$REPO/$IMG:latest \
34     --region $REGION --gpu 1 --gpu-type nvidia-l4 --no-gpu-zonal-redundancy \
35     --cpu 8 --memory 32Gi \
36     --max-retries 0 --task-timeout 3600 \
37     --set-env-vars WASB_WEIGHTS=/gcs/models/wasb_badminton_best.pth.tar \
38     --add-volume name=serving,type=cloud-storage,bucket=$SERVING_BUCKET \
39     --add-volume-mount volume=serving,mount-path=/gcs
40     ```
41     - **`--no-gpu-zonal-redundancy` is REQUIRED on the CLI**, not just a cost choice (it
also halves
42     the GPU rate ? right for batch jobs): without the flag `gcloud` asks an interactive
Y/n which,
43     in a non-interactive shell, **hangs forever with zero output and nothing created**. An
empty
44     `gcloud` that "did nothing" ? suspect a hidden prompt.
45     - 8 vCPU / 32 GiB: decode + x264 are CPU-side (NVDEC is a deferred D16 knob), so CPU
headroom is
46     what keeps the paid GPU busy. `/tmp` is **RAM-backed** ? the pulled input video lives
in memory,
47     so size `--memory` for the largest expected upload (a 5.8 GB GoPro file ? 6 GiB of
tmpfs).
48
49     ## 3. Wire the dispatcher (control plane)
50     The control plane runs with the **cloud-serving** profile (`DEPLOYMENT_PROFILE=cloud-
serving` ?
51     `compute_target=cloud_run`). `get_compute_backend(config)` then builds `CloudRunCompute`

```

```

from env:
52     ``bash
53     export DEPLOYMENT_PROFILE=cloud-serving
54     export CLOUD_RUN_JOB=rally-worker CLOUD_RUN_REGION=$REGION
GOOGLE_CLOUD_PROJECT=$PROJECT
55     # A `worker infer` (or the API job path) with a gs:// video now DISPATCHES to the Job
instead of

```

31. user (2026-07-04T12:20:48.496Z)

```

1     ""Generate the control-plane Cloud Run *service manifest* (COMPUTE_DECOUPLED_SERVING
M8/M9, #473).
2
3     The A1 alpha deploy drove `rally-control-plane` from a hand-generated YAML that lived in
a
4     session scratchpad ? this commits the generator so the deploy is reproducible from a
clean
5     clone. It emits a Knative service manifest for::
6
7         python deploy/cloudrun/gen_service_yaml.py --edge-auth off -o service.yaml
8         gcloud run services replace service.yaml --region asia-southeast1
9
10    Why a MANIFEST (`services replace`) and not `gcloud run deploy` flags: the container
command is
11    a /bin/bash wrapper carrying shell metacharacters (pipes, redirects, &&) ? on Windows
the gcloud
12    cmd-shim mangles those through `--command/--args`, while a YAML round-trips them
verbatim. Run
13    gcloud from PowerShell, not git-bash (MSYS rewrites `/tmp`-style paths).
14
15    The app is configured through a config.json baked INTO the manifest: the wrapper
base64-decodes
16    it to `/tmp/apphome/config.json` and exports `RALLY_APP_HOME=/tmp/apphome` before
exec'ing the
17    server, so one image serves every config shape (config.json wins over the cloud-serving
preset ?
18    backend/config/loader.py). `--edge-auth` is REQUIRED and explicit: `off` is only for the
19    identity-token smoke/e2e window; redeploy with `on` (the Cf-Access JWT verify, M9)
before the
20    service faces testers again.
21
22    Deliberate manifest choices (learned on the A1 deploy):
23    * `run.googleapis.com/cpu-throttling: "false"` + minScale 1 ? the in-process job
worker drains
24    on a background thread; without always-allocated CPU a 202'd compile stalls (README
$3a).
25    * maxScale DEFAULTS TO 1 ? the jobs/videos DB is per-instance ephemeral SQLite, so a
second
26    instance splits the queue and `/api/jobs/{id}` polls (issue #471, option 1). Raise
it only
27    once state is shared.
28    * `CLOUD_RUN_JOB` is NOT set ? it's a reserved runtime env name and the manifest is
rejected
29    with it; the code default ("rally-worker") is already correct.
30    * No secrets here: bucket/project/aud/team-domain are the same non-secret identifiers
the
31    committed runlogs carry; the Gemini key stays in Secret Manager (the alpha runs
32    `skip_ai_handoff=true` and never needs it).
33    ""
34
35    from __future__ import annotations
36
37    import argparse
38    import base64
39    import json

```

```

40     import sys
41
42     DEFAULT_PROJECT = "gen-lang-client-0306026826"
43     DEFAULT_REGION = "asia-southeast1"
44     DEFAULT_SERVICE = "rally-control-plane"
45     DEFAULT_SERVING_BUCKET = "gs://khehsutra-rally-serving"
46     DEFAULT_CF_TEAM_DOMAIN = "https://khehsutra.cloudflareaccess.com"
47     DEFAULT_CF_ACCESS_AUD =
"\"c980da67ed888fff935fcd913ae1eb4da63f9d65805d6e1c5ca76f92f517082b\""
48
49
50     def build_config(args: argparse.Namespace) -> dict:
51         """The app config the wrapper materializes at RALLY_APP_HOME/config.json."""
52         bucket = args.serving_bucket.rstrip("/")
53         return {
54             "deployment": {"profile": "cloud-serving", "compute_target": "cloud_run"},
55             # WASB-only alpha: candidate windows ARE the rallies; no Gemini key in this
service.
56             "indexing": {"skip_ai_handoff": True},
57             "storage": {
58                 "backend": "gcs",
59                 "output_root": bucket,
60                 "input_backend": "gcs",
61                 "input_root": f"{bucket}/videos",
62             },
63             "security": {
64                 "edge_auth_enabled": args.edge_auth == "on",
65                 "cf_team_domain": args.cf_team_domain,
66                 "cf_access_aud": args.cf_access_aud,
67                 "allowed_video_root": "/tmp/apphome/output",
68             },
69         }
70
71
72     def build_yaml(args: argparse.Namespace) -> str:
73         """Render the Knative service manifest for `gcloud run services replace`."""
74         cfg_b64 = base64.b64encode(
75             json.dumps(build_config(args), separators=(",", ":")).encode("utf-8")
76         ).decode("ascii")
77         # mkdir BEFORE the redirect (the redirect can't create the directory), then exec so
the
78         # server is PID 1 and Cloud Run's SIGTERM reaches it directly.
79         wrapper = (
80             "mkdir -p /tmp/apphome && "
81             f"printf %s '{cfg_b64}' | base64 -d > /tmp/apphome/config.json && "
82             "export RALLY_APP_HOME=/tmp/apphome && "
83             "exec python3 -m backend.main --server --host 0.0.0.0 --port 8080"
84         )
85         return f"""\
86 apiVersion: serving.knative.dev/v1
87 kind: Service
88 metadata:
89   name: {args.service}
90   labels:
91     cloud.googleapis.com/location: {args.region}
92 spec:
93   template:
94     metadata:
95       annotations:
96         autoscaling.knative.dev/minScale: "{args.min_instances}"
97         autoscaling.knative.dev/maxScale: "{args.max_instances}"
98         run.googleapis.com/cpu-throttling: "false"
99   spec:
100     timeoutSeconds: {args.timeout}
101     containers:

```



```

102         - image: {args.image}
103         command: ["/bin/bash"]
104         args: ["-c", "{wrapper}"]
105         ports:
106             - containerPort: 8080
107         resources:
108             limits:
109                 cpu: "{args.cpu}"
110                 memory: {args.memory}
111         env:
112             - name: DEPLOYMENT_PROFILE
113               value: cloud-serving
114             - name: CLOUD_RUN_REGION
115               value: {args.region}
116             - name: GOOGLE_CLOUD_PROJECT
117               value: {args.project}
118             - name: RALLY_ALLOWED_HOSTS
119               value: ""
120     """
121
122
123 def main(argv: "list[str] | None" = None) -> int:
124     p = argparse.ArgumentParser(description=__doc__.splitlines()[0])
125     p.add_argument(
126         "--edge-auth",
127         choices=("on", "off"),
128         required=True,
129         help="REQUIRED and explicit: 'off' only for the identity-token smoke/e2e window;
130
131         "redeploy 'on' (M9 Cf-Access verify) before testers touch it again.",
132     )
133     p.add_argument("--project", default=DEFAULT_PROJECT)
134     p.add_argument("--region", default=DEFAULT_REGION)
135     p.add_argument("--service", default=DEFAULT_SERVICE)
136     p.add_argument(
137         "--image",
138         default=None,
139         help="Container image; default = the project's rally-worker:latest (post-#470 it
140
141         "carries cloud-run + auth, so the worker image IS the control-plane image).",
142     )
143     p.add_argument("--serving-bucket", default=DEFAULT_SERVING_BUCKET)
144     p.add_argument("--cf-team-domain", default=DEFAULT_CF_TEAM_DOMAIN)
145     p.add_argument("--cf-access-aud", default=DEFAULT_CF_ACCESS_AUD)
146     p.add_argument("--min-instances", type=int, default=1)
147     p.add_argument(
148         "--max-instances",
149         type=int,
150         default=1,
151         help="Default 1: per-instance SQLite state splits across instances (issue
152 #471).",
153     )
154     p.add_argument("--cpu", type=int, default=2)
155     p.add_argument("--memory", default="4Gi")
156     p.add_argument("--timeout", type=int, default=300)
157     p.add_argument("-o", "--output", default="-", help="Output path, or '-' for
158 stdout.")
159     args = p.parse_args(argv)
160     if args.image is None:
161         args.image = f"{args.region}-docker.pkg.dev/{args.project}/rally/rally-
162 worker:latest"
163
164     text = build_yaml(args)
165     if args.output == "-":
166         sys.stdout.write(text)

```

```

162         else:
163             with open(args.output, "w", encoding="utf-8", newline="\n") as f:
164                 f.write(text)
165         return 0
166
167
168     if __name__ == "__main__":
169         raise SystemExit(main())
170

```

32. assistant (2026-07-04T12:22:08.201Z)

I have everything I need. Let me check `docs/README.md` for the doc index conventions, then write the review.

33. user (2026-07-04T12:22:09.611Z)

```

2-[Omitted long context line]
3-[Omitted long context line]
4:[Omitted long matching line]
5-- [EVALUATION.md](EVALUATION.md) ? How we measure and improve quality.
6-- [EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md) ? Design (P1 SHIPPED) for A/B + shadow + SxS
experimentation on rally-detection variants with zero production regression ? experiments
are decoupled from deploy (new cues default OFF, promote only on measured LOVO lift, rollback =
flip a flag). Composes the existing `backend/eval/` machinery (`calibration`, `regression`,
`metrics`); implemented in `backend/eval/experiment.py`.
--
12-
13-## ? Documentation status & governance
14:[Omitted long matching line]
15-- [DOC_STATUS.md](DOC_STATUS.md) ? living Documentation Status & Archival Register:
every foundational/tracked doc's lifecycle + archive-readiness, the archival due-diligence
checklist (honestly update ? then archive), and the open doc-hygiene findings. Review here;
update rows as docs complete.
16-
17-## Product / platform plans
18:[Omitted long matching line]
19:[Omitted long matching line]
20:[Omitted long matching line]
21-[Omitted long context line]
22-[Omitted long context line]

```

34. user (2026-07-04T12:22:19.107Z)

```

14     - [AUDIT_REPORT_khelsutra-guru_2026-07-02 (? COMPLETE /
ARCHIVED)](archives/past_projects/AUDIT_REPORT_khelsutra-
guru_2026-07-02-COMPLETED-2026-07-02.md) ? the five-repo `khelsutra-guru` workspace code audit +
its folded-in Remediation tracker (P0?P28 all closed). Archived 2026-07-02 once every
remediation wave landed and the last residuals were closed or filed as issues (#443 CORS, #444
corpus Phase-2, #445 GCS eval; + #301/#327/#332/#384). Absorbed the prior
`CODE_CLEANUP_AUDIT_2026-06-24` + June-2026 hardening sweep (both archived).
15     - [DOC_STATUS.md](DOC_STATUS.md) ? living Documentation Status & Archival
Register: every foundational/tracked doc's lifecycle + archive-readiness, the archival due-
diligence checklist (honestly update ? then archive), and the open doc-hygiene findings.
Review here; update rows as docs complete.
16
17     ## Product / platform plans
18     - [ALPHA_LAUNCH_READINESS.md](ALPHA_LAUNCH_READINESS.md) ? Active tracked project
('DRAFT', created 2026-07-03): cross-repo alpha readiness tracker after the golden corpus
reached 15 videos. Separates the hosted product-alpha lane (shareable upload?process?pick?reel)
from the owned-model/commercial rally-detection gate (ship target + WASB-C3 still unresolved).
$7 is the source of truth for the concrete next PR sequence: corpus portability, n=15 baseline
refresh, ablation/fusion reruns, hosted serving, alpha copy/launch review.
19     - [COMPUTE_DECOUPLED_SERVING/](COMPUTE_DECOUPLED_SERVING/README.md) ? Active tracked

```

project (2026-06-21, `DRAFT` owner-gated):** the productionization successor to DECOUPLED_COMPUTE (its deferred M3/M4). **One pure core (DETECT?WINDOW?STITCH) that never branches on deployment profile**, with a `ComputeBackend` (where it runs) + `StorageBackend` (what bytes) selected by **one startup config**. Ships **truly-local** (local/USB + local GPU, in-process stitch, zero cloud) + **cloud-serving** (upload<2GB / gdrive / bucket ? a **Cloud Run GPU job runs detect AND stitch**, metered into a per-job cost ledger) + **cpu-mock** (CI). Includes [TESTING_STRATEGY.md](COMPUTE_DECOUPLED_SERVING/TESTING_STRATEGY.md) (profile-PARITY proof for \$0), [architecture.svg](COMPUTE_DECOUPLED_SERVING/architecture.svg), a [runlogs/](COMPUTE_DECOUPLED_SERVING/runlogs/) posterity dir, and **ADR-014** ([lineage ADR-009?010?011?012?013?014](archives/decisions/DECISIONS.md)). \$7 M0?M10 tracker is the source of truth.

20 - [DECOUPLED_COMPUTE/](archives/past_projects/DECOUPLED_COMPUTE/README.md) ? ?
DELIVERED + ARCHIVED (2026-06-21). ** Lifted the rally pipeline onto a **rented cloud-native GPU doing bucket-driven `train` + `infer`: storage/compute seams (PRs #246?#259) + the **M3.5 native WASB runner** ; **M0?M2 + M3.5 proven on a GCP L4** (M2 ? `weights/0.1.0-l4/`). Delivered on **GCP** ? DigitalOcean dropped for compute ([ADR-013](archives/decisions/DECISIONS.md)). **M3/M4 (serving endpoint + per-video billing + scale-to-zero) deferred ? the "Cloud Run + GPU serving" subproject** ([ADR-012](archives/decisions/DECISIONS.md)). Snapshot under `docs/archives/past\_projects/DECOUPLED\_COMPUTE/` (incl. `DEPLOY\_REPORT\_GCP\_2026-06-20.md`).
21 - PACKAGING_PLAN.md ? **? Active tracked project (`IN PROGRESS`, 11 PRs merged 2026-06-21/22):** ship the engine as a **downloadable batteries-included Python app** ? `pip install` ? `indexer --app` boots a local webserver and opens the **bundled KhelSutra SPA** single-origins, **torch-free** (LIGHT tier), cross-platform, no native component (decision D10/O1). Built ON the decoupled-compute seam. **Supersedes the `DESKTOP\_APP\_PLAN.md` narrative.** \$7 tracker = source of truth (P0a?e ?, P1a/b ?, P2-launcher/P2a/P2d/P2e ?). **? Next:** P3 GPU validation + screencast ? P2b-install ? P2c wizard (cross-repo) ? P4 (ONNX/train/annotate = north star). Memory `[packaging-app-plan]` / `[next-session-packaging-app]`.

22 - DESKTOP_APP_PLAN.md ? **? Superseded narrative (by `PACKAGING\_PLAN.md`; D10 = self-hosted webserver, NOT a desktop app). ** Original scoping doc (2026-06-18, `[design]`): package the local pipeline as a **cross-platform desktop app** for non-technical players ? plug in the camera's USB/SD card, click once, get back **only the highlight reels**, fully on-device (no upload, no original-video bloat). The centerpiece is the **de-WSL native-compute port** (run WASB natively per-OS: Linux/Windows CUDA · macOS MPS/CPU ? drop WSL2), and its **make-or-break unknown**: is CPU acceptable on a typical *no-GPU* enthusiast laptop (WASB is ~3.4x real-time on a laptop 2070S)? \$7 tracker = **P0 compute-feasibility spike ? P1 native detector ? P2 app shell ? P3 packaging/installers**. Realizes the **L0 fully-local** tier of VIDEO_LOCALITY_MODEL.md; is the **`LocalDesktop` ComputeBackend** of PLATFORM_ARCHITECTURE.md §3.2; the smaller/export-clean owned model (OWNED_MODEL_IMPLEMENTATION_PLAN.md) is one answer to the compute crux.

23

24 ## Practical / ongoing

25 - CREATING_REELS.md ? **? End-user guide:** install the LIGHT-tier app ? `indexer --app` ? process a video ? pick rallies ? download a watermarked reel. Task-first usage walkthrough (web UI + the API chain), with a \$6 pointer to the cloud compute option. LIGHT tier only (offline `motion` / optional Gemini; no on-device GPU or one-click cloud). Install mechanics ? root [README.md](../README.md); packaging roadmap ? PACKAGING_PLAN.md.

26 - CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md ? **? Operator runbook:** `gs://` bucket videos ? highlight reels on the approved **L4 Cloud Run** (scale-to-zero, \$0 idle). Copy-pasteable: build image ? create the GPU Job ? cloud-serving control plane ? process(cloud)?query?compile?download ? prove the path (`verify\_compute\_path.py`) ? \$0 teardown. **Verified e2e 2026-06-23.** Raw deploy knobs ? ../deploy/cloudrun/README.md.

27 - SETUP_NEW_MACHINE.md ? **? The canonical "spin up a box ? ready for training & inference" runbook. ** Copy-pasteable, human-followable, **verified end-to-end on DigitalOcean (2026-06-20)** ? but **compute is now GCP-only (ADR-013)** , see [deploy/gcp/README.md](../deploy/gcp/README.md): credentials ? provision ? bootstrap ? suite (922 passed) ? secret-free CPU inference (no GPU/keys) ? real Gemini inference ? Gen-0 training, with pointers to the GPU/native-detector + bucket paths. The `deploy/digitalocean/\*` scripts are provider-agnostic and reused on GCP.

28 - DATA_IN_GCS.md ? **? The corpus source-of-truth map: where every piece of data lives in `gs://kheIsutra-rally-corpus`** (canonical layout · current contents +

drift · how `backend.worker` consumes it · the still-local-only **gap** · migration plan). Read this so a session pulls data from the bucket, not from someone's `C:\F:\Drive` ? the goal is to remove the local box as a blocker. Pairs with the two runbooks below.

```
29 -
[`deploy/gcp/nightly_regression/README.md`](../deploy/gcp/nightly_regression/README.md) ? **?
Nightly engine-regression OPERATIONS MANUAL (one-stop shop):** enable · run on demand · change
any setting (email recipients/frequency/clips/config) · observe · pause/disable/remove. The full
WASB-pipeline reel-count+cost+email nightly on an ephemeral GCP GPU VM (design/status:
[`NIGHTLY_ENGINE_REGRESSION.md`](NIGHTLY_ENGINE_REGRESSION.md)). The lighter CPU cached-
trajectory tripwire is `scripts/nightly_regression.py` (local Windows Scheduled Task, PR #202).
30 - [TRACKNET_WSL_SETUP.md](TRACKNET_WSL_SETUP.md)
31 - Evaluation harness and calibration drivers (in `backend/eval/`)
32
33 ## Data layer (collection · annotation · golden sets ? no ML)
34 - [data_pipeline/](data_pipeline/README.md) ? the isolated data pillar: how
footage is recorded (`VIDEO_RECORDING_GUIDELINES`), golden sets are created + which videos
(`GOLDEN_SET_IMPLEMENTATION` · `GOLDEN_SET_PHASE1_VIDEOS` · `GOLDEN_VIDEOS`), and data is
shared/ingested (`GOLDEN_DATA_SHARING`). Deliberately no segmentation/stitching/model detail
? that's the ML pillar (top level + `COMPUTE_DECOUPLED_SERVING/`). The annotation-vendor
(Roora)** engagement is archived under `archives/roora-vendor/`.
35
36 ## Conventions & templates
37 - [PROJECT_DOC_TEMPLATE.md](PROJECT_DOC_TEMPLATE.md) ? Reusable skeleton for a tracked
project doc**: status header · decisions-locked · progress tracker (??/?) · definition-of-done
· archived on completion. Standardizes the emergent pattern in `RALLY_QUALITY_RESEARCH.md` §7
and `GOLDEN_REGRESSION_FIXTURES.md`. Use for any substantial design/project work.
38
39 ## Historical documents
40 Completed plans, old research, ADRs, and checkpoints live in [archives/](archives/).
41
42 - June-2026 test-hardening sweep ? COMPLETED (2026-06-14; T1?T5 + full loop +
verification per #162) and archived; the former top-level pointer-stubs were removed
2026-07-02 (absorbed into the umbrella's Prior audits). Full records:
[archives/CODE_AUDIT_AND_TEST_HARDENING-
COMPLETED-2026-06-14.md](archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md),
[archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md](archives/DEFERRED_HARDENING_PLAN-
COMPLETED-2026-06-14.md), [archives/HARDENING_LOOP_HANDOFF-
COMPLETED-2026-06-14.md](archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md).
43 - CODE_CLEANUP_AUDIT (2026-06-24) ? the prior deep engine audit (35 findings; Phase
1 complete, Phase 2 largely shipped, remaining = the `main.py` god-module split). Absorbed
into the umbrella** (see its Prior audits (absorbed)) and archived ? [archives/past_projects/
CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-
02.md](archives/past_projects/CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md).
```

35. assistant (2026-07-04T12:24:26.359Z)

I have everything. Now I'll write the review doc.

36. assistant (2026-07-04T12:29:36.892Z)

Alpha Launch Review ? A13 (2026-07-04)

```
> Status: REPORT ? review DELIVERED 2026-07-04; GO/NO-GO awaiting owner** ·
Owner: @avidullu · Created: 2026-07-04
> Lifecycle: terminal record ? archive with the alpha-launch cluster once the owner records
GO/NO-GO on the tracker's Definition of done.
> Tracking anchors: `docs/ALPHA_LAUNCH_READINESS.md` §7 row A13; registered in
`docs/DOC_STATUS.md` §3c.
> Relation to existing docs: the A13 deliverable of
[ALPHA_LAUNCH_READINESS.md](ALPHA_LAUNCH_READINESS.md); builds on the two A1 runlogs
([a1-cuj](COMPUTE_DECOUPLED_SERVING/runlogs/DEPLOY_REPORT_cloud-serving_a1-cuj_2026-07-04.md),
[a1-e2e](COMPUTE_DECOUPLED_SERVING/runlogs/DEPLOY_REPORT_cloud-serving_a1-e2e_2026-07-04.md)).
> Honesty note: [verified] = checked against the repos or the live stack on 2026-07-04.
"live battery, 2026-07-04" = measured this session against the deployed service in a
```

controlled test window (`allUsers` invoker removed first ? public probes 403 during the window ? then edge-auth-off revisions; restored to **rev 00010**, digest `31475f27?`, edge-auth ON, `allUsers` re-added; restore posture re-verified: health(token)=200, health(public)=200 (deliberately exempt), process-sans-JWT=401, `api.khelsutra.guru`?302 CF login, `gemini\_key\_present=false`). `[analysis]` = derived, not observed. `[unverified]` = stated but not checked ? every one is tagged. "Session records" = non-repo operational notes from the deploy sessions. Issue states (#471/#472/#301/#332/#326/#328/#447/#443/#327/#175/#331/#349/#172 open; #465/#476/#470 merged) were re-confirmed via `gh` on 2026-07-04. This is not legal advice.

0. TL;DR + recommendation

The A1 stack is live and end-to-end green on the final image (48-segment CUJ, signed-URL download, edge auth restored ? e2e runlog §3/§6), and this session's live battery adds: the hosted-mode input jail holds at the API surface against foreign-bucket and separator-boundary attacks, upload/compile misuse fails with clean 4xx, a missing gs:// object fails cleanly and client-visibly, three concurrent cloud jobs all completed with no stranded state, and the domain-mapping cert is healthy today. Auth posture, rollback levers, and data-deletion mechanics are documented below. Training capture is hard-denied in code ? stronger than the privacy policy promises.

What is **not** done: nobody has ever completed the CUJ **in** a browser through Cloudflare Access¹ (both runlogs and the battery drove it with curl + tokens/JWTs ? the tracker's own DoD #1 is unchecked); the Studio SPA's first-run wizard tells testers **"your videos never leave this device"**, which is **false** on the hosted alpha, and the public privacy policy says footage is processed on **"our own infrastructure"** while it actually runs on Google Cloud in Singapore (both filed as **khelsutra #154**); A10 (ship-gate target) and A11 (WASB-C3 provenance) remain owner-gated, so **no** quality or owned-model claims are allowed in any copy²; the runtime service account is over-privileged for a public-invoker service; and retention/deletion is manual-only (#301, #447).

Recommendation (the decision is the owner's ? this is the evidence-supported shape):

- **Conditional GO** for the CURRENT 4-email allowlist³, with two conditions to clear **before** testers touch it⁴:

1. Fix the khelsutra copy blockers (**khelsutra #154**): suppress/fix the SetupWizard "never leaves this device" claim for the hosted build, correct the policy's "our own infrastructure" line, and add minimum upload-moment disclosure (§4.4 draft copy is ready to ship).

2. Owner completes the **5-minute browser-CUJ checklist** (§1.1) ? closing the tracker's DoD #1 with a real browser pass.

Rationale: the blast radius is 4 known, owner-invited people; the stack, auth, failure modes, and rollback are verified; the two conditions are the only tester-facing falsehoods/unknowns left. Recording GO also requires the owner to **explicitly defer A11** with the documented "no commercial model sharing" copy guardrail (`docs/ALPHA\_LAUNCH\_READINESS.md:110`) ? the current testers are already non-owner users, which sits in tension with the Dockerfile's "no public, non-owner serving" warning (`deploy/cloudrun/Dockerfile:14-16`); that tension must be owned, not implied.

- **NO-GO** for any wider invite wave⁵ until: the owner closes **A10** (ship-gate target + allowed claims) and **A11** (WASB-C3 resolve-or-defer), the runtime SA is replaced with a **least-privilege SA** (§5.5 ? today it holds project `roles/editor` on a service with an `allUsers` invoker), and the **retention/deletion over-promise** is addressed (automation via #301 or softened policy wording). The no-content-screening gap (#332) is acceptable at 4 trusted users and not beyond.

1. Scorecard ? the tracker's Definition of done, mapped to evidence

Against `docs/ALPHA\_LAUNCH\_READINESS.md` §10 (lines 224?235):

#	DoD item	Verdict	Evidence
1	Invited tester completes the hosted upload?reel CUJ without SSH	NOT DONE ? browser leg never run	Both runlogs drove the CUJ via curl with identity tokens/CF-JWT checks, not a browser (a1-cuj §3, a1-e2e §3); the live battery was also API-driven. Mitigating live fact: the

control plane serves a bundled SPA at its root (`GET /` ? 200 `text/html`, live battery) ? the tester path is **single-origin**, so the CF-Access CORS-preflight worry applies only to a split-origin deploy. Close it with §1.1 |

| 2 | Alpha service gated behind chosen auth posture + documented disable/rollback path | **DONE** (as of this review) | Edge auth ON verified live: 401 sans CF JWT, `api.khelsutra.guru` ? CF login 302 (e2e runlog §6), re-verified at rev 00010 after the battery window (restore posture, header above). Disable/rollback path: §5.6 of this doc (R1?R5) |

| 3 | n=15 corpus portable, no stale absolute paths | **DONE** | #452; tracker row A2 + A2 smoke command (`docs/ALPHA\_LAUNCH\_READINESS.md:101,123-130`) |

| 4 | Nightly baseline + heuristic floor refreshed with recorded commands | **DONE** | #453/#454; recorded outputs at `docs/ALPHA\_LAUNCH\_READINESS.md:132-168` |

| 5 | R2/tolF1 gate promoted or rejected with recorded ablation | **DONE** | #455/#458; A6/A8 records (`docs/ALPHA\_LAUNCH\_READINESS.md:170-177,200-206`) ? R2/tolF1 supported for the baseline-off? served contrast, strict split-count guard still failing |

| 6 | Fusion/person/Gate-A served-default decisions explicit | **DONE** | #458; A8 record (`docs/ALPHA\_LAUNCH\_READINESS.md:200-206`) ? person fusion not promoted, audio shadow-only, Gate A served default **rejected** |

| 7 | Ship-gate target + allowed alpha claims recorded | **NOT DONE** ? A10 GATED | Owner decision open (#172); Gen-0 `ship=False` (sign 9/15, p=0.3036, `docs/ALPHA\_LAUNCH\_READINESS.md:214`) |

| 8 | WASB C3 status recorded and reflected in alpha messaging | **NOT DONE** ? A11 GATED | Recorded (Dockerfile warning, tracker row), **not** reflected in messaging: the SPA wizard still sells a Gemini-key step and no copy carries the no-model-sharing guardrail (§4, khelsutra #154) |

| 9 | P10/P11 reopened or explicitly deferred | **NOT DONE** ? A12 GATED | No recorded owner deferral; recommend recording "deferred for alpha-plus" to close this box honestly (§6.3) |

| 10 | Owner records GO/NO-GO for the first alpha wave | **NOT DONE** | This review is the input; §0 is the recommendation |

1.1 The 5-minute owner checklist that closes DoD #1

From an allowlisted address, in a normal browser (no gcloud, no SSH):

1. Open `https://api.khelsutra.guru` ? complete the Cloudflare Access **email-OTP** login (no Cloudflare account needed). The Studio SPA should render from the service root (single-origin ? live battery confirmed `` serves HTML).
2. Upload a real clip that meets the floor: `.mp4`/`.mov`, **1280x720, 29.9 fps** (a 640x360 clip is rejected by `resolution\_fps\_check` ? e2e runlog §3.2).
3. Submit processing with **cloud compute** (the cost-acknowledgment checkbox is the only pre-upload gate today).
4. Watch the job status progress (`validating ? dispatching (cloud\_run) ? done`; expect ~14 min for a full match, small clips faster ? a1-cuj §3).
5. Select segments ? compile a reel.
6. Download the reel ? expect a 307 ? `storage.googleapis.com` signed URL; play the file.

If any step fails: capture job_id + timestamp + a screenshot, then triage per §3. Note: until khelsutra #154 lands, step 1?2 will show the first-run wizard's false "never leaves this device" copy ? run the checklist anyway, but do not invite testers before the copy fix.

2. Operator runbook ? live A1 alpha (Cloud Run asia-southeast1)

> Scope note: this is an **operations** runbook for the stack that is already live. It does not imply launch GO ? A10 (ship-gate target) and A11 (WASB-C3 provenance) are still **GATED** owner rows (`docs/ALPHA\_LAUNCH\_READINESS.md:109-110`).

Live topology (e2e runlog §1/§6; revision advanced to **rev 00010** by this session's live battery, same digest, posture re-verified): project `gen-lang-client-0306026826`, region `asia-southeast1`; control-plane **Service** `rally-control-plane` at rev 00010 (edge-auth ON, `maxScale=1`, image digest `31475f27` = master `3c97117`); L4 GPU **Job** `rally-worker` (nvidia-l4, 8 vCPU/32Gi, GCS-FUSE `khelsutra-rally-serving`/?/gcs`); storage `gs://khelsutra-rally-serving`; WASB-only (`indexing.skip_ai_handoff=true` ? no Gemini key in this service, `deploy/cloudrun/gen_service.yaml.py:55-56`). `api.khelsutra.guru` fronts the service via a Cloud Run domain mapping behind a Cloudflare Access app (4-email allowlist). The control plane also serves the bundled SPA at `` (live battery) ? the tester path is single-origin.

2.1 Deploy / redeploy the control plane

Canonical path is the **committed manifest generator** (``deploy/cloudrun/README.md:72-79``; generator defaults `bake project/region/service/bucket/aud` at ``deploy/cloudrun/gen_service_yaml.py:42-47``):

```
```powershell
python deploy/cloudrun/gen_service_yaml.py --edge-auth on -o service.yaml
gcloud run services replace service.yaml --region asia-southeast1
```
```

- ``--edge-auth`` is **required and explicit**; ``off`` is only for smoke/e2e windows (§2.4) (``gen_service_yaml.py:124-131``).
- Run `gcloud` from **PowerShell**, not `git-bash` ? MSYS mangles ``/tmp``- and ``/gcs``-style values before `gcloud` sees them (``deploy/cloudrun/README.md:129-133``; ``gen_service_yaml.py:12-13``).
- Do NOT add ``CLOUD_RUN_JOB`` to the manifest ? it's a reserved runtime env name and the manifest is rejected; the code default ``rally-worker`` is already correct (``gen_service_yaml.py:28-29``).

Gotcha A ? ``services replace`` with an unchanged manifest is a NO-OP: it does not re-resolve ``:latest`` after a rebuild. The no-op and its remedy are recorded as an observation in the e2e runlog §5 (``?a1-e2e_2026-07-04.md:83-85``): force a new revision with ``gcloud run services update rally-control-plane --image asia-southeast1-docker.pkg.dev/gen-lang-client-0306026826/rally/rally-worker:latest --region asia-southeast1``, or stamp a revision suffix (a generator ``--revision-suffix`` flag is a filed follow-up candidate). ``[analysis]`` The exact flag expansion above follows session records, not a committed command transcript.

Gotcha B ? a redeploy WIPES per-instance state (issue #471, open): the jobs/videos DB is ephemeral SQLite in the instance's ``/tmp``, so any redeploy invalidates in-flight jobs, job history, and past ``/api/highlights`` download URLs ? the handler 404s on the DB record check (``backend/main.py:1052`` per the e2e runlog) even though the reel mirror still exists in `gs://`; re-processing regenerates state (e2e runlog §5). Corollary: **keep ``maxScale=1`` ?** >1 instance splits the queue and ``/api/jobs/{id}`` polls (#471; encoded as the generator default, ``gen_service_yaml.py:145-150``). Operationally: redeploy only when no tester job is in flight.

2.2 Rebuild the image + re-pin the L4 job

One image serves both the worker Job and the control plane (post-#470 it carries the ``cloud-run`` + ``auth`` extras; ``gen_service_yaml.py:136-140``).

```
```powershell
```

### 1. Rebuild (~8-10 min heavy CUDA build; ``gcloud builds submit`` has NO -f flag, hence

```
gcloud builds submit --config=deploy/cloudrun/cloudbuild.yaml --project gen-lang-client-0306026826 .
```
```

```
(`deploy/cloudrun/cloudbuild.yaml:3-6,26`; `deploy/cloudrun/README.md:23-26`)
```

```
```powershell
```

### 2. Re-pin the Job to the new digest

```
gcloud run jobs update rally-worker --image asia-southeast1-docker.pkg.dev/gen-lang-client-0306026826/rally/rally-worker:latest --region asia-southeast1
```

### 3. Re-point the service (Gotcha A: replace won't re-resolve :latest)

```
gcloud run services update rally-control-plane --image asia-southeast1-docker.pkg.dev/gen-lang-client-0306026826/rally/rally-worker:latest --region asia-southeast1
```
```

The re-pin step is what the e2e deploy did after each build ("the Job re-pinned via ``gcloud run jobs update``", ``?a1-e2e_2026-07-04.md:18``); the full flag string follows session records [unverified against a committed doc]. The same ``jobs update`` mechanism (with ``--set-secrets``/``--remove-secrets``) was exercised live in this session's battery (§5.4).

- **First execution after a push** sits minutes in "Container image import in progress" ? visible in ``gcloud run jobs executions describe <exec> --region asia-southeast1`` ``status.conditions``, NOT in logs; it's progress, not a hang (``deploy/cloudrun/README.md:134-137``).

2.3 Add / remove an alpha tester (Cloudflare Access)

Tester auth is entirely CF-side ? ****no engine redeploy needed**** (the app's aud/team-domain stay constant in the manifest; `gen\_service\_yaml.py:46-47`).

- Where: Cloudflare Zero Trust dashboard, team `khelsutra.cloudflareaccess.com` ? Access ? Applications ? *****KhelSutra Studio (alpha)***** (one self-hosted app covering both `app.khelsutra.guru` and `api.khelsutra.guru`, aud `c980da67?082b`) ? edit the Allow policy's ****Emails**** rule (session records; app shape + email-OTP policy per `khelsutra/deploy/DEPLOY\_ALPHA.md:111-115`; the a1-cuj runlog confirms the same app/allowlist fronts the Cloud Run service, `?a1-cuj\_2026-07-04.md:80-83`).
- Current allowlist is ****4 addresses**** (owner-held list; count per the a1-cuj runlog). Free tier covers ?50 users; one-time-PIN means testers need no Cloudflare account (`khelsutra/deploy/DEPLOY\_ALPHA.md:113`).
- API alternative: an ****owner-held scoped Cloudflare API token**** with Access-Apps edit exists for scripted allowlist edits (session records).
- Remove = delete the email from the same policy. [unverified] whether an already-issued CF session survives until expiry after removal ? also revoke the user's active Access session (\$5.6 R3).
- Triage quirk to know: testers' uploads below ****1280x720 are rejected**** by validation (`resolution\_fps\_check`) ? a real e2e first-attempt failure, not an auth problem (`?a1-e2e\_2026-07-04.md:41-42`).

2.4 Edge-auth OFF/ON windows

****OFF is only for identity-token smoke/e2e tests**** (`deploy/cloudrun/README.md:80-81`; `gen\_service\_yaml.py:18-20,124-131`). Why it's dangerous: the service carries `allUsers` run.invoker so Cloudflare's proxy can reach it ? with edge-auth ON the app-level Cf-Access JWT check is what keeps the direct `\*.run.app` URL at 401 (`?a1-cuj\_2026-07-04.md:94-96`); with edge-auth OFF that last gate is gone.

****Improved practice (used in this session's live battery ? recommended as the standard):**** bracket every auth-off window by ****removing the `allUsers` invoker first**** and re-adding it on restore. During the window, public probes get ****403 at the Google Front End**** (zero public exposure ? measured) while identity-token calls still work. Cost: `api.khelsutra.guru` is ****down for testers for the duration**** (CF-proxied traffic rides that binding), so keep windows short and announced.

```
```powershell
```

### 0. close the public door for the window

```
gcloud run services remove-iam-policy-binding rally-control-plane --region asia-southeast1 --member=allUsers --role=roles/run.invoker
```

### 1. open the smoke window

```
python deploy/cloudrun/gen_service_yaml.py --edge-auth off -o service.yaml
gcloud run services replace service.yaml --region asia-southeast1
```

### 2. drive it with an identity token during the window

```
curl -H "Authorization: Bearer $(gcloud auth print-identity-token)" https://rally-control-plane-2bosgeukpa-as.a.run.app/api/health
```

### 3. restore: edge-auth on + re-add the invoker

```
python deploy/cloudrun/gen_service_yaml.py --edge-auth on -o service.yaml
gcloud run services replace service.yaml --region asia-southeast1
gcloud run services add-iam-policy-binding rally-control-plane --region asia-southeast1 --member=allUsers --role=roles/run.invoker
```
```

(base commands: `deploy/cloudrun/README.md:77-79`; invoker bracket: live battery, 2026-07-04. The replace here is NOT a no-op because the edge-auth flip changes the baked config. Remember Gotcha B: each flip wipes SQLite state.)

****Posture verification after restore**** (exactly what the e2e verified at rev 00007 and the battery re-verified at rev 00010):

1. `GET /api/health` on the run.app URL ? ****200**** (deliberately exempt; the no-auth liveness probe, `backend/main.py:321-325`).

2. `POST /api/process` with valid JSON but no CF JWT ? ****401**** `edge auth: missing Cf-Access-Jwt-Assertion header`.
3. `curl -i https://api.khelsutra.guru/api/health` with no CF session ? ****302**** to `https://khelsutra.cloudflareaccess.com/cdn-cgi/access/login/api.khelsutra.guru??`.
4. `GET /api/capabilities` ? `gemini\_key\_present=false` (the WASB-only posture).
5. Triage quirk: a ****malformed-JSON**** body gets FastAPI's 422 ***before*** the edge-auth dependency ? a 422 is not an auth bypass (`a1-e2e\_2026-07-04.md:91-93`).

2.5 Health / capabilities checks and logs

- `GET /api/health` ? `200 {"status":"ok",?}`; `GET /api/capabilities` ? `wasb\_hybrid` present, `gemini\_key\_present=false` (correct for the WASB-only alpha), ffmpeg/ffprobe present (`a1-e2e\_2026-07-04.md:25-27`). Both endpoints are app-level no-auth probes (`backend/main.py:321-325,359-363`) but sit behind the CF login on `api.khelsutra.guru`.
- ****Job (L4) logs****: `gcloud run jobs executions list --job=rally-worker --region=asia-southeast1`, then `gcloud run jobs executions logs read <execution> --region=asia-southeast1` ? adapted from the committed nightly-job pattern (`deploy/gcp/nightly\_regression/README.md:111`, which targets `nightly-regression-launcher` in asia-south1; the adaptation to `rally-worker/asia-southeast1` is ours). Worker execution logs were read live during this session's battery (they carried the per-window Gemini traffic in §5.4), so the log path works; the exact CLI invocation used is not transcribed [unverified].
- ****Cloud Logging filters**** [unverified ? no committed filter string exists; standard resource labels]:
 - service: `gcloud logging read "resource.type=cloud\_run\_revision AND resource.labels.service\_name=rally-control-plane" --project gen-lang-client-0306026826 --limit 50`
 - job: `gcloud logging read "resource.type=cloud\_run\_job AND resource.labels.job\_name=rally-worker" --project gen-lang-client-0306026826 --limit 50`
- Polling `gcloud logging read` ~every 60s and grepping `worker|wasb|rally|ERROR|Traceback|CUDA|OOM|Killed` is the established practice from the live runs (session records).
- ****Standing owner rule**** (session records): any GPU execution expected to exceed ~20 min must be actively tailed with a ~5-min heartbeat, alerting on stall ? not just awaited.
- Dispatch completion is a ****bucket marker**** (`<out\_uri>/\_dispatch/<token>.done`), not Cloud Run API status; a failed Job writes a failure marker with the real rc so polls terminate fast (`deploy/cloudrun/README.md:109-115`).

2.6 Cost ? what the standing stack costs

- ****What runs idle:**** the control plane holds `min-instances 1` + `cpu-throttling: false` (`gen\_service\_yaml.py:96-98`) at ****2 vCPU / 4Gi**** (`gen\_service\_yaml.py:151-152`) ? one always-warm CPU instance. (NB `deploy/cloudrun/README.md:92` says "~1-vCPU CPU instance" ? that note predates the committed manifest; the deployed value is 2 vCPU. Both agree it is NOT the D7 warm ****GPU**** pool.)
- ****Per run:**** the L4 Job is scale-to-zero and bills only per execution (`deploy/cloudrun/README.md:5`). Cost precedent: M1/M2/M3.5 ran on an L4 for ****~\$12/run**** (`docs/archives/decisions/DECISIONS.md:675`); one full CUJ ? 14 min of L4 + 8 vCPU (a1-cuj \$7). At 4 testers processing a handful of matches each, per-run GPU spend is dollars, not tens of dollars `[analysis]`.
- ****Exact monthly total**** for the always-warm CP + tester runs: ****[unverified ? owner billing-console check]****. Do this check before the G0 decision; it is a single console visit.
- ****Budget guardrail:**** a ****\$100 budget alert**** was provisioned (`deploy/gcp/README.md:18`; D17 per `deploy/cloudrun/README.md:8-9`). [unverified] whether the alert threshold was raised after the owner lifted the pre-A1 spend gate to \$300 (session records) ? align the alert with the real gate during the same console visit.
- ****Teardown**** (if the alpha pauses): `gcloud run services delete rally-control-plane --region asia-southeast1` and `gcloud run jobs delete rally-worker --region asia-southeast1`; the bucket/secret/Artifact-Registry repo are retained (a1-cuj \$7). Owner directive as of 2026-07-04: keep the stack running.
- ****GCP VM always-DELETE rule**** (applies to any GPU-VM diagnostics, not the Cloud Run stack): a STOPPED VM still bills its disk; `deploy/gcp/teardown.sh` defaults to `delete` (`deploy/gcp/README.md:248-257`, `deploy/gcp/teardown.sh:9-12`); also audit `gcloud compute disks list` for orphans.

2.7 Watch items ? owner + mechanism

Documented commands are not reminders. Each watch item needs a dated mechanism, or it will be missed across parallel sessions:

```
Watch item	Owner	Recommended mechanism	Check
**#472 cert renewal** on `api.khelsutra.guru` (Google-managed cert renews ~90 days; validation can stall behind the proxied CF CNAME)	@avidullu	A dated GitHub issue comment/due-date or calendar entry for **~2026-09-30**	`gcloud beta run domain-mappings describe --domain api.khelsutra.guru --region asia-southeast1` ? cert status not stalled. **Healthy today** (live battery, 2026-07-04: `Ready;CertificateProvisioned;DomainRoutable = True;True;True`). If it stalls: temporarily grey-cloud (DNS-only) the CNAME so validation passes, then re-proxy; or switch to a Cloudflare Origin CA cert (issue #472 body). CF Access is unaffected either way
**Billing / budget-alert alignment**	@avidullu	Monthly calendar entry; first check **before the G0 decision**	Billing console: actual monthly spend for the standing stack; confirm the budget-alert threshold matches the real ($300) spend gate ($2.6, both [unverified])
```

3. Observability & job-failure triage

Scope: the live A1 stack ? Cloud Run service `rally-control-plane` + L4 Job `rally-worker`, bucket `gs://khelsutra-rally-serving`. All engine paths below are relative to `badminton-highlight-indexer`.

3.0 Live failure-mode battery (measured 2026-07-04)

Every probe below was executed against the deployed service (identity token, rev 00008 of the battery window) ? these are observed responses, not code-derived predictions:

```
Probe	Result	
`process` `gs://evil-bucket/x.mp4` (foreign bucket)	**400** `hosted mode: '?' is not under the allowed input root 'gs://khelsutra-rally-serving/videos'` ? the #465 jail holds at the API surface	
`process` `gs://khelsutra-rally-serving/videos-evil/x.mp4` (separator-boundary attack)	**400** same jail message ? the prefix-boundary check holds	
`process` a nonexistent object under the input root	**202** job queued ? job `failed`, error `NotFound: 404 ? No such object: ?/videos/definitely-not-here.mp4` ? clean **async** failure, client-visible via the job poll	
`process` with a bad segmenter	**400** `Segmenter 'not_a_segmenter' not found. Available segmenters: [?]	`
`GET /api/jobs/deadbeef`	**404**	
`compile` with a bad video_id	**404** `Indexed video record not found.`	
`GET /api/highlights/x/.%2F.%2Fconfig.json` (traversal)	**404** ? rejected, no file disclosure	
`GET /api/highlights/<missing>/reel.mp4`	**404**	
upload init `filename=malware.exe`	**400** `Extension '.exe' not allowed. Allowed: ['mov', 'mp4']`	
upload init `total_size` 9999999999	**413** `File exceeds the 4096 MB upload limit.`	
upload PUT `part/5` first (out-of-order)	**409** `Out-of-order part: expected 0, got 5`; DELETE cleanup ? 200	
```

****Concurrency smoke (#468, live):**** three distinct 720p clips (4s/5s/6s) uploaded and submitted `compute=cloud` back-to-back ? ****all three jobs claimed and completed `success`**** (0 segments each ? correct for synthetic content). No false-empty result, no stranded job; the project GPU quota of 1 serialized the L4 executions, but the drain claimed everything.

3.1 Reading a job's state

`GET /api/jobs/{job\_id}` returns `{job\_id, status, progress, video\_id, error, result, reports, created\_at, started\_at, finished\_at}` (`backend/main.py:1008-1027`). Two failure channels, deliberately distinct (`backend/api/job\_worker.py:148-153`):
- `status='done'` + `result.status='failed'` ? the ****pipeline's verdict**** (validation, staging,

```
dispatch, cloud rc?0, stitch?) with a human `result.message`;  
- `status='failed'` + `error="<ExcType>: <msg>"` ? the **worker itself crashed**  
(`backend/api/job_worker.py:191-193`).
```

The `progress` field is the first triage signal ? the worker stamps every stage: `starting` ? hashing ? validating ? staging upload to bucket ? dispatching (cloud_run) ? ingesting (cloud_run) / segmenting (<name>) ? finalizing; compile: `resolving` ? stitching ? uploading reel ? finished` (`backend/infrastructure/database_jobs.py:105-118`;
`backend/orchestration.py:287,324,363,447,516,559,608,793,816,851,861`). A stuck job's last `progress` value tells you which stage it died in.

3.2 Triage table

"CP logs" = `rally-control-plane` service logs (Cloud Logging, `cloud\_run\_revision`); "worker logs" = `rally-worker` Job execution logs (`cloud\_run\_job`). The whole backend logs via stdout `logging` only (`backend/main.py:26-30`) ? there is no metrics/tracing layer. Rows marked ****(battery ?)**** were re-verified live in §3.0.

| Failure class | CLIENT sees | OPERATOR sees (log pattern ? where) | First debugging step |
|--|--|--|----------------------|
| Edge-auth reject (M9) | 401 {"detail":"edge auth: missing Cf-Access-Jwt-Assertion header ?"} | (or verification-failure detail) on process/compile/annotations/assets
(`backend/main.py:910-913,1107-1110`; `backend/api/auth.py:134-147`) Nothing app-level ? only the request-log 401 on CP. NB: malformed JSON gets FastAPI's **422 before** edge-auth runs (e2e runlog §5) Confirm the caller came through `api.khelsutra.guru` (CF Access) not the raw run.app URL; check the 4-email allowlist in CF | |
| Submit-time input validation **(battery ?)** | 400 unsupported scheme / null byte
(`backend/main.py:923-933`); 404 "Video file not found at '?'" (`main.py:940-943`); 400 "Segmenter '?' not found. Available: [?]" (`main.py:953-958`); upload-init 400 bad filename/extension (`backend/api/uploads.py:91-99`) No app log ? HTTPException raised before any logging Reproduce the request; the `detail` string is self-describing | **(battery ?)** 400 {"detail":"hosted mode: 'gs://?'"} is not under the allowed input root 'gs://?'" ? a URI outside `storage.input_root`, or any URI when no uri-root is configured (`backend/utils/paths.py:97-103`, wired at `main.py:915-933`; PR #465 merged) No app log (pre-work 400) Compare the submitted URI against `storage.input_root` in the deployed config (`GET /api/config` is redacted but shows storage roots) | |
| Missing gs:// object under the jail root **(battery ?)** | job `done` with `result.status='failed'` or `status='failed'` with `error` `NotFound: 404` ? No such object: ` ? a clean async failure surfaced through the job poll (live battery)   CP: the ingest/staging traceback for the NotFound   `gsutil ls` the submitted URI; typo vs deleted object | In-worker video validation failure (e.g. resolution floor ?1280x720) job `done`, `result.status='failed'`, `message="Video failed validation: resolution_fps_check: ?" + per-validator `results` (`backend/orchestration.py:68-76,520-536`)   CP: `Running validations for <path>...` (`orchestration.py:517`) then the terminal `[COMPUTE] video_id=? status=failed` line (`orchestration.py:632-640`)   Read `result.results[].details` (ffprobe width/height/fps are stored); the 640x360 e2e reject is a known floor (e2e runlog §3.2) | |
| Upload sequencing / size / disk **(battery ? : 409/413)** | 404 unknown upload_id
(`uploads.py:134`); 409 "Out-of-order part: expected N, got M" (`uploads.py:136-139`); 409 "Expected N part(s), received M" on complete (`uploads.py:181-185`); 413 per-part/total overflow ? **upload aborted, re-init** (`uploads.py:157-161`) or declared-size
(`uploads.py:100-104`); 507 "Not enough free disk space for upload?" (`uploads.py:109`;
`backend/utils/freespace.py:77-83`) Only the 507 logs: `free-space guard (upload): ?`
(`freespace.py:70-76`). 409/413 log **nothing** app-side. Upload state is in-process ? an instance restart silently orphans it ? next part 404s (`uploads.py:6-8`) For 409 after a gap: assume instance recycle ? client re-inits. For 507: check CP instance disk | Staging-to-bucket failure (#461 path) job `done`, `result.status='failed'`, `message="cloud-serving: failed to stage the uploaded file to <input_root>: <e>"` ; `compute.path='not_run'`, `dispatched=false` (`orchestration.py:296-307,198-221`) CP: `[API] cloud-serving: failed to stage <src> -> <staged_uri>: <e>` with traceback (`orchestration.py:297-303`) Check the CP runtime SA's write access to `storage.input_root` and that the bucket exists; retry the job | |
| Cloud dispatch failure (SDK missing / project unset / Run API error) | job `done`, `result.status='failed'`, `message="cloud dispatch error: <e>"` (`orchestration.py:348-350`). | Distinct pre-flight RuntimeErrors: `GOOGLE_CLOUD_PROJECT` is unset? and `google-cloud-run` is required? (`backend/compute/cloud_run.py:93-106`) CP: `[API] cloud dispatch error: <e>` with | |

```

traceback (`orchestration.py:349`) | If pre-flight: fix CP env or install the SDK. Else check
IAM (run.jobs.run) + `gcloud run jobs executions list --job rally-worker` |
| L4 execution failure (worker rc?0 / garbled marker) | job `done`, `result.status='failed'`,
`message="cloud job failed (returncode N)." (`orchestration.py:351-352`). A missing/garbled
marker defaults to rc=1, never success (`cloud_run.py:216-226`) | CP: `cloud-run: dispatched
job=rally-worker exec=<execution>; bucket-polling <done_uri>` then `cloud-run: job=? done
(video_id=? rc=N)` (`cloud_run.py:200-205,229-231`). **Worker logs** carry the real failure; the
worker writes the failure marker itself so the poll fails fast
(`backend/worker/___main___py:236-241`) | `gcloud run jobs executions describe <execution>` then
read that execution's logs; inspect `gs://khelsutra-rally-serving/outputs/<video_id>/` and the
`_dispatch/<token>.done` marker JSON |
| Bucket-poll timeout (default budget 1800s @10s cadence ? DEFAULT_POLICY, hard-wired:
`backend/infrastructure/execution_policy.py:49-54`; `backend/compute/___init___py:39,76`) | job
`done`, `result.status='failed'`, `message="cloud dispatch error: cloud-run-infer not ready
within max_wait_sec=1800.0s [cloud-run job=rally-worker execution=<name>; cancel requested ?
verify the execution state in the GCP console]"` (`cloud_run.py:281-285` ?
`orchestration.py:348-350`) ? the execution name IS in the client-visible message | CP: `cloud-
run-infer: still not ready after Ns (max_wait) ? giving up` (`execution_policy.py:184-189`);
then either `done-marker found on the final post-timeout check ? rescuing`
(`cloud_run.py:261-267`) or `requested cancel of execution=?` / `best-effort cancel ? failed`
(`cloud_run.py:270-280`) | Describe the execution: if it completed just after cancel, the GPU
run is lost ? re-submit; if it hung, read worker logs for the stall stage. Confirm the cancel
landed (otherwise L4 bills to `--task-timeout 3600s`) |
| Result-ingest failure (missing/corrupt `intervals.csv`) | job `done`,
`result.status='failed'`, `message="cloud-serving: could not read the result from <outputs_uri>:
<e>"; partial rows pruned, prior segments preserved (`orchestration.py:369-376,107-117`) | CP:
`[API] cloud ingest failed: <e>` with traceback (`orchestration.py:372`) | `gsutil ls
gs:///outputs/<video_id>/` ? if `intervals.csv` is absent the worker died between marker and
artifact push: read worker logs |
| Compile / ffmpeg failure ** (battery ? : submit-time 404) ** | Submit-time: `404 "Indexed video
record not found."` / `400` bad segments/filename/min_shot_count floor
(`backend/main.py:1111-1120`; `orchestration.py:699-745`). Run-time: job `done`,
`result.status='failed'`, `message="Highlight stitching failed: <e>"`
(`orchestration.py:829-837`) | CP: `[compile] stitching failed for video_id=? : <e>` with
traceback (`orchestration.py:830-832`). Mirror-upload failure is separate and non-fatal:
`[compile] could not mirror reel to <uri>: <e>` (`orchestration.py:856`) | Check ffmpeg presence
via `GET /api/capabilities` (`main.py:384-389`); a mirror-only failure means the download will
hit the 32 MiB local-fallback trap below |
| Signed-URL failure (#475?#476) | On success: `307` to a v4 signed URL. On failure: silent
fallback to serving the local file ? for a >32 MiB reel that resurfaces the Cloud Run response-
cap edge-500 (e2e runlog $4) | CP, post-#476 (merged): `[gcs] keyless signBlob path failed
(<e>); falling back to key-based signing` or `[gcs] keyless signing skipped: runtime SA email
unresolved` (`backend/storage/gcs.py:104-113`), then `[highlights] signed-url path failed for
<video_id>; serving local: <e>` (`backend/main.py:1062-1063`) | Grep CP logs for `[gcs]`; verify
the SA still holds self-`roles/iam.serviceAccountTokenCreator` and
`iamcredentials.googleapis.com` is enabled (the #475 root-cause pair, e2e runlog $4) |
| Job stuck `queued`/`waiting` | `GET /api/jobs/{id}` stays `queued` (progress null) or
`waiting` indefinitely; no timeout ever flips it | Usually **no log at all** ? known risk: a
hung job on the drain blocks everything behind it; there is no per-job wall-clock timeout
(`job_worker.py:40-48`). Parked jobs do log `Job <id> parked for Ns (resume when due).`
(`job_worker.py:189`); a broken claim path logs `Failed to claim a due job: <e>`
(`database_jobs.py:394-396`) | Check the head job's `progress`/`started_at` ? a hung GPU/poll
call in the head job explains everything queued behind it. A restart converts the backlog to a
clean state (next row) |
| Server restart / instance recycle mid-job | **Self-hosted (persistent DB):** process job ?
`status='failed'`, `error="interrupted by server restart"`; compile job ? silently re-queued and
finishes (`database_jobs.py:420-450`). **Cloud Run:** SQLite is per-instance ephemeral ? the new
instance has a fresh DB, so `GET /api/jobs/{id}` ? `404 "Unknown job_id"` and past download URLs
404 on the DB check (`main.py:1012-1014,1052`; issue #471 OPEN; e2e runlog $5) | Startup log:
`[JOBS] Recovered stale jobs from a previous run ? N compile re-queued, M process failed.`
(`main.py:1475-1480`) ? on Cloud Run this line reports 0/0 because the DB is fresh `[analysis]`
| Correlate the client's job_id timestamp with CP revision/instance-start events in Cloud
Logging; the client must re-submit (re-process regenerates state ? outputs and reels in gs://
survive) |
| Any other worker crash | `status='failed'`, `error="<ExcType>: <msg>"` | CP: `Job <job_id>

```

crashed: <e>` + full traceback (`job_worker.py:191-193`) ? the one log line guaranteed to carry the job\_id | Grep CP logs for `Job <job_id>` |

3.3 Correlating a client `job\_id` to a Cloud Run execution

There is no single trace id ? the chain is: **job_id ? video_id ? execution name**, stitched from the job row plus two log lines:

1. `job\_id ? video\_id`: the worker stamps `video\_id` onto the job row as soon as the result carries one (`job\_worker.py:177-178`); it's in every `GET /api/jobs/{id}` response.
2. **Dispatch token**: each dispatch mints `done\_uri` = `<output_root>/_dispatch/<uuid4-token>.done`` (`cloud\_run.py:179-182`) and logs `cloud-run: dispatched job=<job> exec=<execution>; bucket-polling <done\_uri>` (`cloud\_run.py:200-205`) ? this CP line is the token?execution join point. The worker's side logs `[worker] wrote completion marker -> <done\_uri>` in the **execution's** logs (`backend/worker/\_\_main\_\_.py:188`), closing the loop from the other end.
3. **Provenance on the result**: a successful cloud job's `result.compute` = `{path:"cloud\_run", job, region, execution, outputs\_uri}` and the CP logs `[COMPUTE] cloud\_run dispatched: execution=? job=? region=? out=? video\_id=?` (`orchestration.py:384-400`); the docstring's own cross-check is `gcloud run jobs executions describe <execution>` (`orchestration.py:386`). Every pipeline run also ends with `[COMPUTE] video\_id=? requested=? actual=? dispatched=? status=?` (`orchestration.py:632-640`).
4. On a **poll timeout** the execution name is carried inside the failure message itself (`cloud\_run.py:281-285`), so it reaches the client. But on other dispatch failures the `compute` dict in the failed result has **no execution field** (`orchestration.py:206-221`) ? the CP `dispatched` log line is then the only record. Note the token/execution are **not persisted** on the job row; if the instance recycles, this mapping survives only in Cloud Logging.

3.4 GAPS ? what an operator cannot currently see

1. **No request tracing/replay** ? issue **#326** (open). No request id crosses the API?worker?L4 boundary.
2. **No per-video runlog surface** ? issue **#328** (open). `emit\_indexer\_report` writes per-video reports on every run (`orchestration.py:645-660`) and the paths ride the job result (`main.py:1023`), but they are **server-local paths** on an ephemeral instance ? unreachable by client or operator after recycle.
3. **`job\_id` is absent from nearly all pipeline logs** ? `[COMPUTE]`, `[API] cloud-serving?`, `cloud-run: dispatched?` key on video_id/execution, not job_id (`orchestration.py:325-329, 393-400, 632-640`; `cloud\_run.py:200-205`). Only crash/park lines carry it (`job\_worker.py:189-193`). Triage from a user's job_id starts with an extra DB/API hop that dies with the instance.
4. **Job history is ephemeral** (issue #471) ? the `error` column and `result` JSON, the primary triage record, vanish on every deploy/recycle; only what was logged survives.
5. **Stuck-queue blindness** ? no per-job wall-clock timeout and no log when a job wedges the single drain lane (`job\_worker.py:40-48`); the failure mode is pure silence.
6. **Client 4xx are unlogged app-side** ? upload 409/413, jail 400s and auth 401s raise HTTPException without a logger call (`uploads.py:127-170`; `main.py:910-913`), leaving only status codes in request logs.
7. **No metrics/alerting** ? zero counters; failure detection is entirely reactive log-grepping.

Top 3 cheapest wins (all log-only, no infra):

1. **One structured terminal line per job** in `\_run\_claimed\_job` ? `[JOB] id=? kind=? status=? video\_id=? compute=? execution=? wall=?s error=?` at done/failed/parked (`job\_worker.py:177-193`). Single choke point; makes every triage class above greppable by the id the client actually has, and closes gap 3 for ~10 lines.
2. **Persist/log the execution name + dispatch token at dispatch time** keyed by job_id ? e.g. thread job_id into the dispatch logs, or stamp `execution` into `progress` ("dispatching (cloud_run exec=?)"). Cost: a few lines; makes job?execution correlation survive crash-before-result.
3. **Mirror the per-video report JSON** to `gs://?/outputs/<video\_id>/` next to `intervals.csv` (the worker already pushes artifacts there ? `worker/\_\_main\_\_.py:327`) and/or serve it via a small authenticated endpoint ? a minimal first slice of #328 that also survives instance recycling (gaps 2 and 4).

Launch-facing note: these gaps are operability debt, not A13 blockers by themselves ? but with

A10/A11 still owner-gated, gaps 4+5 (silent stuck jobs on an ephemeral store) are the ones most likely to burn an alpha tester: a tester whose job wedges gets silence, and a redeploy erases the evidence. Acceptable at 4 users with email support (\$4.6); not beyond.

4. Privacy & consent copy audit

Verdict: the marketing site has a real, decent privacy policy in all 6 locales ? but the Studio SPA (the surface testers actually upload through) contains **zero** privacy/consent copy, and two published claims are **false** on the hosted alpha: the first-run wizard's "Everything runs on this device ? your videos never leave it." and the policy's "We process your video on our own infrastructure" (it runs on rented Google Cloud in Singapore). Both, plus the missing SPA disclosure, are filed as **khelsutra** issue #154 ? the copy launch blockers. One code-level guarantee (training capture hard-denied) is **stronger** than the copy promises; storage location, retention reality, and deletion mechanics are **weaker**.

4.1 What the product says today

Marketing site (`khelsutra` repo, `/privacy`, localized in en/kn/hi/es/da/id ? all 6 catalogs carry the policy keys; English authoritative per `locales/en/common.json:574`):

- Full plain-language policy, "Last updated: 22 June 2026" (`locales/en/common.json:491-577`). Key lines:
 - "We do **not** use your footage to train or improve our models **unless** you give us clear, explicit, opt-in permission ? and even then, never footage involving minors." (:523)
 - "We process your video on our own infrastructure" (:528) ? **false** on the hosted alpha (Google Cloud Run + GCS, Singapore). Same defect class as the wizard line; khelsutra #154.
 - "We may use **trusted** third-party AI and cloud services" (for example, Google's Gemini API) to detect rallies." (:529) ? conditional wording, so not false in the alpha, but stale (\$4.2).
 - "We **never** publish your footage" (:530); processors named: Cloudflare, Google (Gemini API), email delivery (:538-540). **Google Cloud Storage / Cloud Run is not named**; no storage location is given.
 - Retention: "keep your information and footage only as long as we need it to run the alpha? then delete or anonymise it. You can ask us to delete? at any time" (:551); rights include "Delete your data and uploaded footage" (:558); minors never trained/published (:564); "gated behind an authenticated, invite-only sign-in" (:568).
 - Footer trust line: "Run from Bengaluru, India" (:488) ? in tension with all footage being stored/processed in **Singapore**; a counsel-review nuance alongside the India?Singapore transfer.
- Access-request form: email-consent checkbox + "We never publish your footage; footage involving minors is never used to improve the tool." with a `/privacy` link (:180-181); capture checklist: "Please record adults with their okay." (:280).

Studio SPA (`khelsutra/web/src` ? what alpha testers upload through):

- **No privacy copy at all**: a repo-wide search of `web/src` for privacy/consent strings and `/privacy` links returns nothing (grep, 0 matches). No link to the policy, no upload-moment disclosure.
- The only pre-upload acknowledgment is about **cost**, not data: "I understand cloud processing may cost a little." (`locales/en/common.json:110-112`, rendered `web/src/components/IngestPanel.tsx:211-224`).
- **False claim**: the first-run wizard ? shown once per browser before the ingest panel (`web/src/App.tsx:347-349`) ? unconditionally renders "Everything runs on this device ? your videos never leave it." (`web/src/components/SetupWizard.tsx:58`, string at `locales/en/common.json:612`). On the hosted alpha the cloud path uploads footage to GCS. Must be fixed or suppressed for the hosted deployment **before** any tester uploads (khelsutra #154). Wizard step 1 also advertises connecting a Gemini key (:604-608), which the alpha build never uses.

4.2 Engine reality vs. copy

| Alpha reality | Evidence | Disclosed today? |
|---|---|---|
| Uploads staged to `gs://khelsutra-rally-serving/videos/?` in **asia-southeast1** (Singapore);

reels mirrored to `~/highlights/?` in the same bucket |
`deploy/cloudrun/gen\_service\_yaml.py:42-45,57-62`; e2e runlog §3 | ****No ? contradicted.**** Policy affirmatively claims "our own infrastructure" (`:529` context, line `:528`); never names Google Cloud or a region. India?Singapore transfer undisclosed |
| Reel downloads are ****bearer signed URLs, ~1 h expiry**** ? anyone holding the link can fetch | e2e runlog §3.5 (`X-Goog-Expires=3600`) | No |
| ****Per-instance SQLite**** ? a redeploy wipes job/reel records while the bytes ****persist** in the bucket** (orphaned data; old download URLs die) | e2e runlog §5 (#471) | No |
| ****No retention automation**** ? a sweep tool exists (7-day default) but nothing schedules it; ****no user-facing delete endpoint**** (no DELETE routes in `backend/main.py`) |
`tools/cleanup\_caches.py:10-12`; ****#301 open**** (re-confirmed via `gh`, 2026-07-04) | ****Over-promised.**** Policy promises deletion on request + limited retention; today fulfillment is manual operator gsutil work (§5.6 R5) |
| ****Training capture HARD-DENIED****: `consent\_attested()` returns `False` for every job, and `flywheel.enabled` defaults off ? two independent guards; claimed consent fields are ignored |
`backend/flywheel.py:44-52,144-151`; `tests/test\_flywheel.py:38-55` | Copy's "never train without explicit opt-in" (`:523`) is true and ****code-enforced stronger**** today. A launch asset ? say it |
| ****Gemini/AI handoff DISABLED**** (`indexing.skip\_ai\_handoff=true`; `gemini\_key\_present=false` re-verified live at rev 00010) ? no footage reaches any third-party AI in the alpha config |
`gen\_service\_yaml.py:56`; live battery restore posture | Policy's conditional "may use Gemini" over-discloses (fine); SPA wizard still sells a Gemini step (`:604-608`) ? stale for the alpha |
| Reachability = ****Cloudflare Access, same 4-email allowlist policy****, on `api.khelsutra.guru`; edge-auth verified live | a1-cuj §6; e2e runlog §6; battery restore | Matches policy's "invite-only sign-in" (`:568`) |
| ****No content-safety/NSFW pre-flight**** ? nothing screens what a tester routes into the bucket/GPU | ****#332 open**** (re-confirmed via `gh`, 2026-07-04) | No acceptable-use line anywhere in tester-facing copy |
| Upload validation floor ?1280×720 (rejects smaller) | e2e runlog §3.2 | No (expectation-setting, minor) |

Launch-facing constraint: ****A10 (ship-gate/claims) and A11 (WASB-C3 provenance / "no commercial model sharing" copy guardrail) are OWNER-GATED and open**** (`docs/ALPHA\_LAUNCH\_READINESS.md:109-110`) ? alpha copy must make no model-quality claims and no model-sharing/commercial claims until resolved.

4.3 What we must tell testers (checklist)

1. ****Where footage goes:**** uploads and finished reels are stored in a private Google Cloud bucket in ****Singapore**** (asia-southeast1); Google Cloud must be named as a processor alongside Cloudflare.
2. ****Who can reach it:**** invite-only ? a Cloudflare Access email allowlist (currently 4 people) gates the app; operators can access stored footage.
3. ****Download links are shareable:**** reel URLs work for ~1 hour for anyone who has them ? don't forward them if the footage is sensitive.
4. ****Retention is manual today:**** footage is kept while the alpha runs; deletion is on request via hello@khelsutra.guru (automated cleanup not live ? #301); a redeploy may kill in-flight jobs and old links while the footage bytes remain until swept.
5. ****Training is OFF, in code:**** no footage is captured for model training in this build ? hard-disabled, not just policy ? and any future training will be a separate explicit opt-in, never involving minors.
6. ****No third-party AI in this build:**** nothing is sent to Gemini or any external AI (`skip\_ai\_handoff`).
7. ****Tester responsibility:**** adults-only, with their okay, badminton footage only ? there is ****no automated content screening**** yet (#332), so uploads are trusted.
8. ****Quality/limits framing:**** ?720p required; no model-quality or model-sharing claims (A10/A11 gated); see §5.3 for the honest expectation-setting line.
9. Ship the above in ****all 6 locales**** (or explicitly English-only with the `:574` authoritative-English caveat).

4.4 Draft alpha disclosure copy (landing page / pre-upload)

> ****Before you upload ? what happens to your footage.**** This alpha runs in the cloud: when you add a match video, it's uploaded to KhelSutra's private storage on Google Cloud (Singapore region) so a rented GPU can find your rallies, and your finished reel is stored there too.

Access is invite-only, and we never publish or share your footage. We keep uploads and reels only while the alpha runs and delete them whenever you ask (email hello@khelsutra.guru ? automatic cleanup is still being built). Your footage is **not** used to train our models ? training capture is switched off in this build, in the code, and would only ever happen later with your separate, explicit opt-in (never footage involving minors).

>

> - **Stored:** private Google Cloud bucket, Singapore region ? uploads + reels, invite-only access.

> - **Not sent to AI services:** this build uses no Gemini or other third-party AI on your footage.

> - **Not training data:** capture is hard-disabled; future training = separate opt-in, adults only.

> - **Download links:** work for about 1 hour and for anyone who has them ? share with care.

> - **Deletion:** on request today, automatic later; a redeploy may cancel in-progress jobs and old links (just re-process).

> - **Please upload only** badminton footage of adults who are okay being recorded ? there's no automated screening yet, so we're trusting you.

4.5 Gaps, ranked

1. **Two false published claims ? khelsutra #154 (launch blockers):** (a) SPA wizard **"your videos never leave it"** (``SetupWizard.tsx:58``); (b) policy **"our own infrastructure"** (``locales/en/common.json:528``). Same standard, same severity: fix or suppress both for the hosted deployment **before any tester uploads**.
2. **Zero privacy surface in the Studio SPA** ? no policy link, no upload-moment disclosure; the only checkbox is about cost (``:110-112``). Add the \$4.4 copy + link at the upload panel (the third khelsutra #154 item).
3. **Storage processor + location undisclosed** ? Google Cloud / Singapore appear nowhere in the policy (``:538-540`` names only Cloudflare/Gemini/email); India?Singapore transfer + the "Run from Bengaluru" footer line (``:488``) are counsel-review items.
4. **Retention/deletion over-promise** ? policy promises timely deletion (``:551,558``) but there is no automation (#301), no delete endpoint (#447), and redeploys orphan bucket data; either land minimum automation or soften/date the promise. A wider-invite gate (\$0).
5. **No acceptable-use/content-screening line** (#332 open) ? mitigated by the 4-email allowlist, but the copy should state the trust assumption before the allowlist grows.
6. **Signed-URL shareability undisclosed** (1 h bearer links).
7. **Stale Gemini copy** ? SPA wizard Gemini step (``:604-608``) and policy ``:529`` overstate third-party AI for this build; a one-line "this build sends nothing to third-party AI" is both accurate and a selling point.
8. **A11 copy guardrail** ? until WASB-C3 is resolved, keep all "own your model" marketing (``:287``) free of commercial-sharing implications (owner-gated, ``docs/ALPHA_LAUNCH_READINESS.md:110``).

4.6 Tester onboarding one-pager (send with the invite)

Everything an invited tester needs, in one message ? assemble from:

- **URL + login:** ``https://api.khelsutra.guru`` ? enter your invited email ? one-time PIN from Cloudflare (no account creation).
- **What to upload:** badminton match footage, ``mp4/.mov``, **at least 1280x720 and ~30 fps** (phone/GoPro defaults are fine; lower resolutions are rejected), up to 4 GB declared ? in practice keep files a few GB or less.
- **What to expect:** processing a full match takes **~15-20 minutes** (cloud GPU: ~14 min process + 3-5 min compile measured, ai-cuj \$3); the app finds ***candidate rally segments*** ? expect roughly-right boundaries with some rallies split in two (\$5.3); pick the segments you want and compile a reel.
- **Known rough edges:** if a job or download link dies after we ship an update, just re-process (a deploy resets in-flight jobs ? #471); download links expire after ~1 hour.
- **Privacy:** the \$4.4 disclosure + a link to ``/privacy``.
- **Reporting problems:** email **hello@khelsutra.guru** with the approximate time, what you did, and the job id from the status panel if visible. (There is no in-app feedback yet ? A12 is deferred ? so email is the loop.)

5. Known limits + rollback/disable steps

5.1 Locked alpha shape (the limits that are *by design*)

These are deliberate, config-pinned constraints of the live A1 stack ? launch copy and tester expectations must match them.

```
Locked decision	Evidence
**WASB-only detection ? no Gemini anywhere in serving.** `indexing.skip_ai_handoff: true` is	
baked into the control-plane manifest ("candidate windows ARE the rallies; no Gemini key in this	
service")	`deploy/cloudrun/gen_service_yaml.py:55-56`; live-verified
`gemini_key_present=false` (e2e runlog §2; re-verified at rev 00010, live battery)	
**Single region, single GPU.** Everything is in asia-southeast1 (`gen_service_yaml.py:43`);	
the project's GPU quota is `GPUS_ALL_REGIONS=1` ? one L4 execution at a time, globally (the live	
concurrency smoke §3.0 confirmed executions serialize)	
`docs/archives/decisions/DECISIONS.md:675`; worker Job shape e2e runlog §1	
**One control-plane instance (`maxScale=1`).** Deliberate #471 option-1 mitigation ? per-	
instance SQLite state means >1 instance splits the job queue	
`gen_service_yaml.py:25-27,97,145-150`; e2e runlog §1	
**Upload size cap: 4096 MB whole-file, 95 MB parts.** `SecurityConfig.max_upload_mb: int =	
4096`, `max_part_mb: int = 95` (`backend/config/models.py:370-371`), enforced at	
`backend/api/uploads.py:57-58`; the 413 rejection was observed live (§3.0). Parts stay ?95 MB	
because free-tier edge proxies cap bodies ~100 MB (`models.py:368-369`)	files cited; live
battery	
? `[analysis]` the 4 GB cap is theoretical on this deploy: uploads land under	
`/tmp/apphome/output/uploads` (a1-cuj §5.2) on a **4Gi-memory** instance	
(`gen_service_yaml.py:152`) whose `/tmp` is RAM-backed (`deploy/cloudrun/README.md:46-47`) ? a	
multi-GB upload would exhaust instance memory well before the configured cap. Not observed live;	
treat effective cap as ?4 GB	`[analysis]`
**Upload validation floor: ?1280x720 and ?29.9 fps.** A 640x360 clip was rejected live by	
`resolution_fps_check`	e2e runlog §3.2/§5; defaults at `backend/config/models.py:25-27`
**Auth posture:** Cloudflare Access at `api.khelsutra.guru`, "KhelSutra Studio (alpha)" Access	
app, **4-email allowlist**, plus in-app CF-JWT verification (401 without `Cf-Access-Jwt-
Assertion`) | a1-cuj §6; e2e runlog §6; battery restore posture |
```

5.2 Known-limits table (open issues + live rough edges)

```
Limit	Impact on the alpha	Source
**Control-plane state is per-instance ephemeral SQLite** ? a redeploy/restart wipes		
jobs/videos/segments; `/api/highlights` then 404s on the DB check (`backend/main.py:1052`) even		
though the reel still exists in gs://. A deploy invalidates in-flight jobs and past download		
URLs	Any redeploy mid-tester-session breaks their job polls and download links; re-processing	
regenerates them	#471 (open); e2e runlog §5	
**Google-managed cert on `api.khelsutra.guru` can stall at ~90-day renewal** behind the		
proxied CF CNAME. **Healthy today** (live battery, 2026-07-04:		
`Ready;CertificateProvisioned;DomainRoutable = True;True;True`); verify again before ~2026-09-30		
(§2.7)	Silent future outage risk; owner watch item with a dated mechanism (§2.7)	#472
(open); live battery		
**CORS default is `allow_origins=['*']`** ? the deployed manifest sets no		
`RALLY_CORS_ALLOW_ORIGINS` (env block `gen_service_yaml.py:111-119`), so the wildcard is active		
on the hosted service too `[analysis]`. **Live mitigating fact:** the CP serves the bundled SPA		
at its own root (`GET /` ? 200 `text/html`, live battery), so the tester path is **single-		
origin** ? no cross-origin API call, no CF-Access OPTIONS-preflight dependency; the wildcard		
residual and the preflight concern apply only to a split-origin deploy (e.g. a separately-hosted		
`app.khelsutra.guru`, state [unverified this session])	Residual cross-origin exposure only if	
a split-origin front-end is (re)introduced; unresolved audit residual (R1-12)	#443 (open);	
live battery		
**No content-safety/NSFW pre-flight** before processing. The WASB-only alpha removes the send-		
to-Gemini ToS risk, but abusive uploads still burn paid L4 time and produce garbage	Accepted	
for a 4-email allowlist, not for wider invites	#332 (open)	
**Path-jail vs storage schemes**: gs:// ingest through the jail was unblocked by PR #465 (live		
battery: foreign-bucket and boundary attacks 400 at the API surface), but the general scheme-		
exempt design (s3://, gdrive-api://) remains open	gs:// works today; other cloud schemes 400	
#327 (open); live battery		
**No automated upload retention** ? the sweep tool exists but nothing schedules it, and it		
```

targets the `*local*` upload dir. Staged tester uploads persist in ``gs://khelsutra-rally-serving/videos/<video_id>/?`` indefinitely; no bucket lifecycle policy found in-repo [unverified live] | Tester footage accumulates in the bucket until manually deleted (\$5.6 R5) | #301 (open); staging path ``backend/orchestration.py:284`` |

| ****Observability gaps****: no user-facing per-video trace endpoint (#328); no request capture/replay, and CF-edge failures (302s at Access) never reach engine logs so they're invisible in-app (#326) | Tester-reported failures need manual CF-dashboard + Cloud Logging triage | #328, #326 (open) |

| ****Performance****: hosted processing is CPU-bound (ffmpeg/cv2) ? measured ~65% of wait is software encode (#331); WASB decode is single-threaded in streaming mode so the paid L4 sits ~80% idle, ~2x realtime detection (#349). Live A1 numbers: ~13m45s process (incl. cold pull) + ~3m26s74m54s compile for a ~7-min match | Set tester latency expectations in minutes, not seconds; GPU cost per match is inflated | #331, #349 (open); a1-cuj \$3; e2e \$3.4 |

| ****Golden eval data is machine-split**** (GPU box ? dev sessions, sync incomplete) | Slows quality-lane reproduction; no direct tester impact | #175 (open) |

| ****`gcloud run services replace` with an unchanged manifest is a NO-OP**** ? it does not re-resolve ``:latest`` after a rebuild; force a new revision with ``gcloud run services update --image`` | Operator footgun: a "redploy" can silently ship the old image | e2e runlog \$5 |

| ****422-before-401 quirk****: malformed-JSON bodies get FastAPI's 422 before edge-auth runs; well-formed requests without a CF JWT still 401 | No exposure; don't misread a 422 during auth triage | e2e runlog \$5 |

| ****No video-delete API**** ? no supported way to remove a video/job record; only manual DB/bucket surgery | See \$5.6 R5 for the actual deletion procedure | #447 (open) |

5.3 GATED items that constrain launch copy (A10 / A11) ? and what testers WILL see

****A11 ? WASB-C3 provenance (weights NOT commercially cleared).**** The worker image itself carries the warning: `"the pretrained badminton weights are academic-data-trained ? provenance NOT cleared for commercial sharing? Resolve WASB-C3 before any public, non-owner serving"` (``deploy/cloudrun/Dockerfile:14-16``). The tracker keeps A11 GATED on owner/legal (``docs/ALPHA_LAUNCH_READINESS.md:110``) and names C3 a ****hard model-release gate**** (\$5 risk table, ``ALPHA_LAUNCH_READINESS.md:85``). Consequences:

- ****Who may use it:**** only the current 4-email invite-only allowlist. Honestly noted for the GO/NO-GO: those testers are already **non-owner** users, in tension with the Dockerfile's "before any public, non-owner serving" ? recording GO requires the owner to either resolve C3 or ****explicitly defer it with the documented "no commercial model sharing" copy guardrail**** (that is exactly what the A11 row requires). No wider invite wave until then.
- ****What copy must NOT claim:**** an "owned" or commercially cleared rally-detection model, any model sharing/download, or any commercial license posture. Tracker guardrail: `"Keep all alpha copy scoped to invite-only processing, not owned-model claims"` (``ALPHA_LAUNCH_READINESS.md:80``; also \$0 TL;DR and D1). Mitigating fact for the record: the weights are ****not baked into the image or distributed**** ? they're staged at runtime from the bucket and used server-side only (``Dockerfile:15-16``; ``deploy/cloudrun/README.md:15,29-30``).

****A10 ? ship-gate target not owner-ratified (#172).**** The F1@tIoU targets in ``gate_verdict`` are explicitly provisional, the A10 row is GATED on the owner decision (``ALPHA_LAUNCH_READINESS.md:109``), and the latest Gen-0 shadow run is ``ship=False`` (sign test 9/15, p=0.3036; ``ALPHA_LAUNCH_READINESS.md:214``). Gate-A served default was ****rejected**** (served mean ?F1 ?0.0328, ``ALPHA_LAUNCH_READINESS.md:205``). Consequence: ****the alpha can make no quality claims**** ? no accuracy numbers, no "best-in-class", not even the internal R2/tolF1 contrast as a headline (its strict per-video guard still blocks blanket promotion language, ``ALPHA_LAUNCH_READINESS.md:206``). Allowed framing until A10 is ratified: invite-only alpha that detects candidate rally segments and compiles reels, quality under active measurement.

****What testers will actually experience (the honest internal expectation ? NOT publishable as a quality claim):**** the WASB-only served path measures ****tolF1 0.364 / F1@0.5 0.472**** vs baseline-off ****0.138 / 0.237****, merge rate 0.135 vs 0.638 (``docs/ALPHA_LAUNCH_READINESS.md:142-153,206``) ? i.e. ****roughly-right rally boundaries, clearly better than nothing, with residual over-splitting**** (the strict per-video split-count guard still fails: expect some rallies delivered as 2+ segments). Concretely: 48 candidate segments on the ~7-min e2e match (e2e runlog \$3.1). This is the owner's alpha-burn-risk calibration: testers will mostly get usable reels but will need to deselect/merge split segments by hand. No number in this paragraph may appear in tester-facing copy until A10 closes.

5.4 Validated beyond the alpha config (one-off, restored) ? the Gemini handoff leg

The alpha **stays WASB-only**; this subsection records that the alternative config was proven live and then rolled back (live battery, rev 00009 of the test window):

```
- Prep finding (fixed live): the `gemini-api-key` secret existed in Secret Manager but had
no accessor grant ? the documented secret-mount path had never actually been exercisable.
Fixed: `gcloud secrets add-iam-policy-binding gemini-api-key
--member=serviceAccount:492532266377-compute@developer.gserviceaccount.com
--role=roles/secretmanager.secretAccessor`. The grant was left in place (inert: the secret
is only mounted when a config asks for it).
- Job side: `gcloud run jobs update rally-worker --set-secrets GEMINI_API_KEY=gemini-api-
key:latest` worked; CP side: a manifest with `skip_ai_handoff=false` + the secret env booted
clean (the `AI_HANDOFF_KEY_MISSING` startup check passed; `gemini_key_present=true` in
`/api/capabilities`).
- Run: the same ~7-min match, `wasb_hybrid`, `compute=cloud` ? success, 43 play segments
(vs 48 in WASB-only mode ? the AI pass genuinely refined/filtered). Worker logs prove real
third-party traffic: `[Gemini Handoff 1.48] Submitting multimodal request to gemini-flash-
latest` ? `POST ?/models/gemini-flash-latest:generateContent "HTTP/1.1 200 OK"` for all 48
windows; `Phase 3 completed. AI confirmed 43 semantic play segments.` Execution `rally-worker-
khx58`; wall comparable to WASB-only, ~4?11 s per handoff window.
- Restored after: secret unmounted from the Job (`--remove-secrets`), alpha manifest back
(`skip_ai_handoff=true`), `gemini_key_present=false` re-verified at rev 00010.
```

Implications: (a) the Gemini leg is **deployable in hours, not days**, if the owner ever flips the alpha posture ? but flipping it changes the privacy story (§4: "no third-party AI in this build" stops being true, and footage windows WOULD reach Google's Gemini API), so it is a copy + consent change, not just a config change; (b) the secret-accessor gap is exactly the class of "documented but never exercised" path this review exists to catch.

5.5 Runtime service-account blast radius (fix before wider invites)

The control plane and worker run as the **project default compute SA** (`492532266377-compute@developer.gserviceaccount.com`). What is known about its privileges: it holds self-`roles/iam.serviceAccountTokenCreator` and uses it for keyless signBlob (e2e runlog §4, repo-cited), it received `roles/secretmanager.secretAccessor` on `gemini-api-key` this session (§5.4), and per session records it holds **project-wide `roles/editor`** [unverified against live IAM ? one `gcloud projects get-iam-policy` check]. That SA backs a service carrying an **`allUsers` run.invoker** binding (required for CF-proxied traffic; the CF-JWT check is the auth). Net: an application-level RCE or SSRF in the CP would inherit near-project-wide write access `[analysis]`.

Recommendation (a NO-GO condition for any wider invite wave, §0): create a dedicated least-privilege SA for the stack ? roughly `storage.objectAdmin` scoped to `gs://kheIsutra-rally-serving`, `run.developer` (or narrower: jobs run + executions view) on `rally-worker`, self-`iam.serviceAccountTokenCreator` for signing, and per-secret `secretAccessor` only where mounted ? then re-deploy service + job with it and re-run the §2.4 posture checks plus a signed-URL download (the #475/#476 signing path is SA-sensitive). Acceptable to defer at 4 trusted testers; not beyond.

5.6 Rollback / disable inventory

All commands run from PowerShell (not git-bash ? MSYS mangles paths/flags, `deploy/cloudrun/README.md:129-133`), region `asia-southeast1`, project `gen-lang-client-0306026826` (`gen\_service\_yaml.py:42-43`).

R1. Disable the whole alpha (fast, reversible, keeps GCS data). **Two independent switches ?** throwing either one cuts testers off:

- Edge off:** disable (or delete the policy of) the "KheIsutra Studio (alpha)" CF Access application in Zero Trust ? Access ? Applications ? the login path dies at the edge (app identified at a1-cuj §6; "disable the Access application" is the documented teardown lever, `kheIsutra/deploy/DEPLOY\_ALPHA.md:156` ? that runbook is RETIRED but kept as the CF-mechanics reference). Dashboard click-op; exact click path [unverified].
- Origin off:** remove the public invoker that CF-proxied traffic rides in on:

```

...
gcloud run services remove-iam-policy-binding rally-control-plane --region asia-southeast1
--member=allUsers --role=roles/run.invoker
...

Requests then **403 at the Google Front End** ? this exact lever was exercised in the live
battery (public probes 403 for the whole test window, then re-added). `min-instances 1` CPU cost
continues until you also scale down or delete.
3. *Full teardown* (state is ephemeral anyway ? but staged videos/reels in gs:// survive):
...

gcloud run services delete rally-control-plane --region asia-southeast1
gcloud run jobs delete rally-worker --region asia-southeast1
...

(a1-cuj §7: bucket/secret/repo retained.)

```

****R2. Roll back to a previous image digest.**** Do NOT use `services replace` with an unchanged manifest ? it's a no-op (e2e §5). Pin the digest explicitly:

```

...
gcloud run services update rally-control-plane --region asia-southeast1 `
--image asia-southeast1-docker.pkg.dev/gen-lang-client-0306026826/rally/rally-
worker@sha256:<digest>
gcloud run jobs update rally-worker --region asia-southeast1 `
--image asia-southeast1-docker.pkg.dev/gen-lang-client-0306026826/rally/rally-
worker@sha256:<digest>
...

```

The `jobs update` re-pin is the exact mechanism used in the e2e run (e2e §1); the service image path/name from `gen\_service\_yaml.py:157` (one image serves both roles, #470). ****Last-known-good digest: `sha256:31475f27?`**** (running at rev 00010 as of 2026-07-04, edge-auth ON). Do not roll back to `sha256:ed2fe448?`: it predates #476 and reintroduces the broken >32 MiB signed-URL download (#475, e2e §4). Caveats: any new revision ****wipes the per-instance SQLite state**** ? in-flight jobs and existing download URLs die (#471); and if config (not code) is what regressed, regenerate the manifest instead: `python deploy/cloudrun/gen\_service\_yaml.py --edge-auth on --image <pinned-digest-ref> -o service.yaml` then `gcloud run services replace service.yaml --region asia-southeast1` (`gen\_service\_yaml.py:6-8,136-140`).

****R3. Revoke a tester.**** Zero Trust ? Access ? Applications ? "KhelSutra Studio (alpha)" ? edit the Allow policy's *Emails* rule and remove the address (policy mechanics per `khelsutra/deploy/DEPLOY\_ALPHA.md:113`; same-app/allowlist confirmed live at a1-cuj §6). Also revoke the user's active Access session in Zero Trust so an existing cookie doesn't outlive the policy edit ? [unverified: exact dashboard path; not in repo docs]. Defense-in-depth already holds: every `/api/process` re-verifies the CF JWT in-app (`backend/config/models.py:374-379`; e2e §6 shows 401 without it), so a revoked user can't fall back to hitting the run.app URL directly. There is no app-side account/data scoping to clean up (per-user subdirs are a future item, `models.py:364-366`).

****R4. Emergency-stop a running L4 execution.****

```

...
gcloud run jobs executions list --job rally-worker --region asia-southeast1
gcloud run jobs executions cancel <execution-name> --region asia-southeast1
...

```

Execution names look like `rally-worker-k49kg` (a1-cuj §3.2). The `cancel` verb is standard gcloud but not exercised in any repo runlog [unverified in-repo]. Known consequence chain, from repo sources: a killed container writes ****no completion/failure marker****, so the control plane's bucket-marker poll runs to its bounded timeout ? `ExecutionPolicy.max\_wait\_sec` default ****1800 s**** (`backend/infrastructure/execution\_policy.py:54`; `backend/compute/cloud\_run.py:245`) ? then surfaces a clean `cloud dispatch error` job failure, not a hang (`deploy/cloudrun/README.md:111-115`). GPU billing ****should**** stop when the execution stops, but the app-side cancel is ****best-effort**** (`backend/compute/cloud\_run.py:268-277`) and the cancel verb is unexercised in-repo ? ****[unverified]****: always confirm the execution actually terminated in the console, otherwise the L4 bills up to `--task-timeout 3600s`. The job row on the CP resolves ~30 min later or dies with the instance on the next redeploy (#471).

****R5.** Delete a user's data for a given `video\_id`.** Bucket is the durable store; the DB is ephemeral.

- ***Bucket objects*** (all three prefixes, plus the user's own source object if they supplied a raw gs:// path outside `videos/<video\_id>/`):

```
gcloud storage rm -r gs://khelesutra-rally-serving/videos/<video_id>/ # staged upload ?
orchestration.py:284 (input_root gen_service_yaml.py:61)
```

```
gcloud storage rm -r gs://khelesutra-rally-serving/outputs/<video_id>/ # worker outputs ?
a1-cuj §3.2
```

```
gcloud storage rm -r gs://khelesutra-rally-serving/highlights/<video_id>/ # mirrored reels ?
orchestration.py:748-755
```

Also sweep the dispatch done-markers (`<out\_uri>/\_dispatch/<token>.done`, `deploy/cloudrun/README.md:109-110`) ? exact prefix in the serving bucket [unverified].

- ***DB reality:** rows live in per-instance ephemeral SQLite (#471) with ****no delete API**** (#447 open ? the only known method is manual row surgery, impractical inside a Cloud Run instance). Practically: after the bucket `rm`, any surviving DB record is inert (`/api/highlights` 404s once the mirror check fails / on the `main.py:1052` record check after the next restart), and ****any redeploy wipes all DB rows anyway**** ? a blunt but real "forget everything" lever. This is adequate for a 4-email alpha but is not a real GDPR-style deletion story; flagged alongside #447/#301 in §4.5.

****Cross-cutting caution for every lever above:**** rollback ? data rollback. `gs://khelesutra-rally-serving` (weights, staged videos, outputs, reels) is untouched by service/job changes; state loss is confined to the CP's ephemeral SQLite (#471).

6. Open gates & owner questions

6.1 The three gated tracker rows

| Row | State | What closing it takes |
|--|---|--|
| **A10 ? ship-gate target + allowed claims** | GATED (owner decision, #172 open) | Owner ratifies metric/corpus-slice/threshold/significance + records which claims are allowed. Current evidence: Gen-0 `ship=False` (9/15, p=0.3036), Gate-A served default rejected, R2/tolF1 contrast strong but split-count-guarded (`docs/ALPHA_LAUNCH_READINESS.md:200-214`). Until then: zero quality claims (§5.3) |
| **A11 ? WASB-C3 provenance** | GATED (owner/legal) | Resolve C3, or **explicitly defer it** with the documented "no commercial model sharing" copy guardrail (`ALPHA_LAUNCH_READINESS.md:110`). The current 4 testers are already non-owner users ? the G0 record must own that tension (§0, §5.3) |
| **A12 ? P10/P11 corpus flywheel** | GATED (owner D13 reopen, `khelesutra/docs/STUDIO_MEDIA_LIFECYCLE.md`) | Reopen with a committed sequence, or record "explicitly deferred for alpha-plus" ? the record itself is what closes DoD #9. Practical alpha consequence today: no in-app feedback loop; tester feedback arrives by email only (§4.6) |

6.2 The tracker's open owner questions (§9), current status

- *Manual/offline corpus promotion for the first alpha?*** ? Still open; de facto the alpha runs with ****no**** promotion path (A12 gated). Recording "manual promotion, alpha-plus for automation" would close it and DoD #9 together.
- *Which hosted compute target?*** ? ****De facto answered by A1****: Cloud Run (service + L4 Job), live since 2026-07-04. Recommend recording it as answered in the tracker.
- *Acceptable alpha wording while C3 unresolved?*** ? Open and now ****urgent****: this is exactly the khelesutra #154 + §5.3 copy-guardrail package awaiting the owner's A11 defer/resolve decision.
- *Portable corpus manifest form?*** ? Open, engineering; no tester impact.
- *Close A9 on the Gen-0 non-serving bundle vs implement the TemporalMaxer swap?*** ? A9 is marked DONE on the Gen-0 bundle (`ALPHA\_LAUNCH\_READINESS.md:108,214`); the scorer-swap remains an open follow-up candidate. No alpha impact.

6.3 The khelesutra #154 copy blockers (cross-repo, filed 2026-07-04)

1. SetupWizard `"Everything runs on this device ? your videos never leave it."` ? false on the hosted alpha (``khelsutra/web/src/components/SetupWizard.tsx:58``).
2. Privacy policy `"We process your video on our own infrastructure"` ? false on Google Cloud (``khelsutra/locales/en/common.json:528``); Google Cloud/Singapore never named as processor/location.
3. The Studio SPA has `**zero consent/privacy copy**` at the upload moment ? ship \$4.4 (+ a ``/privacy`` link) at the upload panel.

These three are the `**GO conditions**` for the current allowlist (\$0). The remaining \$4.5 items (retention over-promise, acceptable-use line, signed-URL note, Bengaluru/Singapore tension, 6-locale rollout) are wider-invite gates or counsel items, not 4-tester blockers.

Changelog

- 2026-07-04 ? Created: A13 review assembled from the two A1 runlogs, a live test battery (failure modes, concurrency, Gemini-leg one-off, cert + posture checks), a four-section research fan-out with adversarial critique, and the tracker's A-series records. Delivered for owner GO/NO-GO.

37. user (2026-07-04T12:29:36.930Z)

File created successfully at: `C:\Users\avidu\Projects\khelsutra-guru\wt-a13\docs\ALPHA_LAUNCH_REVIEW_2026-07-04.md` (file state is current in your context ? no need to Read it back)

38. user (2026-07-04T12:29:52.475Z)

The file `C:\Users\avidu\Projects\khelsutra-guru\wt-a13\docs\ALPHA_LAUNCH_READINESS.md` has been updated successfully. (file state is current in your context ? no need to Read it back)

39. user (2026-07-04T12:29:55.726Z)

The file `C:\Users\avidu\Projects\khelsutra-guru\wt-a13\docs\ALPHA_LAUNCH_READINESS.md` has been updated successfully. (file state is current in your context ? no need to Read it back)

40. user (2026-07-04T12:30:00.328Z)

The file `C:\Users\avidu\Projects\khelsutra-guru\wt-a13\docs\ALPHA_LAUNCH_READINESS.md` has been updated successfully. (file state is current in your context ? no need to Read it back)

41. user (2026-07-04T12:30:13.507Z)

The file `C:\Users\avidu\Projects\khelsutra-guru\wt-a13\docs\DOC_STATUS.md` has been updated successfully. (file state is current in your context ? no need to Read it back)

42. user (2026-07-04T12:30:18.297Z)

The file `C:\Users\avidu\Projects\khelsutra-guru\wt-a13\docs\DOC_STATUS.md` has been updated successfully. (file state is current in your context ? no need to Read it back)

43. user (2026-07-04T12:30:27.195Z)

M docs/ALPHA_LAUNCH_READINESS.md
A docs/ALPHA_LAUNCH_REVIEW_2026-07-04.md
M docs/DOC_STATUS.md
[check-md-links] clean ? no broken internal links in live docs.

44. user (2026-07-04T12:32:22.334Z)

..... [100%]
===== warnings summary =====

```
tests/test_api_cloud_dispatch.py::test_full_api_process_to_query_via_cloud
tests/test_api_cloud_dispatch.py::test_local_override_survives_api_boundary
tests/test_api_cloud_dispatch.py::test_cloud_override_survives_api_boundary_on_local_server
tests/test_api_cloud_dispatch.py::test_full_api_process_job_result_carries_cloud_provenance
tests/test_api_endpoints.py::test_process_nonlocal_uri_skips_exists_and_fetches
tests/test_api_endpoints.py::test_process_remote_input_reports_use_durable_output_dir
tests/test_api_endpoints.py::test_process_bucket_uri_yields_queryable_video_id[gs://bucket/video
s/x.mp4]
tests/test_api_endpoints.py::test_process_bucket_uri_yields_queryable_video_id[gs://bucket/vide
os/x.mp4]
```

```
C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\site-
packages\google\auth\_default.py:113: UserWarning: Your application has authenticated using end
user credentials from Google Cloud SDK without a quota project. You might receive a "quota
exceeded" or "API not enabled" error. See the following page for troubleshooting:
https://cloud.google.com/docs/authentication/adc-troubleshooting/user-creds.
warnings.warn(_CLOUD_SDK_CREDENTIALS_WARNING)
```

```
-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
1659 passed, 10 skipped, 8 warnings in 103.81s (0:01:43)
```

45. user (2026-07-04T12:32:30.289Z)

All checks passed!

46. assistant (2026-07-04T12:32:36.291Z)

All gates green. Final publication scrub check on the new doc.

47. user (2026-07-04T12:32:38.534Z)

```
3:> **Status:** `REPORT` ? review **DELIVERED 2026-07-04**; **GO/NO-GO awaiting owner** .
**Owner:** @avidullu . **Created:** 2026-07-04
162:[Omitted long matching line]
163:| **Billing / budget-alert alignment** | @avidullu | Monthly calendar entry; first check
**before the GO decision** | Billing console: actual monthly spend for the standing stack;
confirm the budget-alert threshold matches the real ($300) spend gate ($2.6, both [unverified])
|
```

48. assistant (2026-07-04T12:33:12.951Z)

All clean ? the only matches are the owner's public GitHub handle (@avidullu), which the tracker already uses. Work complete; nothing committed.

Report

```
**Files changed** (worktree `C:\Users\avidu\Projects\khelsutra-guru\wt-a13`, branch
`docs/a13-launch-review` at `122f174`, no commits made):
```

```
- **NEW** `C:\Users\avidu\Projects\khelsutra-guru\wt-a13\docs\ALPHA_LAUNCH_REVIEW_2026-07-04.md`
? the A13 owner-facing launch review (~6 sections: TL;DR + recommendation, DoD scorecard +
5-minute browser-CUJ checklist, runbook incl. cost + watch-items, observability/triage incl. the
live failure-mode battery, privacy/consent audit, known limits + rollback incl. the Gemini-leg
validation, SA blast radius, tester one-pager, open gates). Registered via `git add -N` (intent-
to-add, content unstaged) so the `git ls-files`-based link gate actually scans it.
- `docs\ALPHA_LAUNCH_READINESS.md` ? A13 row ? `GATED`, PR column ? `review DELIVERED 2026-07-04`
? docs\ALPHA_LAUNCH_REVIEW_2026-07-04.md; GO/NO-GO awaiting owner`; $10 auth-posture/rollback
box ? `[x]` with citations (e2e runlog $6 + review $5.6); browser-CUJ box left `[ ]`; changelog
line added. All edits were in-place so the review's `ALPHA_LAUNCH_READINESS.md:<line>` citations
stay valid.
- `docs\DOC_STATUS.md` ? new $3c REPORT row for the review; $3a ALPHA_LAUNCH_READINESS row's
status/next-action refreshed (A0?A9 done, A13 delivered, owner actions listed).
```

```
**Gate outputs (verbatim final lines, run from the worktree root):**
```

```
1. `python -m tools.check_md_links` ? `[check-md-links]` clean ? no broken internal links in live
```

docs.`

2. `python -m pytest tests/ -q -p no:cacheprovider` ? `1659 passed, 10 skipped, 8 warnings in 103.81s (0:01:43)`

3. `python -m ruff check .` ? `All checks passed!`

****Review verdict/emphases (5 bullets):****

- ****Conditional GO for the current 4-email allowlist**** ? conditions: fix khelsutra #154 copy blockers (SetupWizard "never leaves this device" + policy "our own infrastructure" are **false** on the hosted alpha; SPA has zero consent copy) and owner completes the 5-minute browser-CUJ checklist; ****NO-GO** for any wider wave** until A10/A11 close plus least-privilege SA and retention fixes. Recording GO also requires the owner to explicitly defer A11 (current testers are already non-owner users vs the Dockerfile's C3 warning).
- ****DoD scorecard: 5 of 10 boxes done**** ? the browser CUJ has **never** been run through a real browser (all evidence is curl+token); the live SPA-served-at-root fact makes the tester path single-origin (kills the CF-Access CORS-preflight worry), but only the owner checklist closes the box.
- ****Live battery folded in as authoritative****: 11-probe failure-mode table (jail holds against foreign-bucket/boundary attacks, clean 4xx/async failures), 3-job concurrency smoke all-green, cert healthy today with a dated ~2026-09-30 watch item + owner/mechanism, and the improved auth-off-window practice (remove `allUsers` for the window) now the runbook standard.
- ****Gemini leg recorded as "validated beyond the alpha config****: real gemini-flash-latest calls, 48?43 segments, secret-accessor grant gap found+fixed live, fully restored to WASB-only ? deployable in hours if ever wanted, but flagged as a privacy-copy change, not just config.
- ****Every critique fix applied and the doc scrubbed for repo publication****: cost subsection (idle 2vCPU/4Gi CP, ~\$1?2/L4-run precedent, monthly \$ `[unverified ? owner billing check]`), honest tester-expectation numbers (tolF1 0.364/F1@0.5 0.472 vs 0.138/0.237, over-splitting persists, no public claims until A10), SA `roles/editor` blast-radius paragraph, tester onboarding one-pager, all five contradiction fixes, and zero secret paths / tester emails / session-memory citations.